# AOS-CX 10.08 IP Routing Guide

## 4100i, 6000, 6100, 6200 Switch Series

a Hewlett Packard
Enterprise company

Published: September 2021
Edition: 2

**Open Source Code**

This product includes code licensed under the GNU General Public License, the GNU Lesser General Public License, and/or certain other open source licenses. A complete machine-readable copy of the source code corresponding to such code is available upon request. This offer is valid to anyone in receipt of this information and shall expire three years following the date of the final distribution of this product version by Hewlett Packard Enterprise Company. To obtain such source code, send a check or money order in the amount of US $10.00 to:

Hewlett Packard Enterprise Company
6280 America Center Drive
San Jose, CA 95002
USA

**Notices**

The information contained herein is subject to change without notice. The only warranties for Hewlett Packard Enterprise products and services are set forth in the express warranty statements accompanying such products and services. Nothing herein should be construed as constituting an additional warranty. Hewlett Packard Enterprise shall not be liable for technical or editorial errors or omissions contained herein.

Confidential computer software. Valid license from Hewlett Packard Enterprise required for possession, use, or copying. Consistent with FAR 12.211 and 12.212, Commercial Computer Software, Computer Software Documentation, and Technical Data for Commercial Items are licensed to the U.S. Government under vendor's standard commercial license.

Links to third-party websites take you outside the Hewlett Packard Enterprise website. Hewlett Packard Enterprise has no control over and is not responsible for information outside the Hewlett Packard Enterprise website.

**Acknowledgments**

Intel®, Itanium®, Optane™, Pentium®, Xeon®, Intel Inside®, and the Intel Inside logo are trademarks of Intel Corporation in the U.S. and other countries.

Microsoft® and Windows® are either registered trademarks or trademarks of Microsoft Corporation in the United States and/or other countries.

Adobe® and Acrobat® are trademarks of Adobe Systems Incorporated.

Java® and Oracle® are registered trademarks of Oracle and/or its affiliates.

UNIX® is a registered trademark of The Open Group.

All third-party marks are property of their respective owners.

# Contents

# Chapter 1
# About this document

This document describes features of the AOS-CX network operating system. It is intended for administrators responsible for installing, configuring, and managing Aruba switches on a network.

## Applicable products

This document applies to the following products:

- Aruba 4100i Switch Series (JL817A, JL818A)
- Aruba 6000 Switch Series (R8N85A, R8N86A, R8N87A, R8N88A, R8N89A)
- Aruba 6100 Switch Series (JL675A, JL676A, JL677A, JL678A, JL679A)
- Aruba 6200 Switch Series (JL724A, JL725A, JL726A, JL727A, JL728A)

## Latest version available online

Updates to this document can occur after initial publication. For the latest versions of product documentation, see the links provided in Support and Other Resources.

## Command syntax notation conventions

| Convention | Usage |
|---|---|
| `example-text` | Identifies commands and their options and operands, code examples, filenames, pathnames, and output displayed in a command window. Items that appear like the example text in the previous column are to be entered exactly as shown and are required unless enclosed in brackets (`[ ]`). |
| **example-text** | In code and screen examples, indicates text entered by a user. |
| Any of the following:<br>■ *<example-text>*<br>■ <example-text><br>■ *example-text*<br>■ *example-text* | Identifies a placeholder—such as a parameter or a variable—that you must substitute with an actual value in a command or in code:<br>■ For output formats where italic text cannot be displayed, variables are enclosed in angle brackets (< >). Substitute the text—including the enclosing angle brackets—with an actual value.<br>■ For output formats where italic text can be displayed, variables might or might not be enclosed in angle brackets. Substitute the text including the enclosing angle brackets, if any, with an actual value. |
| \| | Vertical bar. A logical OR that separates multiple items from which you can choose only one.<br>Any spaces that are on either side of the vertical bar are included for readability and are not a required part of the command syntax. |
| { } | Braces. Indicates that at least one of the enclosed items is required. |

| Convention | Usage |
|---|---|
| `[ ]` | Brackets. Indicates that the enclosed item or items are optional. |
| … or<br>`. . .` | Ellipsis:<br>▪ In code and screen examples, a vertical or horizontal ellipsis indicates an omission of information.<br>▪ In syntax using brackets and braces, an ellipsis indicates items that can be repeated. When an item followed by ellipses is enclosed in brackets, zero or more items can be specified. |

# About the examples

Examples in this document are representative and might not match your particular switch or environment.

The slot and port numbers in this document are for illustration only and might be unavailable on your switch.

## Understanding the CLI prompts

When illustrating the prompts in the command line interface (CLI), this document uses the generic term `switch`, instead of the host name of the switch. For example:

`switch>`

The CLI prompt indicates the current command context. For example:

`switch>`

Indicates the operator command context.

`switch#`

Indicates the manager command context.

`switch(`*`CONTEXT-NAME`*`)#`

Indicates the configuration context for a feature. For example:

`switch(config-if)#`

Identifies the `interface` context.

## Variable information in CLI prompts

In certain configuration contexts, the prompt may include variable information. For example, when in the VLAN configuration context, a VLAN number appears in the prompt:

`switch(config-vlan-100)#`

When referring to this context, this document uses the syntax:

`switch(config-vlan-<VLAN-ID>)#`

Where *`<VLAN-ID>`* is a variable representing the VLAN number.

# Identifying switch ports and interfaces

Physical ports on the switch and their corresponding logical software interfaces are identified using the format:

*`member/slot/port`*

## On the 4100i Switch Series

- *member*: Always 1. VSF is not supported on this switch.
- *slot*: Always 1. This is not a modular switch, so there are no slots.
- *port*: Physical number of a port on the switch.

For example, the logical interface `1/1/4` in software is associated with physical port 4 on the switch.

### On the 6000 and 6100 Switch Series

- *member*: Always 1. VSF is not supported on this switch.
- *slot*: Always 1. This is not a modular switch, so there are no slots.
- *port*: Physical number of a port on the switch.

For example, the logical interface `1/1/4` in software is associated with physical port 4 on the switch.

### On the 6200 Switch Series

- *member*: Member number of the switch in a Virtual Switching Framework (VSF) stack. Range: 1 to 8. The primary switch is always member 1. If the switch is not a member of a VSF stack, then member is 1.
- *slot*: Always 1. This is not a modular switch, so there are no slots.
- *port*: Physical number of a port on the switch.

For example, the logical interface `1/1/4` in software is associated with physical port 4 in slot 1 on member 1.

# Chapter 2
# Virtual Routing and Forwarding (VRF)

VRF is a technology that allows multiple instances of a routing table to co-exist within the same router. Because the routing instances are independent, the same or overlapping IP addresses can be used without conflicting with each other. Network functionality is improved because network paths can be segmented without requiring multiple routers.

## VRF support

The 4100i, 6000, 6100, and 6200 Switch Series support predefined VRFs:

- **`mgmt`**: Assigned to the management port and is used to isolate management traffic (6200 Switch Series only).
- **`default`**: All other interfaces are automatically assigned to the VRF `default` (6000, 6100 and 6200 Switch Series).

User-defined VRFs are not supported on the 4100i, 6000, 6100, and 6200 Switch Series.

# Chapter 3 Loopback

Loopback interfaces are not supported on the Aruba 6000 and 6100 Switch Series.

Loopback interfaces are not supported on the Aruba 4100i Switch Series.

A loopback interface is a virtual interface supporting IPv4/IPv6 address configuration. The loopback interface can be considered stable because once created, it remains up. The loopback interface can then be configured with an address to use as a reference or identifier independent of the physical interfaces.

- **Device identification**: As long as the router is operational, the state of the loopback interface is always up. Even if only one link to the router is active, the loopback interface can be reached. This functionality makes it possible to identify an active device in the network using the IP address configured on the loopback interface.
- **Routing**: Since the loopback interface is always active, a routing session (such as a BGP session) can continue on an alternate path even if the outbound interface fails. In OSPF, a loopback interface address is advertised as an interface route into the network. This functionality increases reliability by allowing traffic to take alternate paths if there is a link failure. In OSPF and BGP, the router ID can be set to the loopback address to avoid reassignment of the router ID when physical interfaces are added or removed.
- **Device management**: Loopback interface is always reachable and can be used for sending and receiving management information such as logs and SNMP traps without interruption.

## Loopback commands

### interface loopback

```
interface loopback <INSTANCE>
no interface loopback <INSTANCE>
```

#### Description

Creates a loopback interface and enters loopback configuration mode.

The `no` form of this command deletes a loopback interface.

| Parameter | Description |
|---|---|
| *<INSTANCE>* | Selects the loopback interface ID. Range: 0 to 4 |

#### Examples

```
switch(config)# interface loopback 1
switch(config-loopback-if)#
```

#### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| All platforms | `config` | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## ip address

```
ip address <IPV4-ADDR/MASK> [secondary]
no ip address <IPV4-ADDR/MASK> [secondary]
```

### Description

Sets the IPv4 address for a loopback interface.

The `no` form of this command reverses the set of the IPv4 address for a loopback interface.

| Parameter | Description |
|---|---|
| *<IPV4-ADDR>* | Specifies an IP address in IPv4 format (`x.x.x.x`), where `x` is a decimal number from 0 to 255. |
| *<MASK>* | Specifies the number of bits in the address mask in CIDR format (`x`), where `x` is a decimal number from 0 to 128. |
| *secondary* | Indicates that the IPv4 address is a secondary address. |

### Examples

```
switch(config)# interface loopback 1
switch(config-loopback-if)# ip address 16.93.50.2/24
switch(config-loopback-if)# ip address 20.1.1.1/24 secondary
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| All platforms | `config` | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

# ipv6 address

`ipv6 address <IPV6-ADDR/MASK>`

### Description

Sets the IPv6 address for a loopback interface.

| Parameter | Description |
|---|---|
| *<IPV6-ADDR>* | Specifies an IP address in IPv6 format (`xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx`), where `x` is a hexadecimal number from 0 to F. |
| *<MASK>* | Specifies the number of bits in the address mask in CIDR format (`x`), where `x` is a decimal number from 0 to 128. |

### Examples

```
switch(config)# interface loopback 1
switch(config-loopback-if)# ipv6 address fd00:5708::f02d:4df6/64
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| All platforms | `config` | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

# show interface loopback

`show interface loopback [brief | instance <ID>]`

### Description

This command displays the configuration and status of loopback interfaces.

| Parameter | Description |
|---|---|
| `brief` | Displays brief information about all configured loopback interfaces. |
| `instance` *<ID>* | Displays the configuration and status of a loopback interface ID. Range: 1-255 |

### Examples

```
switch# show interface loopback

 Interface loopback1 is up
 IPv4 address 192.168.1.1/24

 Interface loopback2 is up
 IPv4 address 182.168.1.1/24
```

```
switch# show interface loopback brief

---------------------------------------------------------------------
Loopback      IP Address                                    Status
Interface
---------------------------------------------------------------------

loopback1   10.1.1.1/24                                        up
loopback1   1111:2222:3333:4444::6666/128                      up
```

```
switch# show interface loopback 1
 Interface loopback1 is up
 IPv4 address 192.168.1.1/24
```

## Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

## Command Information

| Platforms | Command context | Authority |
|---|---|---|
| All platforms | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

# Chapter 4 Static routing

Static routes are manually configured. If the network topology is simple enough, you can use static routes for the network's routing requirements. The proper configuration and usage of static routes can improve network performance and ensure bandwidth for important network applications.

The disadvantage of using static routes is that they cannot adapt to network topology changes. If a fault or topological change occurs in the network, the relevant routes will be unreachable and a network administrator must modify the static routes manually.

## Default route

Without a default route, a packet that does not match any routing entries is discarded and an ICMP destination-unreachable packet is sent to the source. A default route is used to forward packets that do not match any routing entry.

The network administrator can configure a default route with the destination as 0.0.0.0 and the mask as 0. The router forwards any packet whose destination address fails to match any entry in the routing table to the next hop of the default static route.

## Recursive static routes

A recursive static route is a route for which the next hop is learned from another routing look up (for example, dynamic protocol or from another another static route). For example, the following commands create an unsupported recursive static route:

- `ip route 99.0.0.0/24 30.0.0.2` - Main static route with gateway 30.0.0.2.
- `ip route 30.0.0.0/24 20.0.0.2` - Static route to reach the nexthop of the previously configured route.

## Configuration concepts

Before configuring a static route, you must understand the following concepts:

- **Destination address and mask:** In the `ip route` command, an IPv4 address is in dotted-decimal format. A mask can be in the form of mask length - the number of consecutive 1s in the mask.
- **Output interface and next hop address:** When configuring a static route, specify the output interface or next hop address. The next hop address cannot be a local interface IP address or the route configuration will not take effect.
- **Other attributes**: You can configure different priorities and administrative distance for different static routes to make route management policies more flexible. For example, specifying the same priority for different routes to the same destination enables load sharing, but specifying different priorities for these routes enables route backup.

## Configuration example procedure

Before configuring a static route, complete the following tasks:

- Configure the physical parameters for related interfaces.
- Configure the link-layer attributes for related interfaces.
- Configure the IP addresses for related interfaces.

**6000 and 6100 Switch Series:** The number of IPv4 and IPv6 static routes that can be configured is limited to 512 combined.

**6200 Switch Series:** The number of IPv4 and IPv6 static routes that can be configured is limited to 1024 combined. 16K routes are not supported.

6000, 6100 and 6200 Series Switches do not support IPv6 addresses with prefixes greater than 64. Prefixes between 65 and 127 will be software-forwarded.

## Basic static route configuration example

The IP addresses and masks of the switches and hosts are displayed here. Static routes are required for interconnection between any two hosts.

### Procedure

1. Configure IP addresses for interfaces (details not shown).
2. Configure static routes.
   a. Configure a default route on Switch A.
   ```
   6300# config
   6300(config)#
   <SwitchA> config
   [SwitchA] ip route 0.0.0.0/0 1.1.4.2
   ```
   b. Configure two static routes on Switch B.
   ```
   <SwitchB> config
   [SwitchB] ip route 1.1.2.0/24 1.1.4.1
   [SwitchB] ip route 1.1.3.0/24 1.1.5.6
   ```
   c. Configure a default route on Switch C.
   ```
   6300# config
   6300(config)#
   [SwitchC] ip route 0.0.0.0/0 1.1.5.5
   ```
3. Configure the hosts. The default gateways for hosts A, B, and C are `1.1.2.3`, `1.1.6.1`, and `1.1.3.1`. The configuration procedure is not shown.
4. Display the configuration.
   a. Display the IP routing table of Switch A.
   b. Display the IP routing table of Switch B.
   c. Use the `ping` command on Host B to check the reachability of Host A, assuming Windows XP runs on the two hosts.
   ```
   C:\Documents and Settings\Administrator>ping 1.1.2.2
   Pinging 1.1.2.2 with 32 bytes of data:
   Reply from 1.1.2.2: bytes=32 time=1ms TTL=255
   Reply from 1.1.2.2: bytes=32 time=1ms TTL=255
   Reply from 1.1.2.2: bytes=32 time=1ms TTL=255
   Reply from 1.1.2.2: bytes=32 time=1ms TTL=255
   15
   Ping statistics for 1.1.2.2:
   Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
   Approximate round trip times in milli-seconds:
   Minimum = 1ms, Maximum = 1ms, Average = 1ms
   ```
   d. Use the traceroute command on Host B to check the reachability of Host A.

```
8320# traceroute
traceroute  Trace the IPv4 route to a device on the network
traceroute6 Trace the IPv6 route to a device on the network
8320# traceroute
A.B.C.D Enter IPv4 address of the device to traceroute
WORD  Enter host name oo fthe device to traceroute
8320# traceroute
```

# Static routing commands

## ip route

```
ip route <DEST-IPV4-ADDR>/<NETMASK> {<NEXTHOP-ADDR> | <NEXTHOP-PORT-LAG-VLAN> | reject |
   blackhole}
no ip route <DEST-IPV4-ADDR>/<NETMASK> {<NEXTHOP-ADDR> | <NEXTHOP-PORT-LAG-VLAN> | reject |
   blackhole}
```

### Description

Adds an IPv4 static route on the default VRF.

The `no` form of this command deletes a IPv4 static route.

| Parameter | Description |
|---|---|
| *<DEST-IPV4-ADDR>/<NETMASK>* | Specifies the IPv4 route destination. |
| *<NEXTHOP-ADDR>* | Specifies the next hop address for reaching the destination in IPv4 format (`x.x.x.x`), where `x` is a decimal number from 0 to 255. |
| *<NEXTHOP-PORT-LAG-VLAN>* | Specifies the next hop as an outgoing interface. |
| `blackhole` | Specifies that packets matching the destination route are silently discarded and no ICMP error notification is sent to the sender. |
| `reject` | Specifies that packets matching the destination route are discarded and an ICMP error notification is sent to the sender. |

### Examples

```
switch(config)# ip route 10.0.0.0/24 blackhole
switch(config)# ip route 10.0.1.0/24 reject
switch(config)# ip route 10.0.2.0/24 20.0.0.2
switch(config)# ip route 10.0.3.0/24 1/1/1
switch(config)# ip route 10.0.3.0/24 1/1/1.110
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| All platforms | `config` | Administrators or local user group members with execution rights for this command. |

## ip route distance

Not supported on the 6200 Switch Series.

```
ip route <DEST-IPV4-ADDR>/<NETMASK> [<NEXT-HOP-IP-ADDR>|<INTERFACE>] distance <VALUE>
no ip route <DEST-IPV4-ADDR>/<NETMASK> [<NEXT-HOP-IP-ADDR>|<INTERFACE>] distance <VALUE>
```

### Description

Configures the administrative distance for the IPv4 static route.

The `no` form of this command deletes the static route.

| Parameter | Description |
|---|---|
| *<DEST-IPV4-ADDR>>* | Specifies an IP address in IPv4 format (`x.x.x.x`), where `x` is a decimal number from 0 to 255. |
| *<MASK>* | Specifies the number of bits in the address mask in CIDR format (`x`), where `x` is a decimal number from 0 to 32. |
| *<NEXT-HOP-IP-ADDR>* | Specifies the next hop IPv4 address in IPv4 format (`x.x.x.x`), where `x` is a decimal number from 0 to 255. |
| *<INTERFACE>* | Specifies the next hop as an outgoing interface. |
| `distance` *<VALUE>* | Specifies the administrative distance to associate with this static route. Default: 1. Range: 1-255. |

### Examples

```
switch(config)# ip route 10.0.2.0/24 20.0.0.2 distance 4
switch(config)# ip route 10.0.3.0/24 1/1/1 distance 6
```

```
switch(config)# no ip route 10.0.3.0/24 1/1/1 distance 6
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 4100i<br>6000<br>6100 | `config` | Administrators or local user group members with execution rights for this command. |

## ip route tag

```
ip route <DEST-IPV4-ADDR>/<NETMASK> {<NEXTHOP-ADDR> | <NEXTHOP-PORT-LAG-VLAN> | reject |
   blackhole} [tag] <1-4294967295>
no ip route <DEST-IPV4-ADDR>/<NETMASK> {<NEXTHOP-ADDR> | <NEXTHOP-PORT-LAG-VLAN> | reject |
   blackhole} [tag] <1-4294967295>
```

### Description

Configures tag for IPv4 static route.

The `no` form of this command deletes tag for IPv4 static route.

| Parameter | Description |
|---|---|
| *<DEST-IPV4-ADDR>/<NETMASK>* | Specifies the IPv4 route destination. |
| *<NEXTHOP-ADDR>* | Specifies the next hop address for reaching the destination in IPv4 format (`x.x.x.x`), where `x` is a decimal number from 0 to 255. |
| *<NEXTHOP-PORT-LAG-VLAN>* | Specifies the next hop as an outgoing interface. |
| `reject` | Specifies that packets matching the destination route are discarded and an ICMP error notification is sent to the sender. |
| `blackhole` | Specifies that packets matching the destination route are silently discarded and no ICMP error notification is sent to the sender. |
| `tag` | Specifies and assigns tag for the route. |

### Examples

```
switch(config)# ip route 10.1.1.1/32 20.1.1.2 tag 10
switch(config)# ip route 10.1.1.5/32 1/1/1 tag 20
```

```
switch(config)# no ip route 10.1.1.1/32 20.1.1.2 tag 10
switch(config)# no route 10.1.1.5/32 1/1/1 tag 20
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| All platforms | config | Administrators or local user group members with execution rights for this command. |

## ipv6 route

```
ipv6 route <DEST-IPV6-ADDR>/<NETMASK> {<NEXTHOP-ADDR> | <NEXTHOP-PORT-LAG-VLAN> | reject |
   blackhole}
no ipv6 route <DEST-IPV6-ADDR>/<NETMASK> {<NEXTHOP-ADDR> | <NEXTHOP-PORT-LAG-VLAN> |
   reject | blackhole}
```

### Description

Adds an IPv6 static route.

The `no` form of this command deletes an IPv6 static route on the default VRF.

| Parameter | Description |
|---|---|
| *<DEST-IPV6-ADDR>* | Specifies the route destination in IPv6 format (`xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx`), where `x` is a hexadecimal number from 0 to F. |
| *<NETMASK>* | Specifies the number of bits in the address mask in CIDR format (`x`), where `x` is a decimal number from 0 to 128. |
| *<NEXTHOP-ADDR>* | Specifies the next hop in IPv6 format (`xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx`), where `x` is a hexadecimal number from 0 to F. |
| *<NEXTHOP-PORT-LAG-VLAN>* | Specifies the next hop as an outgoing interface. |
| `reject` | Specifies that packets matching the destination route are discarded and an ICMP error notification is sent to the sender. |
| `blackhole` | Specifies that packets matching the destination route are silently discarded and no ICMP error notification is sent to the sender. |

### Usage

- Access platforms for IPv6 routes above /64 are not supported.
- IPv6 addresses with prefixes 65-127 will not be configured in the ASIC route table.

### Examples

```
switch(config)# ipv6 route 120::/124 blackhole
switch(config)# ipv6 route 121::/124 blackhole
switch(config)# ipv6 route 122::/124 1/1/1
switch(config)# ipv6 route 122::/124 1/1/1.110
```

```
switch(config)# no ipv6 route 122::/124 1/1/1.110
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| All platforms | `config` | Administrators or local user group members with execution rights for this command. |

## ipv6 route distance

```
ipv6 route <DEST-IPV6-ADDR>/<MASK> [<NEXT-HOP-IP-ADDR>|<INTERFACE>] distance <VALUE>
no ipv6 route <DEST-IPV6-ADDR>/<MASK> [<NEXT-HOP-IP-ADDR>|<INTERFACE>] distance <VALUE>
```

### Description

Configures the administrative distance for the IPv6 static route

The `no` form of this command deletes the static route.

| Parameter | Description |
|---|---|
| *<DEST-IPV6-ADDR>* | Specifies the route destination address in IPv6 format (`xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx`), where `x` is a hexadecimal number from 0 to F. |
| *<MASK>* | Specifies the number of bits in the address mask in CIDR format (`x`), where `x` is a decimal number from 0 to 128. |
| *<NEXT-HOP-IP-ADDR>* | Specifies the next hop in IPv6 format (`xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx`), where `x` is a hexadecimal number from 0 to F. |
| *<INTERFACE>* | Specifies the next hop as an outgoing interface. |
| `distance` *<VALUE>* | Specifies the administrative distance to associate with this static route. Range: 1 to 255. Default: 1. |

### Examples

```
switch(config)# ipv6 route 122::/124 1/1/1 distance 5
switch(config)# ipv6 route 123::/124 120::1 distance 6
```

```
switch(config)# no ipv6 route 123::/124 120::1 distance 6
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| All platforms | `config` | Administrators or local user group members with execution rights for this command. |

## ipv6 route tag

Not supported on the 6200 Switch Series.

```
ipv6 route <DEST-IPV6-ADDR>/<NETMASK> {<NEXTHOP-ADDR> | <NEXTHOP-PORT-LAG-VLAN> | reject |
   blackhole} [tag] <1-4294967295>
no ipv6 route <DEST-IPV6-ADDR>/<NETMASK> {<NEXTHOP-ADDR> | <NEXTHOP-PORT-LAG-VLAN> |
   reject | blackhole} [tag] <1-4294967295>
```

### Description

Configures tag for IPv6 static route.

| Parameter | Description |
|---|---|
| *<DEST-IPV6-ADDR>* | Specifies the route destination in IPv6 format (`xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx`), where `x` is a hexadecimal number from 0 to F. |
| *<NETMASK>* | Specifies the number of bits in the address mask in CIDR format (`x`), where `x` is a decimal number from 0 to 128. |
| *<NEXTHOP-ADDR>* | Specifies the next hop in IPv6 format (`xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx`), where `x` is a hexadecimal number from 0 to F. |
| *<NEXTHOP-PORT-LAG-VLAN>* | Specifies the next hop as an outgoing interface. |
| `reject` | Specifies that packets matching the destination route are discarded and an ICMP error notification is sent to the sender. |
| `blackhole` | Specifies that packets matching the destination route are silently discarded and no ICMP error notification is sent to the sender. |
| `tag` | Specifies and assigns tag for the route. |

### Examples

```
switch(config)# ipv6 route 3001::1/128 1/1/1 tag 10
switch(config)# ipv6 route 3002::1/128 1000::2 tag 20
```

```
switch(config)# no ipv6 route 3001::1/128 1/1/1 tag 10
switch(config)# no ipv6 route 3002::1/128 1000::2 tag 20
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 4100i<br>6000<br>6100 | `config` | Administrators or local user group members with execution rights for this command. |

## show ipv4 rib

`show ip rib <FILTER> [vrf <VRF-NAME>]`

### Description

Shows the IPv4 Routing Information Base (RIB) of VRF with name (`<VRF-NAME>`). If VRF name is not specified, default VRF routes are displayed.

| Parameter | Description |
|---|---|
| *<FILTER>* | Selects filter, see Usage section. |
| vrf *<VRF-NAME>* | Specifies the VRF name. |

### Usage

There are sub-options available within this command:

- **A.B.C.D:** Shows longest prefix match.
- **A.B.C.D/M:** Shows exact route match.
- **all-vrfs:** Shows all VRF information.
- **bgp:** Shows BGP routes only.
- **connected:** Shows connected routes only.
- **local:** Shows local routes only.
- **ospf:** Shows OSPF routes only.
- **rip:** Shows RIP routes only.
- **static:** Shows static routes only.
- **summary:** Shows aggregate count of routes per routing protocol.
- **vrf:** Specifies the VRF name.
- **selected:** Shows routes selected for forwarding only.
- **non-selected:** Shows routes not selected for forwarding only.

### Examples

Showing IPv4 routes in RIB:

```
Origin Codes: R - RIP, O - OSPFv2, B - BGP
              C - connected, S - static
```

```
Type Codes:   E - External BGP, I - Internal BGP, IA - OSPF inter area
              E1 - OSPF external type 1, E2 - OSPF external type 2
* indicates selected for forwarding

VRF: default

Prefix          Nexthop      Interface   VRF       Origin/ Distance/ Age
                                                     Type    Metric
----------------------------------------------------------------------------------
*10.0.0.0/30    -            1/1/1       -         S       [20/0]     0d:10h:01m:41s
*10.0.1.0/30    -            1/1/1       -         B/I     [200/0]    2d:20h:01m:42s
*10.1.64.0/18   -            loopback2   -         C       [0/0]      -
*10.2.64.0/18   10.0.0.3     lag1        -         O/E1    [110/25]   1d:05h:03m:43s
*10.2.64.0/18   20.10.0.1    vlan100     -         O/E1    [110/25]   0d:05h:03m:43s
*20.1.2.3/32    2.2.2.2      1/1/4       vrf_red   B/E     [20/0]     2d:10h:01m:45s
*30.1.3.0/24    -            reject      -         S       [1/0]      33d:10h:01m:43s
*50.10.13.0/24  -            reject      -         S       [1/0]      12d:10h:01m:44s
*61.1.1.2/32    4.4.4.4      1/1/5       -         B/I     [200/0]    1d:11h:01m:45s
*62.1.1.3/32    5.5.5.5      1/1/6       -         B/I     [200/0]    0d:12h:01m:45s
*193.0.0.2/32   50.0.0.2     1/1/2       -         S       [1/0]      0d:04h:01m:43s
 193.0.0.2/32   56.0.0.3     1/1/3       -         O/E1    [110/25]   0d:04h:03m:43s

Total Route Count : 13
```

Showing IPv4 exact route match in RIB:

```
show ip rib 10.0.0.0/30

VRF : default

Prefix          : 10.0.0.0/30            VRF(egress)        :  -
Nexthop         :  -                     Interface          :  1/1/1
Origin          : Connected              Type               :  -
Distance        : 0                      Metric             :  0
Age             : -                      Tag                :  0
Selected        : Yes                    Recursive Nexthop  :  No
```

Showing IPv4 RIB summary:

```
show ip rib summary

IPv4 RIB Table Summary

VRF name :      default
  Protocol      RIB Routes
-------------- -------------
connected      1010
local          1011
static         4
ospfv2         509
bgp            9014
selected       10008
non-selected   1518
```

## Command History

| Release | Modification |
| --- | --- |
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
| --- | --- | --- |
| All platforms | Operator (>) or Manager (#) | Administrators or local user group members with execution rights for this command. |

## show ipv6 rib

```
show ipv6 rib <FILTER> [vrf <VRF-NAME>]
```

### Description

Shows the IPv6 Routing Information Base (RIB) of VRF with name (<VRF-NAME>). If VRF name is not specified, default VRF routes are displayed.

| Parameter | Description |
| --- | --- |
| *<FILTER>* | Selects filter, see usage section. |
| vrf *<VRF-NAME>* | Shows routes in the VRF and specifies VRF name. |

### Usage

There are sub-options available within this command:

- **X:X: :X:X:** Shows longest prefix match.
- **X:X: :X:X/M:** Shows exact route match.
- **all-vrfs:** Shows all VRF information.
- **bgp:** Shows BGP routes only.
- **connected:** Shows connected routes only.
- **local:** Shows local routes only.
- **ospf:** Shows OSPF routes only.
- **rip:** Shows RIP routes only.
- **static:** Shows static routes only.
- **summary:** Shows aggregate count of routes per routing protocol.
- **vrf:** Specifies the VRF name.
- **selected:** Shows routes selected for forwarding only.
- **non-selected:** Shows routes not selected for forwarding only.

### Examples

Showing IPv6 routes in RIB:

```
Origin Codes: R - RIPng, O - OSPFv3, B - BGP
              C - connected, S - static
Type Codes:   E - External BGP, I - Internal BGP, IA - OSPF inter area
```

```
             E1 - OSPF external type 1, E2 - OSPF external type 2
* indicates selected for forwarding

VRF: default

Prefix                Nexthop     Interface  VRF    Origin/ Distance/ Age
                                                      Type    Metric
-----------------------------------------------------------------------------------
*1000::/64            -           1/1/1      -      C       [0/0]     -
*1000::8/128          -           1/1/1      -      L       [0/0]     -
*1001:db8::/32        1000::10    1/1/1      -      B/I     [200/0]   1d:20h:01m:42s
*2000::/64            fe80::3182  vlan100    -      S       [1/0]     2d:05h:03m:43s
 2000::/64            fe80::1241  1/1/1      -      O/E1    [110/25]  0d:05h:03m:43s
*2000::2000:0:0:0/67 fe80::1111  lag1       Green  B/E     [20/0]    1d:10h:01m:45s
*3001::0/64           -           vlan100    -      C       [0/0]     -
*3001::1/128          -           vlan100    -      L       [0/0]     -
*6101::0/64           -           blackhole  -      S       [1/0]     12d:10h:01m:43s

Total Route Count : 9
```

Showing IPv6 exact route match in RIB:

```
show ipv6 rib 2000::2000:0:0:0

VRF : default

Prefix           : 2000::2000:0:0:0/67         VRF(egress)        :  Green
Nexthop          : fe80::1111                  Interface          :  lag1
Origin           : BGP                         Type               :  External
Distance         : 20                          Metric             :  0
Age              : 1d:10h:01m:45s              Tag                :  20
Selected         : Yes                         Recursive Nexthop  :  Yes
```

Showing IPv6 RIB summary:

```
show ipv6 rib summary

IPv6 RIB Table Summary

VRF name :  default
  Protocol      RIB Routes
-------------- -------------
connected      1009
local          1010
static         3
ospfv3         508
bgp            1013

selected       10004
non-selected   1527
```

## Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

## Command Information

| Platforms | Command context | Authority |
| --- | --- | --- |
| All platforms | Operator (>) or Manager (#) | Administrators or local user group members with execution rights for this command. |

# Chapter 5
# Open Shortest Path First version 2 (OSPFv2)

OSPFv2 is not supported on the Aruba 6000 and 6100 Switch Series.

OSPFv2 is not supported on the Aruba 4100i Switch Series.

Open Shortest Path First version 2 (OSPFv2) is a routing protocol described in RFC2328. It is a Link State-based IGP (Interior Gateway Protocol) routing protocol applied to routers grouped into OSPF areas identified by the routing configuration on each routing switch. It is widely used in medium to large-sized enterprise networks.

## Overview

The characteristics of OSPFv2 are:

- Provides a loop-free topology using SPF algorithm.
- Supports load balancing with equal cost routes for the same destination.
- OSPFv2 is a classless protocol and allows for a hierarchical design with VLSM (Variable Length Subnet Masking) and route summarization.
- Provides authentication of routing messages.
- Provides fast convergence with triggered, incremental updates via LSAs.

Some OSPFv2 configuration is done in the global configuration context, others in the router ospf context, or in the interface configuration context, or in the vlink context. OSPFv2 can be configured on L3 ports, VLAN interfaces, LAG interfaces, and loopback interfaces. All such configurations work in the mentioned interfaces context. OSPFv2 mandates the associated interface to be a routed interface.

**Supported features**

- OSPFv2 neighbor adjacency, Hello protocol, multiple areas, Inter-area routing
- AS external routes, stub areas, totally stubby areas, NSSA, ASBR
- Designated Router/Backup Designated Router
- Point-to-point interfaces/broadcast interfaces
- Equal-cost multipath
- Null authentication, Simple password authentication, MD5 authentication
- Graceful Restart - un-planned, restart interval, Graceful Restart Helper
- Stub router advertisement
- SPF throttling
- LSA Throttling
- SPF throttling
- Configuration of interface parameters such as priority, cost, hello-interval, dead-interval, retransmit-interval, transit-delay, etc.

- Configuration of virtual link parameters such as hello-interval, dead-interval, retransmit-interval, transit-delay, etc.
- Route redistribution
- Passive interfaces
- Multi VRF support with each VRF having up to eight OSPF process instances
- Congestion control (prioritizing hello packets, inactivity timer reset, and adjacency throttling)

# How OSPFv2 protocol works

The protocol uses Link State Advertisements (LSAs) transmitted by each router to update neighboring routers regarding its interfaces and the routes available through those interfaces. Each routing switch in an area also maintains a link-state database (LSDB) that describes the area topology. (All routers in a given OSPF area have identical LSDBs.) The routing switches used to connect areas to each other flood summary link LSAs and external link LSAs to neighboring OSPF areas to update them regarding available routes. Through this means, each OSPF router determines the shortest path between itself and a desired destination router in the same OSPF domain (Autonomous System (AS)).

# OSPFv2 concepts

The following sections describe OSPFv2 Link-State types, router types, and area types.

## OSPFv2 Link-state advertisement (LSA) types

OSPFv2 sends routing information in Link-State Advertisements (LSAs). The switches support the following types of LSAs.

- **Router LSA—Type-1 LSA.** Describes the states of the router interfaces to an area, and is flooded throughout a single area only.
- **Network LSA—Type-2 LSA.** Describes the list of routers connected to the network, and is flooded throughout a single area only.
- **Summary LSA—Type-3 LSA.** Describes the route to networks in another OSPF area of the same Autonomous System (AS). Propagated through backbone area to other areas.
- **ASBR summary LSA—Type-4 LSA.** Describes the route to an ASBR in an OSPF normal or backbone area of the same AS. Propagated through backbone area to other areas.
- **AS external LSA—Type-5 LSA.** Describes the route to a destination in another AS (external route.) Originated by ASBR in normal or backbone areas of an AS and propagates through backbone area to other normal areas.
- **NSSA LSA—Type-7 LSA.** Describes the route to a destination in another AS (external route.) Originated by ASBR in NSSA. ABR converts type-7 LSAs to type-5 LSAs for injection into the backbone area.

## OSPFv2 router types

The following router types are supported

### Internal routers

Internal OSPFv2 routers belong to only one area. Internal routers flood type-1 LSAs to all routers in the same area and maintain identical LSDBs.

### Area border routers (ABRs)

Area border routers have membership in multiple areas. ABRs are used to connect the various areas in an AS to the backbone area for that AS. Multiple ABRs can be used to connect a given area to the backbone, and a given ABR can belong to multiple areas other than the backbone.

An ABR maintains a separate LSDB for each area to which it belongs. (All routers within the same area have identical LSDBs.) The ABR is responsible for flooding summary LSAs between its border areas. You can reduce summary LSA flooding by configuring area ranges. An area range enables you to assign an aggregate address to a range of IP addresses. This aggregate address is advertised instead of all the individual addresses it represents.

### Autonomous system boundary router (ASBR)

Autonomous system boundary routers run multiple interior gateway protocols and serve as a gateway to other autonomous systems operating with interior gateway protocols. The ASBR imports and translates different protocol routes into OSPF through redistribution. ASBRs can be used in backbone areas, normal areas, and NSSAs, but not in stub areas.

### Designated routers (DRs)

In an OSPF network having two or more routers, one router is elected to serve as the DR and another router to act as the Backup Designated Router (BDR). All other routers in the area forward their routing information to the DR and BDR, and the DR forwards this information to all routers in the network. This action minimizes the amount of repetitive information that is forwarded on the network by eliminating the need for each individual router in the area to forward its routing information to all other routers in the network. If the area includes multiple networks, each network elects its own DR and BDR.

In an OSPF network with no DR and no BDR, the neighboring router with the highest priority is elected the DR, and the router with the next highest priority is elected the BDR. If the DR goes off-line, the BDR automatically becomes the DR, and the router with the next highest priority then becomes the new BDR. If multiple routing switches on the same OSPF network are declaring themselves DRs, both priority and router ID are used to select the DR and BDRs.

Priority is configurable using the `ip ospf priority` command at the interface level. If two neighbors share the same priority, the router with the highest router ID is elected as the DR. The router with the next highest router ID is elected as the BDR.

## OSPFv2 area types

OSPFv2 is built upon a hierarchy of network areas. All areas for a given OSPF domain reside in the same AS. An AS is defined as a number of contiguous networks that share the same interior gateway routing protocol.

An AS can be divided into multiple areas. Each area represents a collection of contiguous networks and hosts, and the topology of a given area is not known by the internal routers in any other area. Areas define the boundaries to which types 1 and 2 LSAs are broadcast, which limits the amount of LSA flooding that occurs within the AS and also helps to control the size of the LSDBs maintained in OSPF routers. An area is represented in OSPF by either an IP address or a number. Area types include: Backbone, Normal, Not-so-stubby (NSSA), and stub.

### Backbone area

Every AS must have one (and only one) backbone area (identified as area 0 or 0.0.0.0.) The ABRs of all other areas in the same AS connect to the backbone area, either physically through an ABR or through a configured virtual link. The backbone is a transit area that carries the type-3 summary LSAs, type-5 AS external link LSAs and routed traffic between non-backbone areas, as well as the type-1 and type-2 LSAs and routed traffic internal to the area. ASBRs are allowed in backbone areas.

### Normal area

Normal area connects to the backbone area through one or more ABRs (physically or through a virtual link) and supports type-3 summary LSAs and type-5 external link LSAs to and from the backbone area. ASBRs are allowed in normal areas.

### Stub area

Stub area connects to the AS backbone through one or more ABRs. It does not allow an internal ASBR, and does not allow external (type 5) LSAs. A stub area supports these actions:

- Advertise the area summary routes to the backbone area.
- Advertise summary routes from other areas.
- Use the default summary (type-3) route to advertise both of the following:
  - Summary routes to other areas in the AS
  - External routes to other ASs

You can configure the stub area ABR to do the following:

- Suppress advertising from some or all area summarized internal routes into the backbone area.
- Suppress LSA traffic from other areas in the AS by replacing type-3 summary LSAs and the default external route from the backbone area with the default summary route (0.0.0.0/0.)

Virtual links are not allowed for stub areas.

### Not-so-stubby (NSSA) area

NSSA area connects to the backbone area through one or more ABRs. NSSAs are intended for use where an ASBR exists in an area where you want to control the following:

- Advertising the ASBR external route paths to the backbone area
- Allowing LSAs from the backbone area to advertise in the NSSA:
  - Summary routes (type-3 LSAs) from other areas
  - External routes (type-5 LSAs) from other areas as a default external route (type-7 LSAs)

Virtual links are not allowed for NSSAs.

# OSPFv2 configuration task list

## Tasks at a glance

- Configuring OSPF on the routing switch
- Assigning the routing switch to an OSPF area
- Configuring external route redistribution and control
- Influencing route choice by changing the administrative distance
- Configuring graceful restart of OSPF routing
- Configuring OSPF interface settings
- Configuring OSPF interface authentication
- Configuring all OSPF interfaces as passive
- Viewing OSPFv2 information
- Clearing OSPF statistics on a switch

# Configuring OSPF on the routing switch

Create the OSPF instance and enter the OSPF router configuration context. From this, you can proceed with other OSPF configuration tasks.

## Prerequisites

- You must be in the global configuration context, as indicated by the `switch(config)#` prompt to create the OSPF instance and enter the OSPF router configuration context.
- To configure a router ID, create OSPF network areas, or adjust other global OSPF configuration items, you must be in the router configuration context, as indicated by the `switch(config-router)#` prompt.

## Procedure

1. Create the OSPF instance and enter the OSPF router configuration context using the following command. For command details, see router ospf.
   `router ospf <PROCESS-ID>`
   For example, the following command creates OSPF instance 1.

   ```
   switch(config)# router ospf 1
   switch(config-ospf-1)#
   ```

2. Configure a global router ID using the following command. For command details, see router-id.
   `router-id <ROUTER-ADDRESS>`
   For example, the following command sets the router ID to 1.1.1.1.

   ```
   switch(config-ospf-1)# router-id 1.1.1.1
   ```

3. Optionally, if the OSPF process was disabled (it is enabled by default), enable the OSPF process using the following command.
   `enable`
   For command details, see enable. (Refer to disable for disabling the OSPF process).

# Assigning the routing switch to an OSPF area

Create a Normal, Stub, or Not So Stubby (NSSA) area.

## Prerequisites

You must be in the router configuration context, as indicated by the `switch(config-router)#` prompt.

## Procedure

Create an OSPF area for the routing switch using one of the following commands:

- Create a normal area using the following command: `area <area-id>`. For command details, see area.
- Create a stub area using the following command: `area <area-id> stub`. For command details, see area stub.
- Create a Not-So-Stubby-Area using the command: `area <area-id> nssa`. For command details, see area nssa.

For example, the following command creates a normal area with an area identifier of 10.1.1.1. Area identifier could alternatively be entered in decimal format such as 1.

```
switch(config-ospf-1)# area 10.1.1.1
```

# Setting OSPF network for the area

After you define an OSPF area, you can assign one or more networks to it. The OSPF protocol will run on the interface with the configured IPv4 address. The interfaces which have an IP address configured in this network or in a subset of this network, will participate in the OSPF protocol.

## Prerequisites

You must be in the interface configuration context, as indicated by the `switch(config-if)#` prompt.

## Procedure

1. Set an OSPF network for the area using the following command. For command details, see ip ospf area.
   `ip ospf <PROCESS-ID> area <AREA-ID>`
   For example, use the following command to assign interface 1/1/1 to OSPF area 1. The area can alternatively be entered as an IPv4 address.

   ```
   switch(config)# interface 1/1/1
   switch(config-if)# ip ospf 1 area 1
   ```

2. Optionally, you can disable OSPF on the interface using the following command. For command details, see ip ospf shutdown.
   `ip ospf shutdown`

   ```
   switch(config)# interface 1/1/1
   switch(config-if)# ip ospf shutdown
   ```

# Configuring external route redistribution and control

Configuring route redistribution for OSPFv2 establishes the routing switch as an ASBR for importing and translating different protocol routes into OSPFv2. When you configure redistribution for OSPFv2, you can specify that routes external to the OSPFv2 domain are imported as OSPFv2 routes.

1. Enable route redistribution using the `redistribute` command.

   ```
   redistribute {bgp | connected | local loopback | static | rip | ospf <PROCESS-ID>}
                [route-map <ROUTE-MAP-NAME>]
   ```

   For example, setting redistribution of connected routes as OSPFv2 routes:

   ```
   switch(config-ospf-1)# redistribute connected
   ```

   If a route map is specified, then only the routes that pass the match clause specified in the route map are redistributed to OSPFv2. For example, setting redistribution of local loopback routes, that only match the routes specified by the route map:

```
switch(config-ospf-1)# redistribute local loopback route-map local_routes
```

2. Optionally, modify the default metric for redistribution using the `default-metric` command.
   `default-metric <METRIC-VALUE>`
   For example, setting the default metric for redistribution to 37.

```
switch(config-ospf-1)# default-metric 37
```

3. Optionally, set the cost of default-summary LSAs using the area command.
   `area <AREA-ID> default-metric <COST>`
   For example, setting the cost of default summary LSAs to 2.

```
switch (config-ospf-1)# area 1 default-metric 2
```

4. Optionally, use the `max-metric router-lsa` command to set the protocol to advertise a maximum metric so that other routers do not prefer this router as an intermediate hop in their shortest path first (SPF) calculations.
   `max-metric router-lsa [on-startup [<ADVERT-TIME>]]`
   For example, setting advertise max-metric router-lsa on startup.

```
switch(config-ospf-1)# max-metric router-lsa on-startup 3000
```

# Influencing route choice by changing the administrative distance

The administrative distance value can be left in its default configuration setting (110) unless a change is needed to improve OSPF performance for a specific network configuration.

## Prerequisites

You must be in the router configuration context, as indicated by the `switch(config-router)#` prompt.

## Procedure

Reconfigure the administrative distance using the following command. For command details, see distance command.
`distance <distance>`
For example, use the following command to set administrative distance to 100.

```
switch(config-ospf-1)# distance 100
```

# Configuring graceful restart of OSPF routing

OSPF routing can be gracefully restarted on the switch without losing packets that are in transit. OSPF neighbors are informed that the router is completing a graceful restart, which allows for maintenance on the switch without interrupting traffic in the network. There is no effect on the saved switch configuration.

## Prerequisites

Graceful restart is only applicable to the 6200-VSF switches.

You must be in the router configuration context, as indicated by the `switch(config-router)#` prompt.

### Procedure

Enable graceful restart of OSPF routing using the following command. For command details, see graceful-restart.

`graceful-restart {restart-interval <seconds> | helper}`

For example, the following command specifies 50 seconds as the maximum interval another router will wait for this router to gracefully restart.

```
switch(config-ospf-1)# graceful-restart restart-interval 50
```

## Configuring OSPF interface settings

You can optionally adjust the following OSPF interface settings.

### Prerequisites

You must be in the interface configuration context, as indicated by the `switch(config-if)#` prompt.

### Procedure

1. Set the interface cost using the following command. For command details, see ip ospf cost.
   `ip ospf cost <interface-cost>`
   For example, the following command sets the cost associated with interface 1/1/1 to 100.
   ```
   switch(config)# interface 1/1/1
   switch(config-if)# ip ospf cost 100
   ```
2. Set the time interval between OSPF hello packets for the OSPF interface using the following command. For command details, see ip ospf hello-interval.
   `ip ospf hello-interval  <seconds>`
3. Set the interval after which a neighbor is declared dead if no hello packet is received on the OSPF interface using the following command. For command details, see ip ospf dead-interval.
   `ip ospf dead-interval <seconds>`
4. Set the time between retransmitting lost link state advertisements for the OSPF interface using the following command. For command details, see ip ospf retransmit-interval.
   `ip ospf retransmit-interval  <seconds>`
5. Sets the transit delay in link state transmission for the OSPF interface using the following command. For command details, see ip ospf transit-delay.
   `ip ospf transit-delay <seconds>`
6. Set the OSPF network type for the interface. For command details, see ip ospf network.
   `ip ospf network {broadcast|point-to-point}`
7. Set the OSPF priority on the interface using the following command. For command details, see ip ospf priority.
   `ip ospf priority <number-value>`

8. Set the OSPF interface as OSPF passive interface. The interface participates in OSPF but does not send or receive packets on that interface. For command details, see ip ospf passive.
   `ip ospf passive`

# Configuring OSPF interface authentication

Configure authentication on the interface. Only one method of authentication can be active on an interface at a time. If one method is already configured on an interface, configuring an alternative method on the same interface automatically overwrites the first method used.

## Prerequisites

You must be in the interface configuration context, as indicated by the `switch(config-if)#` prompt.

## Procedure

1. Set the authentication type that will be used for authentication with the neighbor router using the following command. For command details, see ip ospf authentication.
   `ip ospf authentication {message-digest | simple-text | null | keychain}`

   For example, the following command sets the authentication type to simple-text.

```
switch(config-if)# ip ospf authentication simple-text
```

2. For simple-text authentication, set the password using the following command. For command details, see ip ospf authentication-key.
   `ip ospf authentication-key <PASSWORD>`

   For example:

```
switch(config-if)# ip ospf authentication-key secure
```

3. For MD5 authentication, set the password using the following command. For command details, see ip ospf message-digest-key md5.
   `ip ospf message-digest-key {Key ID} md5 {ciphertext | plaintext}  <KEY>`

   For example:

```
switch(config-if)# ip ospf message-digest-key 1 md5 ciphertext 100
```

# Configuring all OSPF interfaces as passive

OSPF sends LSAs to all other routers in the same AS. To limit the flooding of LSAs throughout the AS, you can configure OSPF to be passive.

## Prerequisites

You must be in the router configuration context, as indicated by the `switch(config-router)#` prompt.

## Procedure

Configure all OSPF interfaces as passive using the following command. For command details, see passive-interface default.
`passive-interface default`

```
switch(config-ospf-1)# passive-interface default
```

# Configuring SPF throttling timers

SPF calculation is throttled with default timer values (start-time 200ms, hold-time 1000ms, max-wait-time 5000ms). You can throttle SPF calculation by configuring non-default timers to improve performance of a specific network configuration.

### Prerequisites

You must be in the router configuration context, as indicated by the `switch(config-router)#` prompt.

### Procedure

Reconfigure SPF throttling timers using the following command. For command details, see timers throttle spf.

```
timers throttle spf start-time <milliseconds> hold-time <milliseconds> max-wait-time
<milliseconds>
```

For example, use the following command to set start-time to 500ms, hold-time to 3000ms and max-wait-time to 9000ms:

```
switch(config-ospf-1)# timers throttle spf start-time 500 hold-time
3000 max-wait-time 9000
```

# Viewing OSPFv2 information

Use these show commands from the Manager context, as indicated by the `switch#` prompt.

To view OSPFv2 information, use the following commands. For command details and examples, click the links.

- To view general OSPF, area, state and configuration information use: show ip ospf.
- To view OSPF routing table entries for Area Border Router (ABR) and Autonomous System Border Router (ASBR) use: show ip ospf border-routers.
- To view OSPF link state database summary for different OSPF LSAs (Link State Advertisement) use: show ip ospf lsdb.

Many different parameters are available with this command to display information for a particular LSA.

- To view information about OSPF-enabled interfaces use: show ip ospf interface.
- To view information about OSPF neighbors use: show ip ospf neighbors.
- To view OSPF routing table information use: show ip ospf routes.
- To view OSPF statistics use: show ip ospf statistics.
- To view OSPF statistics for the OSPF-enabled interfaces use show ip ospf statistics interface.
- To view the current state and parameters of the OSPF virtual-links use show ip ospf virtual-links.

# Clearing OSPF statistics on a switch

Clear OSPF event statics using the following command. For command details, see clear ip ospf statistics .

```
clear ip ospf [<PROCESS-ID>] statistics [interface [<INTERFACE-NAME>]] [all-vrfs | vrf <VRF-
NAME>]
```

For example, the following command clears OSPF event statistics from interface 1/1/1.

```
switch(config-router)# clear ip ospf statistics interface 1/1/1
```

## An example of the OSPFv2 information in the show running-config command

When OSPFv2 is configured, the output of the `show running-config` command includes OSPFv2 information.

For example:

```
switch# show running-config
Current configuration:
!
!Version ArubaOS-CX 10.0X.XXXX
!
lldp enable
timezone set utc
mgmt
led base-loc_fdc on
led base-loc on
led base-hlth_fdc fast_blink
led base-pwr_fdc on
!
!
!
!
!
!
aaa authentication login default local
aaa authorization commands default none
!
!
!
!
router ospf 1 vrf vrf_default
    area 0.0.0.0
vlan 1
    no shutdown
interface lag 44
    no shutdown
    ip address 33.1.1.1/24
    ip ospf 1 area 0.0.0.0
    ip ospf hello-interval 22
interface 1/1/1
    no shutdown
    ip address 33.44.1.1/24
    ip ospf 1 area 0.0.0.0
    ip ospf passive
    ip ospf dead-interval 88
    ip ospf transit-delay 44
    ip ospf priority 88
    ip ospf cost 8
    ip ospf network point-to-point
    ip ospf authentication simple-text
    ip ospf authentication-key kkk
    ip ospf shutdown
interface 1/1/2
interface loopback 2
    ip address 55.55.55.55/32
```

```
ip ospf 1 area 0.0.0.0
```

# OSPFv2 commands

## area

```
area <AREA-ID>
no area <AREA-ID>
```

### Description

Creates a normal area, with `<AREA-ID>` set if not present. If the area is already present and it is not a normal area, then this command changes the area type to normal.

The `no` form of this command deletes the area with the `<AREA-ID>` specified. Area can be of any type (nssa, nssa no-summary, stub, stub no-summary, and default normal area).

| Parameter | Description |
| --- | --- |
| *<AREA-ID>* | Specifies the area ID in one of the following formats.<br>OSPF area identifier in IPv4 format (`x.x.x.x`), where `x` is a decimal number from 0 to 255.<br>OSPF area identifier in decimal format. Range: 0 to 4294967295. |

### Examples

Creating a normal area:

```
switch(config)# router ospf 1
switch(config-ospf-1)# area 1
Switch(config-ospf-1)# show running-config current-context router ospf 1
       router-id 1.1.1.1
       area 0.0.0.1
```

Deleting an area:

```
switch(config)# router ospf 1
switch(config-ospf-1)# no area 1
```

### Command History

| Release | Modification |
| --- | --- |
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
| --- | --- | --- |
| 6200 | `config-ospf-<PROCESS-ID>` | Administrators or local user group members with execution rights for this command. |

# area default-metric

```
area <AERA-ID> default-metric <COST>
no area <AREA-ID> default-metric
```

### Description

Sets the cost of the default route announced to NSSA or stub areas.

The `no` form of this command resets the cost of the default route announced to NSSA or stub areas, to the default value of 1.

| Parameter | Description |
|---|---|
| `<AREA-ID>` | Specifies area ID in one of the following formats.<br>OSPF area identifier in IPv4 format (`x.x.x.x`), where `x` is a decimal number from 0 to 255.<br>OSPF area identifier in decimal format. Range: 0 to 4294967295. |
| `default-metric <COST>` | Sets the cost of default-summary LSAs announced to NSSA or stub areas, to the specified value. Default cost: 1. Range: 0 to 16777215. |

### Examples

Setting cost for default LSA summary:

```
switch(config)# router ospf 1
switch(config-ospf-1)# area 1 default-metric 2
```

Setting cost for default LSA summary to default:

```
switch(config)# router ospf 1
switch(config-ospf-1)# no area 1 default-metric
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config-ospf-<PROCESS-ID>` | Administrators or local user group members with execution rights for this command. |

# area nssa

```
area <AREA-ID> nssa [no-summary]
no area <AREA-ID> nssa [no-summary]
```

### Description

Creates the NSSA area (Not So Stubby Area) with *<AREA-ID>* if not present. If area is present and not NSSA area, this command changes the area type to NSSA area. If `no-summary` is used, area type will be NSSA No-Summary.

The `no` form of this command unsets the area type as NSSA. That is, the configured area will be changed to default normal area. The `no area <AREA-ID> nssa no-summary` command enables sending inter-area routes into NSSA, but will not unset the area as NSSA.

| Parameter | Description |
|---|---|
| *<AREA-ID>* | Specifies the area ID in one of the following formats.<br>OSPF area identifier in IPv4 format (`x.x.x.x`), where `x` is a decimal number from 0 to 255.<br>OSPF area identifier in decimal format. Range: 0 to 4294967295. |
| `nssa [no-summary]` | Specifies Not So Stubby Area (NSSA) area type. If area is present and not NSSA area, parameter changes the area type to NSSA area. If `no-summary` is specified, area type will be NSSA No-Summary, which means do not inject inter-area routes into NSSA. |

### Examples

Creating an NSSA area:

```
switch(config)# router ospf 1
switch(config-ospf-1)# area 1 nssa
switch(config-ospf-1)# area 1 nssa no-summary
```

Unsetting the area as NSSA

```
switch(config)# router ospf 1
switch(config-ospf-1)# no area 1 nssa
switch(config-ospf-1)# no area 1 nssa no-summary
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config-ospf-<PROCESS-ID>` | Administrators or local user group members with execution rights for this command. |

## clear ip ospf neighbors

```
clear ip ospf [<PROCESS-ID>] neighbor [<NEIGHBOR>] [interface [<INTERFACE-NAME>]]
[all-vrfs | vrf <VRF-NAME>]
```

### Description

Resets the neighbor and clears the OSPF neighbor information.

| Parameter | Description |
|---|---|
| *<PROCESS-ID>* | Specifies the OSPFv2 process ID to clear the statistics for the particular OSPFv2 process. Range: 1 to 63. |
| *<NEIGHBOR>* | Specifies the router ID of a neighbor. |
| *<INTERFACE-NAME>* | Specifies the OSPFv2 statistics to clear for the specified interface. |
| `all-vrfs` | Select to clear the OSPFv2 statistics for all VRFs. |
| `vrf` *<VRF-NAME>* | Specifies the name of a VRF. |

### Example

Clearing the OSPFv2 neighbor information:

```
switch# clear ip ospf 1 neighbor 3.3.3.3
switch# clear ip ospf 1 neighbor interface 1/1/1
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## clear ip ospf statistics

`clear ip ospf [<PROCESS-ID>] statistics [interface [<INTERFACE-NAME>]] [all-vrfs | vrf <VRF-NAME>]`

### Description

Clear the OSPF event statistics.

| Parameter | Description |
|---|---|
| *<PROCESS-ID>* | OSPF process ID. Clear the statistics for the particular OSPF process. Range: 1 to 63. |
| *<INTERFACE-NAME>* | Clear the OSPF statistics for the specified interface. |
| `all-vrfs` | Optionally select to clear the OSPF statistics for all VRFs. |

| Parameter | Description |
|---|---|
| `vrf <VRF-NAME>` | Optionally select to clear the OSPF statistics for a particular VRF. If the VRF is not specified, information for the default VRF is cleared. |

### Examples

Clearing the OSPF event statistics:

```
switch# clear ip ospf statistics
switch# clear ip ospf statistics interface 1/1/1
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## default-information originate

```
default-information originate
no default-information originate
```

### Description

Configures OSPF to advertise the default route (0.0.0.0/0) to its neighbors if it is present in the routing table.

The `no` form of this command disables advertisement of the default route.

### Examples

Setting advertisement of the default route:

```
switch(config)# router ospf 1
switch(config-ospf-1)# default-information originate
```

Disabling advertisement of the default route:

```
switch(config)# router ospf 1
switch(config-ospf-1)# no default-information originate
```

### Command History

| Release | Modification |
| --- | --- |
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
| --- | --- | --- |
| 6200 | `config-ospf-<PROCESS-ID>` | Administrators or local user group members with execution rights for this command. |

## default-information originate always

```
default-information originate always
no default-information originate always
```

### Description

Configures OSPF to advertise the default route (0.0.0.0/0) to its neighbors, regardless if it is present in the routing table or not.

The `no` form of this command sets the OSPF administrative distance to the default of 110.

### Examples

Setting advertisement of the default route:

```
switch(config)# router ospf 1
switch(config-ospf-1)# default-information originate always
```

Disabling advertisement of the default route:

```
switch(config)# router ospf 1
switch(config-ospf-1)# no default-information originate always
```

### Command History

| Release | Modification |
| --- | --- |
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
| --- | --- | --- |
| 6200 | `config-ospf-<PROCESS-ID>` | Administrators or local user group members with execution rights for this command. |

## default-metric

```
default-metric <METRIC-VALUE>
no default-metric
```

### Description

Sets the default metric for redistributed routes in the OSPF.

The `no` form of this command sets the default metric to be used for redistributed routes into OSPF to the default of 25.

| Parameter | Description |
|---|---|
| *<METRIC-VALUE>* | Specifies the default metric value to use for redistributed routes. Default: 25. Range: 0-1677214. |

### Examples

Setting default metric for redistributed routes:

```
switch(config)# router ospf 1
switch(config-ospf-1)# default-metric 37
```

Setting default metric for redistributed routes to the default:

```
switch(config)# router ospf 1
switch(config-ospf-1)# no default-metric
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | config-ospf-*<PROCESS-ID>* | Administrators or local user group members with execution rights for this command. |

## disable

`disable`

### Description

Disables the OSPF process.

This command does not remove the OSPF configurations.

### Examples

Disabling OSPF process:

```
switch(config)# router ospf 1
switch(config-ospf-1)# disable
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config-ospf-<PROCESS-ID>` | Administrators or local user group members with execution rights for this command. |

## distance

```
distance <DISTANCE>
no distance
```

### Description

Defines an administrative distance for OSPF. Administrative distance is used as a criteria to select the best route when multiple routes are present from different routing protocols.

The `no` form of this command sets the OSPF administrative distance to the default of 110.

| Parameter | Description |
|---|---|
| *<DISTANCE>* | Specifies OSPF administrative distance. Range: 1-255. Default: 110. |

### Examples

Setting OSPF administrative distance:

```
switch(config)# router ospf 1
switch(config-ospf-1)# distance 100
```

Setting OSPF administrative distance to the default:

```
switch(config)# router ospf 1
switch(config-ospf-1)# no distance
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config-ospf-<PROCESS-ID>` | Administrators or local user group members with execution rights for this command. |

# enable

`enable`

### Description

Enables the OSPF process, if disabled. By default the OSPF process is enabled.

### Examples

Enabling OSPF process:

```
switch(config)# router ospf 1
switch(config-ospf-1)# enable
```

### Command History

| Release | Modification |
| --- | --- |
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
| --- | --- | --- |
| 6200 | `config-ospf-<PROCESS-ID>` | Administrators or local user group members with execution rights for this command. |

# graceful-restart

```
graceful-restart {restart-interval <SECONDS> | helper}
no graceful-restart {restart-interval | helper}
```

### Description

Configures graceful restart parameters for OSPF.

The `no` form of this command sets the restart interval to the default interval of 120 seconds or disables the helper mode, depending on the parameter specified.

| Parameter | Description |
| --- | --- |
| `restart-interval <SECONDS>` | Specifies the time another router should wait for this router to gracefully restart and selects the maximum interval (in seconds) that another router should wait. Default: 120 seconds. Range: 5-1800. |
| `helper` | Specifies that the router will participate in the graceful restart of a neighbor router. |

### Examples

Enabling OSPF nonstop forwarding:

```
switch(config)# router ospf 1
switch(config-ospf-1)# graceful-restart restart-interval 50
switch(config-ospf-1)# graceful-restart helper
```

Setting restart-interval to default, disable helper mode:

```
switch(config)# router ospf 1
switch(config-ospf-1)# no graceful-restart restart-interval
switch(config-ospf-1)# no graceful-restart helper
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config-ospf-<PROCESS-ID>` | Administrators or local user group members with execution rights for this command. |

## ip ospf area

`ip ospf <PROCESS-ID> area <AREA-ID>`
`no ip ospf <PROCESS-ID> area <AREA-ID>`

### Description

Runs the OSPF protocol on the interface with the configured IPv4 address for the area specified. The interfaces which have an IP address configured in this network or in a subset of this network, will participate in the OSPF protocol.

To move an interface to a new area, unmap the existing area and then associate a new area with the interface.

The `no` form of this command disables OSPF on the interface and removes the interface from the area. Interfaces which have an IP address configured on the network or in a subset of the network, stop participating in the OSPF protocol.

| Parameter | Description |
|---|---|
| *<PROCESS-ID>* | Specifies the OSPF process Id. Range: 1 to 63. |
| *<AREA-ID>* | Specifies the OSPF area ID in one of the following formats.<br>Area identifier in IPv4 format (`x.x.x.x`), where `x` is a decimal number from 0 to 255.<br>Area identifier in decimal format. Range: 0 to 4294967295. |

### Examples

Setting OSPF network for the area:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ip ospf 1 area 1
switch(config-if-vlan)# ip ospf 1 area 0.0.0.1
```

Disabling OSPF network for the area:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ip ospf 1 area 1
switch(config-if-vlan)# no ip ospf 1 area 0.0.0.1
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config-if`<br>`config-if-vlan` | Administrators or local user group members with execution rights for this command. |

## ip ospf authentication

```
ip ospf authentication {message-digest | simple-text | null | keychain | hmac-sha-1 | hmac-
sha-256 | hmac-sha-384 | hmac-sha-512}
no ip ospf authentication
```

### Description

Sets the authentication type that will be used for authentication with the neighbor router.

The `no` form of this command deletes the authentication type used for a particular authentication with the neighbor router and sets to null authentication.

| Parameter | Description |
|---|---|
| `message-digest` | Sets authentication type as message-digest. |
| `simple-text` | Sets authentication type as simple-text. |
| `null` | Sets authentication type as null. |
| `keychain` | Sets the authentication type to use the key chain. |
| `hmac-sha-1` | Sets the authentication type to SHA-1. |
| `hmac-sha-256` | Sets the authentication type to SHA-256. |
| `hmac-sha-384` | Sets the authentication type to SHA-384. |
| `hmac-sha-512` | Sets the authentication type to SHA-512. |

### Examples

Setting OSPF authentication type on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ip ospf authentication simple-text
```

Deleting OSPF authentication type on the interface and sets it to null:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ip ospf authentication
```

Setting OSPF authentication type to SHA-384 on the interface:

```
switch(config)# interface vlan 5
switch(config-if-vlan)# ip ospf authentication hmac-sha-384
```

### Command History

| Release | Modification |
| --- | --- |
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
| --- | --- | --- |
| 6200 | `config-if`<br>`config-if-vlan` | Administrators or local user group members with execution rights for this command. |

## ip ospf authentication-key

```
ip ospf authentication-key [{ciphertext | plaintext} <PASSWORD>]
no ip ospf authentication-key
```

### Description

Sets the authentication password used for simple-text authentication. If the password is given in ciphertext it will be decrypted and applied to the protocol.

The `no` form of this command deletes the authentication password used for simple-text authentication.

| Parameter | Description |
| --- | --- |
| `{ciphertext | plaintext}` | Selects the password format. |
| `<PASSWORD>` | Specifies the password. |

When the password is not provided on the command line, plaintext password prompting occurs upon pressing Enter. The entered password characters are masked with asterisks.

### Examples

Setting the OSPF simple-text authentication password in plaintext format:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ip ospf authentication-key plaintext F82#450b
```

Setting the OSPF simple-text authentication password with a prompted plaintext password:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ip ospf authentication-key
Enter the authentication key: ********
Re-Enter the authentication key: ********
```

Setting the OSPF simple-text authentication password in ciphertext format:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ip ospf authentication-key ciphertext AQBaZ...ecopg=
```

Deleting the OSPF simple-text authentication password:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ip ospf authentication-key
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config-if`<br>`config-if-vlan` | Administrators or local user group members with execution rights for this command. |

## ip ospf cost

```
ip ospf cost <INTERFACE-COST>
no ip ospf cost
```

### Description

Sets the cost (metric) associated with a particular interface. The interface cost is used as a parameter to calculate the best routes.

The `no` form of this command sets the cost (metric) associated with a particular interface to the default cost 1.

| Parameter | Description |
|---|---|
| *<INTERFACE-COST>* | Specifies the interface cost value. Range: 1 to 65535. Default: 1. |

### Examples

Setting OSPF interface cost

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ip ospf cost 100
```

Setting the OSPF interface cost to default

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ip ospf cost
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config-if`<br>`config-if-vlan` | Administrators or local user group members with execution rights for this command. |

## ip ospf dead-interval

```
ip ospf dead-interval <INTERVAL>
no ip ospf dead-interval
```

### Description

Sets the interval after which a neighbor is declared dead if no hello packet is received on the OSPF interface.

The `no` form of this command sets the interval after which a neighbor is declared dead, to the default for the OSPF interface. The default value is 40 seconds (generally 4 times the hello packet interval).

| Parameter | Description |
|---|---|
| *<INTERVAL>* | Specifies the time interval for the dead interval, in seconds. Range: 1 to 65535. Default: 40. |

### Examples

Setting OSPF dead interval on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ip ospf dead-interval 30
```

Setting OSPF dead interval to default on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ip ospf dead-interval
```

### Command History

| Release | Modification |
| --- | --- |
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
| --- | --- | --- |
| 6200 | config-if<br>config-if-vlan | Administrators or local user group members with execution rights for this command. |

## ip ospf hello-interval

```
ip ospf hello-interval <INTERVAL>
no ip ospf hello-interval
```

### Description

Sets the time interval between OSPF hello packets for the OSPF interface.

The `no` form of this command sets the time interval OSPF hello packets to the default of 10 seconds for the OSPF interface.

| Parameter | Description |
| --- | --- |
| *<INTERVAL>* | Specifies the time interval for the hello interval, in seconds. Range: 1 to 65535. Default: 10. |

### Examples

Setting OSPF hello interval on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ip ospf hello-interval 30
```

Setting OSPF hello interval to the default on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ip ospf hello-interval
```

### Command History

| Release | Modification |
| --- | --- |
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config-if`<br>`config-if-vlan` | Administrators or local user group members with execution rights for this command. |

## ip ospf keychain

```
ip ospf keychain <KEYCHAIN-NAME>
no ip ospf keychain
```

### Description

Sets the key chain for md5 authentication. A key chain configures rotating keys for packet authenticating, reducing the risk of keys being compromised.

The `no` form of this command deletes the key chain used for md5 authentication.

| Parameter | Description |
|---|---|
| *<KEYCHAIN-NAME>* | Name of key chain to be used for md5 authentication. |

### Examples

Setting OSPFv2 key chain authentication:

```
switch(config)# interface 1/1/1
switch(config-if)# ip ospf keychain ospf_keys
```

Deleting OSPFv2 key chain authentication:

```
switch(config)# interface 1/1/1
switch(config-if)# no ip ospf keychain
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config-if` | Administrators or local user group members with execution rights for this command. |

## ip ospf message-digest-key md5

```
ip ospf message-digest-key <KEY-ID> md5 [{ciphertext | plaintext} <KEY>]
no ip ospf message-digest-key <KEY-ID>
```

### Description

Sets the md5 message digest authentication key. If the md5 key is given in ciphertext, it will be decrypted and applied to the protocol.

The `no` form of this command deletes the md5 authentication key.

| Parameter | Description |
|---|---|
| *<KEY-ID>* | Specifies the md5 key ID. Range: 1 to 255. |
| {ciphertext \| plaintext} | Selects the md5 key format. |
| *<KEY>* | Specifies the md5 authentication key. |

When the authentication key is not provided on the command line, plaintext key prompting occurs upon pressing Enter. The entered key characters are masked with asterisks.

## Examples

Setting the md5 key in plaintext format:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ip ospf message-digest-key 1 md5 plaintext F82#450b
```

Setting the md5 key with a prompted plaintext key:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ip ospf message-digest-key 1 md5
Enter the MD5 authentication key: ********
Re-Enter the MD5 authentication key: ********
```

Setting the md5 key in ciphertext format:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ip ospf message-digest-key 1 md5 ciphertext AQt6e...7qEa4=
```

Deleting the md5 key:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ip ospf message-digest-key 1
```

## Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

## Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | config-if<br>config-if-vlan | Administrators or local user group members with execution rights for this command. |

# ip ospf network

```
ip ospf network {broadcast | point-to-point}
no ip ospf network
```

## Description

Configures the OSPF network type for the interface. Choose one of the following parameters as the interface network type.

The `no` form of this command sets the network type for the interface to the system default which is broadcast network.

| Parameter | Description |
|---|---|
| broadcast | Specifies the OSPF network type as a broadcast multi-access network. |
| point-to-point | Specifies the OSPF network type as a point-to-point network. |

## Examples

Setting OSPF network type for the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ip ospf network broadcast
switch(config-if-vlan)# ip ospf network point-to-point
```

Disabling OSPF network type for the interface to system default of broadcast network:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ip ospf network
```

## Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

## Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | config-if<br>config-if-vlan | Administrators or local user group members with execution rights for this command. |

# ip ospf passive

```
ip ospf passive
```

```
no ip ospf passive
```

### Description

Configures the interface as an OSPF passive interface. With this setting, the interface participates in OSPF but does not send or receive packets on that interface.

The `no` form of this command resets the interface as active. With this setting, the interface starts sending and receiving OSPF packets.

### Examples

Setting the interface as OSPF passive interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ip ospf passive
```

Setting the interface as OSPF active interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ip ospf passive
```

### Command History

| Release | Modification |
| --- | --- |
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
| --- | --- | --- |
| 6200 | `config-if`<br>`config-if-vlan` | Administrators or local user group members with execution rights for this command. |

## ip ospf priority

```
ip ospf priority <PRIORITY-VALUE>
no ip ospf priority
```

### Description

Sets the OSPF priority for the interface. The larger the numeric value of the priority, the higher the chances for it to become the designated router. Setting a priority of zero makes the router ineligible to become a designated router or back up designated router.

The `no` form of this command sets the OSPF priority for the interface to the default of 1.

| Parameter | Description |
| --- | --- |
| *<PRIORITY-VALUE>* | Specifies the OSPF priority value. Range: 0 to 255. Default: 1. |

### Examples

Setting OSPF priority for the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ip ospf priority 50
```

Disabling OSPF priority for the interface to default:

```
switch(config)# interface vlan1
switch(config-if-vlan)# no ip ospf priority
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config-if`<br>`config-if-vlan` | Administrators or local user group members with execution rights for this command. |

## ip ospf retransmit-interval

```
ip ospf retransmit-interval <INTERVAL>
no ip ospf retransmit-interval
```

### Description

Sets the time between retransmitting lost link state advertisements for the OSPF interface.

The `no` form of this command sets the time between retransmitting lost link state advertisements to the default of 5 seconds for the OSPF interface.

| Parameter | Description |
|---|---|
| *<INTERVAL>* | Specifies the retransmit interval, in seconds. Range: 1 to 3600. Default: 5. |

### Examples

Setting OSPF retransmit interval on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ip ospf retransmit-interval 30
```

Setting OSPF retransmit interval to the default on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ip ospf retransmit-interval
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config-if`<br>`config-if-vlan` | Administrators or local user group members with execution rights for this command. |

## ip ospf sha-key sha

```
ip ospf sha-key <KEY-ID> sha [{ciphertext | plaintext} <KEY>]
no ip ospf sha-key <KEY-ID>
```

### Description

Sets the SHA (secure hash authentication) key for the selected interface. If the SHA key is given in ciphertext, it will be decrypted and applied to the protocol. This command accepts a key of up to 64 characters irrespective of the SHA version configured on the interface. OSPF will internally pad zeros to the key to obtain a 64-byte key. For all types of SHA, key length is adjusted to 64 bytes.

The `no` form of this command deletes the SHA authentication key.

| Parameter | Description |
|---|---|
| *<KEY-ID>* | Specifies the SHA key ID. Range: 1 to 255. |
| `{ciphertext | plaintext}` | Selects the SHA key format. |
| *<KEY>* | Specifies the SHA authentication key. |

When the authentication key is not provided on the command line, plaintext key prompting occurs upon pressing Enter. The entered key characters are masked with asterisks.

### Examples

Setting the SHA authentication key in plaintext format:

```
switch(config)# interface 1/1/1
switch(config-if)# ip ospf sha-key 1 sha plaintext F82#450b
```

Setting the SHA authentication key in prompted plaintext format:

```
switch(config)# interface 1/1/1
switch(config-if)# ip ospf sha-key 1 sha
Enter the SHA authentication key: ********
Re-Enter the SHA authentication key: ********
```

Setting the SHA authentication key in ciphertext format:

```
switch(config)# interface 1/1/1
switch(config-if)# ip ospf sha-key 1 sha ciphertext AQapu...C2K47A=
```

Deleting the SHA authentication key:

```
switch(config)# interface 1/1/1
switch(config-if)# no ip ospf sha-key 1
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config-if`<br>`config-if-vlan` | Administrators or local user group members with execution rights for this command. |

## ip ospf shutdown

```
ip ospf shutdown
no ip ospf shutdown
```

### Description

Disables OSPF on the interface. The interface state changes to Down. It does not remove the interface from the OSPF area. To remove the interface, use the command `no ip ospf area`.

The `no` form of this command re-enables OSPF on the interface

### Examples

Disabling OSPF on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ip ospf shutdown
```

Re-enabling OSPF on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ip ospf shutdown
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config-if`<br>`config-if-vlan` | Administrators or local user group members with execution rights for this command. |

## ip ospf transit-delay

```
ip ospf transit-delay <DELAY>
no ip ospf transit-delay
```

### Description

Sets the time delay in link state transmission for the OSPF interface.

The `no` form of this command sets the delay in link state transmission to the default of 1 second for the OSPF interface.

| Parameter | Description |
|---|---|
| *<DELAY>* | Specifies the transit delay in seconds. Range: 1 to 3600. Default: 1. |

### Examples

Setting OSPF transit delay on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ip ospf transit-delay 30
```

Setting OSPF transit delay to the default on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ip ospf transit-delay
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config-if`<br>`config-if-vlan` | Administrators or local user group members with execution rights for this command. |

## keychain

```
keychain <KEYCHAIN-NAME>
no keychain
```

### Description

Sets the key chain for md5 authentication. A key chain configures rotating keys for packet authenticating, reducing the risk of keys being compromised.

The `no` form of this command deletes the key chain used for md5 authentication.

| Parameter | Description |
|---|---|
| *<KEYCHAIN-NAME>* | Name of key chain to be used for md5 authentication. |

### Examples

Setting OSPF virtual link key chain authentication:

```
switch(config)# router ospf 1
switch(config-ospf-1)# area 100 virtual-link 100.0.1.1
switch(config-router-vlink)# keychain ospf_keys
```

Deleting OSPF virtual link key chain authentication:

```
switch# configure terminal
switch(config)# router ospf 1
switch(config-ospf-1)# area 100 virtual-link  100.0.1.1
switch(config-router-vlink)# no keychain
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config-router-vlink` | Administrators or local user group members with execution rights for this command. |

## max-metric router-lsa

```
max-metric router-lsa [on-startup [<ADVERT-TIME>]]
no max-metric router-lsa [on-startup]
```

### Description

Sets the protocol to advertise a maximum metric so that other routers do not prefer this router as an intermediate hop in their shortest path first (SPF) calculations. If the on-startup parameter is used, the router is configured to advertise a maximum metric at startup for the time mentioned in seconds or for a default value of 600 seconds.

The `no` form of this command advertises the normal cost metrics instead of advertising the maximized cost metric. This setting causes the router to be considered in traffic forwarding.

| Parameter | Description |
|---|---|
| `on-startup <ADVERT-TIME>` | Specifies the time in seconds to advertise self as stub-router on startup. If no time is specified, the default time of 600 seconds is used. Range: 5 to 86400. Default: 600. |

### Examples

Setting to maximize the cost metrics for Router LSA:

```
switch(config)# router ospf 1
switch(config-ospf-1)# max-metric router-lsa
switch(config-ospf-1)# max-metric router-lsa on-startup
switch(config-ospf-1)# max-metric router-lsa on-startup 3000
```

Setting to advertise the normal cost metrics instead of advertising the maximized cost metric:

```
switch(config)# router ospf 1
switch(config-ospf-1)# no max-metric router-lsa
switch(config-ospf-1)# no max-metric router-lsa on-startup
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config-ospf-<PROCESS-ID>` | Administrators or local user group members with execution rights for this command. |

## passive-interface default

```
passive-interface default
no passive-interface
```

### Description

Configures all OSPF interfaces as passive.

The `no` form of this command sets all OSPF interfaces as active.

### Examples

Setting OSPF-enabled interfaces as passive:

```
switch(config)# router ospf 1
switch(config-ospf-1)# passive-interface default
```

Setting OSPF-enabled interfaces as active:

```
switch(config)# router ospf 1
switch(config-ospf-1)# no passive-interface
```

### Command History

| Release | Modification |
| --- | --- |
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
| --- | --- | --- |
| 6200 | `config-ospf-<PROCESS-ID>` | Administrators or local user group members with execution rights for this command. |

## redistribute

```
redistribute {bgp | connected | local loopback | static | rip | ospf <PROCESS-ID>}
             [route-map <ROUTE-MAP-NAME>]
no redistribute {bgp | connected | local loopback | static | rip | ospf <PROCESS-ID>}
                [route-map <ROUTE-MAP-NAME>]
```

### Description

Redistributes routes originating from other protocols, or from another OSPFv2 process, to the current OSPFv2 process.

if a route map is specified, then only the routes that pass the match clause specified in the route map are redistributed to OSPFv2. Configuration is not allowed if the referenced route map has not yet been configured.

If you try to redistribute routes from an OSPFv2 process which is not created, you are prompted to allow the OSPFv2 process to be auto-created before proceeding with redistribution. If you confirm at the prompt, the OSPFv2 process is created with defaults and redistribution configuration applied. If you deny at the prompt, redistribution configuration is skipped.

If command `route-redistribute active-routes-only` has been issued, only the routes from other protocols which are selected for forwarding are considered for redistribution into OSPFv2.

The `no` form of this command disables redistribution of routes to the current OSPFv2 process.

| Parameter | Description |
| --- | --- |
| `bgp` | Specifies redistributing BGP (Border Gateway Protocol) routes. |
| `connected` | Specifies redistributing connected (directly attached subnet or host). |
| `local loopback` | Specifies redistributing local routes of the loopback interface. |
| `static` | Specifies redistributing static routes. |
| `rip` | Specifies redistributing RIP routes. |
| `ospf <PROCESS-ID>` | Specifies redistributing routes from the specified OSPFv2 process |

| Parameter | Description |
|---|---|
| | ID. Range: 1 to 63. |
| `route-map <ROUTE-MAP-NAME>` | Specifies redistribution filtering by route map. To create a route map, use command `route-map`. |

### Examples

Redistributing routes to OSPFv2:

```
switch(config)# router ospf 1
switch(config-ospf-1)# redistribute connected
switch(config-ospf-1)# redistribute connected route-map connected_routes
switch(config-ospf-1)# redistribute local loopback
switch(config-ospf-1)# redistribute local loopback route-map local_routes
switch(config-ospf-1)# redistribute static
switch(config-ospf-1)# redistribute static route-map static_networks
switch(config-ospf-1)# redistribute rip
switch(config-ospf-1)# redistribute rip route-map rip-routes
switch(config-ospf-1)# redistribute ospf 2
```

Disabling redistributing routes to OSPFv2:

```
switch(config)# router ospf 1
switch(config-ospf-1)# no redistribute connected
switch(config-ospf-1)# no redistribute connected route-map connected_routes
switch(config-ospf-1)# no redistribute local loopback
switch(config-ospf-1)# no redistribute local loopback route-map local_routes
switch(config-ospf-1)# no redistribute static
switch(config-ospf-1)# no redistribute static route-map static_networks
switch(config-ospf-1)# no redistribute rip
switch(config-ospf-1)# no redistribute rip route-map rip-routes
switch(config-ospf-1)# no redistribute ospf 2
```

### Command History

| Release | Modification |
|---|---|
| 10.08 | Added `route-map` support for supported redistribute source-protocols. Updated information and examples. |
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config-ospf-<PROCESS-ID>` | Administrators or local user group members with execution rights for this command. |

## reference-bandwidth

```
reference-bandwidth <BANDWIDTH>
no reference-bandwidth
```

### Description

Sets the reference bandwidth for OSPFv2. If the OSPFv2 interface cost is not explicitly set. then the cost of all the OSPFv2 interfaces is recalculated based on the reference bandwidth and link speed of the interface.

For VLAN interfaces the link speed value is taken as 1 Gbps (if the OSPFv2 interface cost is not explicitly set).

The `no` form of this command sets the reference bandwidth for OSPF to the default of 100000 Mbps.

| Parameter | Description |
|---|---|
| *<BANDWIDTH>* | Specifies the reference bandwidth used to calculate the cost of an interface in Mbps. Range: 1 to 4000000. Default: 100000. |

### Examples

Setting the reference bandwidth:

```
switch(config)# router ospf 1
switch(config-ospf-1)# reference-bandwidth 40000
```

Setting the reference bandwidth to the default value:

```
switch(config)# router ospf 1
switch(config-ospf-1)# no reference-bandwidth
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config-ospf-<PROCESS-ID>` | Administrators or local user group members with execution rights for this command. |

## rfc1583-compatibility

```
rfc1583-compatibility
no rfc1583-compatibility
```

### Description

Enables OSPF compatibility with RFC1583 (backward compatibility). If RFC1583 compatibility is enabled, then the route cost calculation follows a different method.

The `no` form of this command disables OSPF compatibility with RFC1583 (backward compatibility). By default the RFC1583 compatibility is disabled.

### Examples

Enabling OSPF RFC1583 compatibility:

```
switch(config)# router ospf 1
switch(config-ospf-1)# rfc1583-compatibility
```

Disabling OSPF RFC1583 compatibility:

```
switch(config)# router ospf 1
switch(config-ospf-1)# no rfc1583-compatibility
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config-ospf-<PROCESS-ID>` | Administrators or local user group members with execution rights for this command. |

## router ospf

```
router ospf <PROCESS-ID> [vrf <VRF-NAME>]
no router ospf <PROCESS-ID> [vrf <VRF-NAME>]
```

### Description

Creates an OSPF process (if not created already) on a VRF, and switches to the OSPF router instance context. Up to eight OSPF processes are supported per VRF.

The `no` form of this command removes the OSPF instance.

| Parameter | Description |
|---|---|
| `<PROCESS-ID>` | Specifies an OSPF process ID. Range: 1 to 63. |
| `vrf <VRF-NAME>` | Specifies a VRF name for the OSPF process. Default: default. |

### Examples

```
switch(config)# router ospf 1
switch(config-ospf-1)#
```

```
switch(config)# router ospf 1 vrf vrf_red
```

```
switch(config)# no router ospf 1
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config` | Administrators or local user group members with execution rights for this command. |

## router-id

```
router-id <ROUTER-ADDR>
no router-id
```

### Description

Sets an ID for the router in an IPv4 address format.

The `no` form of this command unconfigures the router-id for the instance and sets the router-id to the default as follows: the router-id is selected dynamically as equal to the highest loopback address on the router, or the highest active interface if there are no loopback addresses. If no IP address is configured on any interfaces on the router, OSPF will not form an adjacency.

| Parameter | Description |
|---|---|
| *<ROUTER-ADDR>* | Specifies the Router ID in IPv4 format (`x.x.x.x`), where `x` is a decimal number from 0 to 255. |

### Examples

Setting router-id in the OSPF context:

```
switch(config)# router ospf 1
switch(config-ospf-1)# router-id 1.1.1.1
```

Unconfiguring router-id:

```
switch(config)# router ospf 1
switch(config-ospf-1)# no router-id
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config-ospf-<PROCESS-ID>` | Administrators or local user group members with execution rights for this command. |

## sha-key sha

```
sha-key <KEY-ID> sha [{ciphertext | plaintext} <KEY>]
no sha-key <KEY-ID>
```

### Description

Sets the SHA (secure hash authentication) key for the selected virtual link. If the SHA key is given in ciphertext, it will be decrypted and applied to the protocol. This command accepts a key of up to 64 characters irrespective of the SHA version configured on virtual link. OSPF will internally pad zeros to the key to obtain a 64-byte key. For all types of SHA, key length is adjusted to 64 bytes.

The `no` form of this command deletes the virtual link SHA authentication key.

| Parameter | Description |
|---|---|
| `<KEY-ID>` | Specifies the virtual link SHA key ID. Range: 1 to 255. |
| `{ciphertext \| plaintext}` | Selects the virtual link SHA key format. |
| `<KEY>` | Specifies the virtual link SHA authentication key. |

When the authentication key is not provided on the command line, plaintext key prompting occurs upon pressing Enter. The entered key characters are masked with asterisks.

### Examples

Setting virtual link SHA authentication key in plaintext format:

```
switch(config)# router ospf 1
switch(config-ospf-1)# area 100 virtual-link 100.0.1.1
switch(config-router-vlink)# sha-key 1 sha plaintext F82#450b
```

Setting the virtual link SHA authentication key in prompted plaintext format:

```
switch(config)# router ospf 1
switch(config-ospf-1)# area 100 virtual-link 100.0.1.1
switch(config-router-vlink)# sha-key 1 sha
Enter the SHA authentication key: ********
Re-Enter the SHA authentication key: ********
```

Setting the virtual link SHA authentication key in ciphertext format:

```
switch(config)# router ospf 1
switch(config-ospf-1)# area 100 virtual-link 100.0.1.1
switch(config-router-vlink)# sha-key 1 sha ciphertext AQapu...C2K47A=
```

Deleting the virtual link SHA authentication key:

```
switch(config)# router ospf 1
switch(config-ospf-1)# area 100 virtual-link 100.0.1.1
switch(config-router-vlink)# no sha-key 1
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config-router-vlink` | Administrators or local user group members with execution rights for this command. |

## show ip ospf

`show ip ospf [<PROCESS-ID>] [all-vrfs | vrf <VRF-NAME>]`

### Description

Displays general OSPF, area, state, and configuration information.

| Parameter | Description |
|---|---|
| *<PROCESS-ID>* | Enter an OSPF process ID to display general OSPF information for a particular OSPF process. Range: 1 to 63. |
| `all-vrfs` | Optionally select to display general OSPF information for all VRFs. |
| `vrf` *<VRF-NAME>* | Specifies the name of a VRF. Default: default. |

### Examples

Showing general OSPF configurations:

```
switch# show ip ospf 1
Routing Process 1 with ID : 1.1.1.1 VRF default
------------------------------------------------

OSPFv2 Protocol is enabled
Graceful-restart is configured
Restart Interval: 120, State: inactive
Last Graceful Restart Exit Status: none
Area Border: false
AS Border: false
SPF: Start Time: 200ms, Hold Time: 1000ms, Max Wait Time: 5000ms
LSA: Start Time: 5000ms, Hold Time: 0ms, Max Wait Time: 0ms
LSA Arrival: 1000ms
Maximum Paths to Destination: 4
Number of external LSAs 0, checksum sum 0
Summary address:
  prefix 20.1.1.0/24 advertise tag 10
```

```
Number of areas is 2, 2 normal, 0 stub, 0 NSSA
Number of active areas is 1, 1 normal, 0 stub, 0 NSSA
BFD is disabled
Reference Bandwidth: 100000 Mbps
Area (0.0.0.0) (Active)
  Interfaces in this Area: 1 Active Interfaces: 1
  Passive Interfaces: 0 Loopback Interfaces: 0
  SPF calculation has run 7 times
  Area ranges:
  Number of LSAs: 3, checksum sum 75093
Area (0.0.0.1) (Inactive)
  Interfaces in this Area: 0 Active Interfaces: 0
  Passive Interfaces: 0 Loopback Interfaces: 0
  SPF calculation has run 2 times
  Area ranges:
    ip-prefix 20.1.1.1/24, inter-area, advertise
  Number of LSAs: 0, checksum sum 0
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show ip ospf border-routers

```
show ip ospf [<PROCESS-ID>]
     border-routers [all-vrfs | vrf <VRF-NAME>]
```

### Description

Displays the OSPF routing table entries for Area Border Router (ABR) and Autonomous System Border Router (ASBR).

| Parameter | Description |
|---|---|
| *<PROCESS-ID>* | Enter an OSPF process ID to display general OSPF information for a particular OSPF process. Range: 1 to 63. |
| `all-vrfs` | Optionally select to display general OSPF information for all VRFs. |
| `vrf` *<VRF-NAME>* | Specifies the name of a VRF. Default: default. |

### Examples

Showing OSPF border routers information:

```
switch# show ip ospf border-routers
OSPF Process ID 1 VRF default Internal Routing Table
-----------------------------------------------------

Codes: i - Intra-area route, I - Inter-area route

i 2.2.2.2 [10], ABR, Area 0.0.0.0, SPF 15 via
    10.1.1.2, Interface 1/1/1
i 2.2.2.2 [10], ABR, Area 0.0.0.1, SPF 15 via
    11.1.1.12, Interface 1/1/2
```

### Command History

| Release | Modification |
| --- | --- |
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
| --- | --- | --- |
| 6200 | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show ip ospf interface

`show ip ospf [<PROCESS-ID>] interface [<interface-name>] [brief] [all-vrfs | vrf <VRF-NAME>]`

### Description

Displays general OSPF, area, state, and configuration information.

| Parameter | Description |
| --- | --- |
| `<PROCESS-ID>` | Enter an OSPF process ID to display general OSPF information for a particular OSPF process. Range: 1 to 63. |
| `<interface-name>]` | Specify the name of an OSPF interface. |
| `brief` | Display a brief overview of OSPF configuration information. |
| `all-vrfs` | Optionally select to display general OSPF information for all VRFs. |
| `vrf <VRF-NAME>` | Specifies the name of a VRF. Default: default. |

### Examples

Showing general OSPF configuration settings for the default VRF:

```
switch# show ip ospf 1
Routing Process 1 with ID : 1.1.1.1 VRF default
------------------------------------------------
```

```
OSPFv2 Protocol is enabled
Graceful-restart is configured
Restart Interval: 120, State: inactive
Last Graceful Restart Exit Status: none
Area Border: false
AS Border: false
SPF: Start Time: 200ms, Hold Time: 1000ms, Max Wait Time: 5000ms
LSA: Start Time: 5000ms, Hold Time: 0ms, Max Wait Time: 0ms
LSA Arrival: 1000ms
Maximum Paths to Destination: 4
Number of external LSAs 0, checksum sum 0
Summary address:
  prefix 20.1.1.0/24 advertise tag 10
Number of areas is 2, 2 normal, 0 stub, 0 NSSA
Number of active areas is 1, 1 normal, 0 stub, 0 NSSA
BFD is disabled
Reference Bandwidth: 100000 Mbps
Area (0.0.0.0) (Active)
  Interfaces in this Area: 1 Active Interfaces: 1
  Passive Interfaces: 0 Loopback Interfaces: 0
  SPF calculation has run 7 times
  Area ranges:
  Number of LSAs: 3, checksum sum 75093
Area (0.0.0.1) (Inactive)
  Interfaces in this Area: 0 Active Interfaces: 0
  Passive Interfaces: 0 Loopback Interfaces: 0
  SPF calculation has run 2 times
  Area ranges:
    ip-prefix 20.1.1.1/24, inter-area, advertise
  Number of LSAs: 0, checksum sum 0
```

Showing OSPF configuration settings for all interfaces:

```
switch# show ip ospf interface
Interface 1/1/1 is up, line protocol is up
-------------------------------------------

IP address 10.1.1.1/24, Process ID 1 VRF default, area 0.0.0.0
    State Backup-dr, Status up, Network type Broadcast
    Link Speed: 10000 Mbps
    Cost Configured NA, Calculated 10
    Transit delay 1 sec, Router priority 1
    Designated Router IP: 10.1.1.2
    Backup Designated Router IP: 10.1.1.1
    Timer Intervals: Hello 10, Dead 40, Retransmit 5
    No authentication
    Number of Link LSAs: 0, checksum sum 0
    BFD is disabled

Interface 1/1/2 is up, line protocol is up
-------------------------------------------

IP address 11.1.1.1/24, Process ID 1 VRF default, area 0.0.0.1
    State Dr, Status up, Network type Broadcast
    Link Speed: 10000 Mbps
    Cost Configured NA, Calculated 10
    Transit delay 1 sec, Router priority 1
    Designated Router IP: 11.1.1.1
    No backup designated router on this network
    Timer Intervals: Hello 10, Dead 40, Retransmit 5
    No authentication
```

```
Number of Link LSAs: 0, checksum sum 0
BFD is disabled
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show ip ospf lsdb

```
show ip ospf [<PROCESS-ID>] lsdb  [all-vrfs | vrf <VRF-NAME>]
     [area <AREA-ID>] [lsid <LINK-STATE-ID>]
     [adv-router {<ROUTER-ID> | self}]
show ip ospf [<PROCESS-ID>] lsdb  [all-vrfs | vrf <VRF-NAME>]
     asbr-summary [area <AREA-ID>] [lsid <LINK-STATE-ID>]
     [adv-router {<router-id> | self}]
show ip ospf [<PROCESS-ID>] lsdb  [all-vrfs | vrf <VRF-NAME>]
     external [lsid <LINK-STATE-ID>] [adv-router {<ROUTER-ID> | self}]
show ip ospf [<PROCESS-ID>] lsdb  [all-vrfs | vrf <VRF-NAME>]
     router [area <AREA-ID>] [lsid <LINK-STATE-ID>]
     [adv-router {<ROUTER-ID> | self}]
show ip ospf [<PROCESS-ID>] lsdb  [all-vrfs | vrf <VRF-NAME>]
     network [area <AREA-ID>] [lsid <LINK-STATE-ID>]
     [adv-router {<ROUTER-ID> | self}]
show ip ospf [<PROCESS-ID>] lsdb  [all-vrfs | vrf <VRF-NAME>]
     summary [area <AREA-ID>] [lsid <LINK-STATE-ID>]
     [adv-router {<ROUTER-ID> | self}]
show ip ospf [<PROCESS-ID>] lsdb  [all-vrfs | vrf <VRF-NAME>]
     nssa-external [area <AREA-ID>] [lsid <LINK-STATE-ID>]
     [adv-router {<ROUTER-ID> | self}]
show ip ospf [<PROCESS-ID>] lsdb  [all-vrfs | vrf <VRF-NAME>]
     database-summary
```

### Description

Shows the OSPF link state database summary for different OSPF LSAs (Link State Advertisement). Use the parameters to get information for a particular LSA.

| Parameter | Description |
|---|---|
| *<PROCESS-ID>* | Enter an OSPF process ID to display general OSPF information for a particular OSPF process. Range: 1 to 63. |
| all-vrfs | Optionally select to display general OSPF information for all VRFs. |
| vrf *<VRF-NAME>* | Specifies the name of a VRF. Default: default.<br>Optionally select one of the following parameters to filter the link |

| Parameter | Description |
|---|---|
| | state database information.<br>`asbr-summary`. Displays ASBR summary link states (LSA type 4).<br>`external`. Displays external link states (LSA type 5). The external parameter does not take the area <AREA-ID> parameter.<br>`router`. Displays router LSAs (LSA type 1).<br>`network`. Displays network LSAs (LSA type 2).<br>`summary`. Displays network-summary link states (LSA type 3).<br>`nssa-external`. Displays NSSA external link states (LSA type 7). |
| `database-summary` | Select to display the count of each type of LSA and each area in the database. |
| `area <AREA-ID>` | Select to display information filtered for the specified area in one of the following formats.<br>OSPF area identifier in IPv4 format (`x.x.x.x`), where `x` is a decimal number from 0 to 255.<br>OSPF area identifier in decimal format. Value: 0 to 4294967295. |
| `lsid <LINK-STATE-ID>` | Select to display information filtered by link state identifier specified in IPv4 address format (A.B.C.D). |
| `adv-router {<ROUTER-ID> \| self}` | Select to display link states for a particular advertising router. Specify either a Router ID of the advertising router or specify `self` to show self-originated link states. |

## Examples

Showing OSPF link state database (LSDB) general information:

```
switch# show ip ospf lsdb
OSPF Router with ID (50.50.50.50) (Process ID 1 VRF default)
============================================================

Router Link State Advertisements (Area 0.0.0.0)
-----------------------------------------------

LSID            ADV Router      Age     Seq#        Checksum    Link Count
--------------------------------------------------------------------------
40.40.40.40     40.40.40.40     930     0x80000004  0x2ea1      3
50.50.50.50     50.50.50.50     935     0x80000002  0x8b52      1
60.60.60.60     60.60.60.60     943     0x800003c5  0x9854      2

Network Link State Advertisements (Area 0.0.0.0)
------------------------------------------------

LSID            ADV Router      Age     Seq#        Checksum
--------------------------------------------------------------
209.165.201.3   60.60.60.60     944     0x80000001  0x7179
192.0.2.1       50.50.50.50     935     0x80000001  0x516a

Inter-area Summary Link State Advertisements (Area 0.0.0.0)
------------------------------------------------------------
```

```
LSID            ADV Router      Age     Seq#        Checksum
--------------------------------------------------------------
209.165.201.1   40.40.40.40     929     0x80000001  0x2498
209.165.201.1   50.50.50.50     928     0x80000001  0x5b2f
209.165.201.1   60.60.60.60     1265    0x800003c3  0xf49b
192.0.2.0       40.40.40.40     943     0x80000001  0x53f3
192.0.2.0       50.50.50.50     935     0x80000001  0x26f8
192.0.2.0       60.60.60.60     930     0x80000001  0x7b51
```

Showing ASBR summary link states:

```
switch# show ip ospf lsdb asbr-summary
OSPF Router with ID (2.2.2.1) (Process ID 1 VRF default)
========================================================

ASBR Summary Link State Advertisements (Area 0.0.0.0)
-----------------------------------------------------

LSID            ADV Router      Age     Seq#            Checksum
------------------------------------------------------------------
209.165.201.3   60.60.60.60     944     0x80000001      0x7179
192.0.2.1       50.50.50.50     935     0x80000001      0x516a
```

Showing external link states:

```
switch# show ip ospf lsdb external
OSPF Router with ID (2.2.2.1) (Process ID 1 VRF default)
========================================================

AS External Link State Advertisements
--------------------------------------

LSID            ADV Router      Age     Seq#            Checksum
------------------------------------------------------------------
209.165.201.3   60.60.60.60     944     0x80000001      0x7179
192.0.2.1       50.50.50.50     935     0x80000001      0x516a
```

Showing database summary:

```
switch# show ip ospf lsdb database-summary
OSPF Router with ID (10.1.1.1) (Process ID 1 VRF default)
==========================================================

Area 0.0.0.0 database summary
------------------------------

LSA Type             Count
---------------------------
Router               2
Network              1
Inter-area Summary   1
ASBR Summary         0
NSSA External        0
Subtotal             4

Process 1 database summary
---------------------------
```

```
LSA Type             Count
---------------------------
Router               2
Network              1
Inter-area Summary   1
ASBR Summary         0
NSSA External        0
AS External          0
Total                4
```

Showing router LSAs:

```
switch# show ip ospf lsdb router
OSPF Router with ID (2.2.2.1) (Process ID 1 VRF default)
========================================================

Router Link State Advertisements (Area 0.0.0.0)
-----------------------------------------------

LSID            ADV Router      Age         Seq#       Checksum Link Count
--------------------------------------------------------------------------
1.1.1.2         1.1.1.2         15          0x80000004 0xf526    1
2.2.2.1         2.2.2.1         14          0x80000005 0x6c5e    2
2.2.2.2         2.2.2.2         104         0x80000004 0xf51a    1
```

Showing network LSAs:

```
switch# show ip ospf lsdb network
OSPF Router with ID (2.2.2.1) (Process ID 1 VRF default)
========================================================

Network Link State Advertisements (Area 0.0.0.0)
------------------------------------------------

LSID            ADV Router      Age         Seq#       Checksum
----------------------------------------------------------------
1.1.1.2         1.1.1.2         141         0x80000001 0xc55e
2.2.2.2         2.2.2.2         230         0x80000001 0xa179
```

Showing network-summary link states:

```
sswitch# show ip ospf lsdb summary
OSPF Router with ID (2.2.2.1) (Process ID 1 VRF default)
========================================================

Inter-area Summary Link State Advertisements (Area 0.0.0.0)
-----------------------------------------------------------

LSID            ADV Router      Age         Seq#       Checksum
----------------------------------------------------------------
1.1.1.0         2.2.2.1         133         0x80000002 0xa089

Inter-area Summary Link State Advertisements (Area 0.0.0.1)
-----------------------------------------------------------

LSID            ADV Router      Age         Seq#       Checksum
```

```
----------------------------------------------------------------
2.2.2.0         2.2.2.1         133         0x80000002 0x7caa
```

Showing NSSA external link states:

```
switch(config-ospf-1)# show ip ospf lsdb nssa-external
OSPF Router with ID (2.2.2.1) (Process ID 1 VRF default)
========================================================

NSSA External Link State Advertisements (Area 0.0.0.1)
------------------------------------------------------

LSID            ADV Router      Age         Seq#       Checksum
----------------------------------------------------------------
8.8.8.0         1.1.1.2         162         0x80000003 0xc7b2
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show ip ospf neighbors

```
show ip ospf [<PROCESS-ID>] neighbors [<NEIGHBOR-ID>]
     [interface <INTERFACE-NAME>] [detail | summary]
     [all-vrfs | vrf <VRF-NAME>]
```

### Description

Displays information about OSPF neighbors.

| Parameter | Description |
|---|---|
| `<PROCESS-ID>` | Enter an OSPF process ID to display OSPF neighbor information for the particular OSPF process. Range: 1 to 63. |
| `neighbors <NEIGHBOR-ID>` | Select to display information about a particular neighbor, specified in IPv4 format (A.B.C.D). |
| `interface <INTERFACE-NAME>` | Select to display neighbor information only for the specified interface. |
| `detail` | Select to display detailed information for all the neighbors. |
| `summary` | Select to display summary information for the neighbors. |

| Parameter | Description |
|---|---|
| `all-vrfs` | Select to display neighbor information for all VRFs. |
| `vrf <VRF-NAME>` | Specify the name of a VRF. Default: default. |

## Examples

Showing OSPF neighbors information for the default VRF:

```
switch# show ip ospf neighbors
OSPF Process ID 1 VRF default
==============================

Total Number of Neighbors: 1

Neighbor ID      Priority  State             Nbr Address       Interface
-------------------------------------------------------------------------
2.2.2.2          1         FULL/DR           10.1.1.2           1/1/1
```

Showing OSPF neighbors information for VRF red:

```
switch# show ip ospf neighbors vrf red
OSPF Process ID 1 VRF red
==========================

Total Number of Neighbors: 1

Neighbor ID      Priority  State             Nbr Address       Interface
-------------------------------------------------------------------------
1.1.1.1          1         FULL/BDR          10.1.1.1           1/1/1
```

Showing OSPF neighbors information for a specific neighbor:

```
switch# show ip ospf neighbors 2.2.2.2
Neighbor 2.2.2.2, interface address 10.1.1.2
--------------------------------------------
Process ID 1 VRF default, in area 0.0.0.0 via interface 1/1/1
Neighbor priority is 1, State is FULL
Options is 0x42
Dead timer due in 00:00:32
Time since last state change 00h:19m:17s
```

Showing OSPF neighbors information for a specific neighbor and interface:

```
switch# show ip ospf neighbors 2.2.2.2 interface 1/1/1
Neighbor 2.2.2.2, interface address 10.1.1.2
--------------------------------------------
Process ID 1 VRF default, in area 0.0.0.0 via interface 1/1/1
Neighbor priority is 1, State is FULL
DR is 10.1.1.2, BDR is 10.1.1.1
Options is 0x42
Dead timer due in 00:00:38
Retransmission queue length 0
Time since last state change 00h:22m:21s
```

Showing detail information for OSPF neighbors:

```
switch# show ip ospf neighbors detail
Neighbor 2.2.2.2, interface address 10.1.1.2
--------------------------------------------
Process ID 1 VRF default, in area 0.0.0.0 via interface 1/1/1
Neighbor priority is 1, State is FULL
DR is 10.1.1.2, BDR is 10.1.1.1
Options is 0x42
Dead timer due in 00:00:38
Retransmission queue length 0
Time since last state change 00h:22m:21s
```

Showing summary information for OSPF neighbors in default VRF:

```
switch# show ip ospf neighbors summary
OSPF Process ID 1 VRF default, Neighbor Summary
================================================

Interface  Down Attempt Init TwoWay ExStart Exchange Loading Full  Total
-------------------------------------------------------------------------
1/1/1      0    0       0    0      0       0        0       1     1
1/1/2      0    0       0    0      0       0        0       0     0
Total      0    0       0    0      0       0        0       1     1
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show ip ospf routes

```
show ip ospf [<PROCESS-ID>] routes
     [<IPV4-ADDR>/<MASK>] [all-vrfs | vrf <VRF-NAME>]
```

### Description

Displays OSPF routing table information.

| Parameter | Description |
|---|---|
| *<PROCESS-ID>* | Enter an OSPF process ID to display OSPF neighbor information for the particular OSPF process. Range: 1 to 63. |
| *<IPV4-ADDR>* | Specify an IP address in IPv4 format (`x.x.x.x`), where `x` is a decimal number from 0 to 255. . |
| *<MASK>* | Specifies the number of bits in the address mask in CIDR format |

| Parameter | Description |
| --- | --- |
| | (x), where x is a decimal number from 0 to 32. |
| all-vrfs | Select to display neighbor information for all VRFs. |
| vrf <VRF-NAME> | Specify the name of a VRF. Default: default. |

### Examples

Showing OSPF routing table information:

```
switch# show ip ospf routes
Codes: i - Intra-area route, I - Inter-area route
       E1 - External type-1, E2 - External type-2

OSPF Process ID 1 VRF default, Routing Table
---------------------------------------------

Total Number of Routes : 2

10.1.1.0/24        (i) area: 0.0.0.0
     directly attached to interface 1/1/1, cost 1 distance 110
20.1.1.0/24        (I)
     via 10.1.1.2 interface 1/1/1, cost 2 distance 110
```

Showing OSPF routing table information for a specific subnet:

```
switch# show ip ospf routes 10.1.1.0/2
Codes: i - Intra-area route, I - Inter-area route
       E1 - External type-1, E2 - External type-2

OSPF Process ID 1 VRF default, Routing Table for prefixes 10.1.1.0/24
----------------------------------------------------------------------

Total Number of Routes : 1

10.1.1.0/24        (i) area: 0.0.0.0
     directly attached to interface 1/1/1, cost 1 distance 110
```

### Command History

| Release | Modification |
| --- | --- |
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
| --- | --- | --- |
| 6200 | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show ip ospf statistics

```
show ip ospf [<PROCESS-ID>] statistics [all-vrfs | vrf <VRF-NAME>]
```

### Description

Displays OSPF statistics.

| Parameter | Description |
|---|---|
| *<PROCESS-ID>* | Enter an OSPF process ID to display OSPF neighbor information for the particular OSPF process. Range: 1 to 63. |
| `all-vrfs` | Select to display OSPF statistics information for all VRFs. |
| `vrf <VRF-NAME>` | Specify the name of a VRF. Default: default. |

### Examples

Showing OSPF statistics:

```
switch# show ip ospf statistics
OSPF Process ID 1 VRF default, Statistics (cleared 1h 16m 24s ago)
-------------------------------------------------------------

Unknown Interface Drops         : 0
Unknown Virtual Interface Drops : 0
Bad Instance ID Drops           : 0
Bad IP Header Length Drops      : 0
Wrong OSPF Version Drops        : 0
Bad Source IP Drops             : 0
Resource Failure Drops          : 0
Bad Header Length Drops         : 0
Total Drops                     : 0
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show ip ospf statistics interface

```
show ip ospf [<PROCESS-ID>] statistics interface [<INTERFACE-NAME>]
     [all-vrfs | vrf <VRF-NAME>]
```

### Description

Displays OSPF statistics for the OSPF-enabled interfaces.

| Parameter | Description |
|---|---|
| *<PROCESS-ID>* | Enter an OSPF process ID to display OSPF-enabled interface statistics information on the specified OSPF process. Range: 1 to 63. |
| *<INTERFACE-NAME>* | Select to display information only for the specified interface. |
| all-vrfs | Select to display OSPF-enabled interface statistics information for all VRFs. |
| vrf *<VRF-NAME>* | Specify the name of a VRF. Default: default. |

## Examples

Showing OSPF-enabled interfaces information:

```
switch# show ip ospf statistics interface 1/1/1
OSPF Process ID 1 VRF default, interface 1/1/1 statistics (cleared 0h 30m 28s ago)
============================================================================

Tx Hello Packets      : 101              Rx Hello Packets       : 99
Tx Hello Bytes        : 101              Rx Hello Bytes         : 99
Tx DD Packets         : 101              Rx DD Packets          : 99
Tx DD Bytes           : 101              Rx DD Bytes            : 99
Tx LS Request Packets : 101              Rx LS Requests Packets : 99
Tx LS Request Bytes   : 101              Rx LS Request Bytes    : 99
Tx LS Update Packets  : 101              Rx LS Update Packets   : 99
Tx LS Update Bytes    : 101              Rx LS Update Bytes     : 99
Tx LS Ack Packets     : 101              Rx LS Ack Packets      : 99
Tx LS Ack Bytes       : 101              Rx LS Ack Bytes        : 99

Total Number of State Changes : 8
Number of LSAs                : 29
LSA Checksum Sum              : 2345
Total Transmit Failures       : 29
Total OSPF Packets Discarded  : 999

Reason                               Packets Dropped
-----------------------------------------------------
Invalid type                         19
Invalid length                       9
Invalid checksum                     0
Invalid version                      23
Bad or unknown source                67
Area mismatch                        1
Self-originated                      19
Duplicate router ID                  9
Interface standby                    0
Total Hello packets dropped          60
  Network Mask mismatch              10
  Hello interval mismatch            10
  Dead interval mismatch             10
  Options mismatch                   10
  MTU mismatch                       10
  Neighbor ignored                   10
Authentication errors                12
  Type mismatch                      6
  Authentication failures            6
Wrong protocol                       0
Resource failures                    0
```

```
Bad LSA length                        0
Others                                0

Total LSAs Ignored : 176
Bad Type           : 10
Bad Length         : 56
Invalid Data       : 55
Invalid Checksum   : 55
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## summary-address

```
summary-address  <IPV4-ADDR>/<MASK> [no-advertise | tag <TAG-VALUE>]
no summary-address <prefix/length> [no-advertise | tag <tag-value>
```

### Description

Summarizes the external routes with the matching address and mask. When advertising this route, its metric is set to the lowest cost path from among the routes that were summarized.

The `no` form of this command disables route summarization.

This command only works for an ASBR (Autonomous System Boundary Router).

| Parameter | Description |
|---|---|
| *<IPV4-ADDR>* | Specifies an IP address in IPv4 format (`x.x.x.x`), where `x` is a decimal number from 0 to 255. |
| *<MASK>* | Specifies the number of bits in the address mask in CIDR format (`x`), where `x` is a decimal number from 0 to 32. |
| `no-advertise` | Do not advertise the aggregate route. Suppress routes that match the specified prefix/mask pair. |
| `tag` *<TAG-VALUE>* | Specify the tag for the aggregate route. The summary prefix will be advertised along with the tag value in External LSAs. Range: 0 to 4294967295 |

### Examples

Setting OSPF route summarization:

```
switch(config)# router ospf 1
switch(config-ospf-1)# summary-address 10.1.0.0/16
```

Disabling route summarization:

```
switch(config)# router ospf 1
switch(config-ospf-1)# no summary-address 10.1.0.0/16
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config-ospf-<PROCESS-ID>` | Administrators or local user group members with execution rights for this command. |

## timers lsa-arrival

```
timers lsa-arrival <DELAY>
no timers lsa-arrival
```

### Description

Configures the minimum delay between receiving the same LSA from a peer. The same LSA is an LSA that contains the same LSA ID number, LSA type, and advertising router ID. If an instance of the same LSA arrives sooner before the delay expires, the LSA is dropped. Generally, the LSA arrival timer should be set to a value less than or equal to the start-time value for the command `timers throttle lsa start` on the neighbor.

The `no` form of this command sets the LSA timers to default values.

| Parameter | Description |
|---|---|
| *<DELAY>* | Specifies the delay in milliseconds. Range: 0 to 600000. Default: 1000. |

### Examples

Setting the LSA arrival timer:

```
switch(config)# router ospf 1
switch(config-ospf-1)# timers lsa-arrival 10
```

Setting the LSA arrival timer to default:

```
switch(config)# router ospf 1
switch(config-ospf-1)# no timers lsa-arrival
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config-ospf-<PROCESS-ID>` | Administrators or local user group members with execution rights for this command. |

## timers throttle lsa

```
timers throttle lsa start-time <START-TIME> hold-time <HOLD-TIME> max-wait-time <WAIT-TIME>
no timers throttle lsa
```

### Description

Configures the timers for LSA generation.

The `no` form of this command sets the LSA timers to default values.

| Parameter | Description |
|---|---|
| `start-time <START-TIME>` | Specifies the initial wait time in milliseconds after which LSAs are generated. When set to 0, the LSAs are generated without any delay. Range: 0 to 600000. Default: 5000. |
| `hold-time <HOLD-TIME>` | Specifies the amount of time, in milliseconds, between regeneration of an LSA. The hold time doubles each time the same LSA must be regenerated, until `max-wait-time` is reached. When set to 0, LSA regeneration time is not increased. Range: 0 to 600000. Default: 0. |
| `max-wait-time <WAIT-TIME>` | Specifies the maximum wait time, in milliseconds, for regeneration of the same LSA. When set to 0, LSA regeneration time is not increased. Range: 0 to 600000. Default: 0. |

### Examples

Setting the LSA timers:

```
switch(config)# router ospf 1
switch(config-ospf-1)# timers throttle lsa start-time 100 hold-time 1000 max-wait-
time 10000
```

Setting LSA timers to default values:

```
switch(config)# router ospf 1
switch(config-ospf-1)# no timers throttle lsa
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config-ospf-<PROCESS-ID>` | Administrators or local user group members with execution rights for this command. |

## timers throttle spf

```
timers throttle spf start-time <START-TIME> hold-time <HOLD-TINME>
max-wait-time <WAIT-TIME>
no timers throttle spf
```

### Description

Configures timers for SPF calculation. There are three timers:

- `start-time` Is the initial delay before an SPF calculation is started. Default is 200 milliseconds.
- `hold-time` Is the progressive backoff time to wait before next scheduled SPF calculation. Default is 1000 milliseconds. If a route change event occurs during this period, the value doubles until it reaches the *max-wait-time*.
- `max-wait-time` Is the maximum time to wait before the next scheduled SPF calculation. Default is 5000 milliseconds. This is used to limit the SPF hold timer and also defines the time to be considered for which the OSPF LSDB has to be stable, after which the SPF throttle mechanism is reset.

The `no` form of this command sets all the configured non-default timers to default value.

| Parameter | Description |
|---|---|
| *<START-TIME>* | Time in milliseconds to set timer for initial SPF delay. Default: 200. |
| *<HOLD-TINME>* | Time in milliseconds to set the minimum hold time between two consecutive SPF calculations. Default: 1000. |
| *<WAIT-TIME>* | Time in milliseconds to set the maximum wait time between two consecutive SPF calculations. Default: 5000. |

### Examples

Setting non-default timer values for SPF throttling:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# timers throttling spf start-time 500 hold-time 3000 max-
```

```
wait-time 9000
Switch(config-ospfv3-1)# show running-config current-context
router ospfv3 1
              area 0.0.0.0
```

Setting default timer values for SPF throttling after configuring non-default values:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# no timers throttling spf
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config-ospfv3-<PROCESS-ID>` | Administrators or local user group members with execution rights for this command. |

## trap-enable

```
trap-enable
no trap-enable
```

### Description

Enables the notification of the events to be sent as traps to the SNMP management stations for OSPF.

The `no` form of this command disables the notification of the events to be sent as traps to the SNMP management stations for OSPF.

### Examples

Enabling sending notification of events as traps:

```
switch(config)# router ospf 1
switch(config-ospf-1)# trap-enable
```

Disabling sending notification of events as traps:

```
switch(config)# router ospf 1
switch(config-ospf-1)# no trap-enable
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config-ospf-<PROCESS-ID>` | Administrators or local user group members with execution rights for this command. |

# Chapter 6

# Open Shortest Path First version 3 (OSPFv3)

OSPFv3 is the IPv6 implementation of Open Shortest Path First protocol (OSPFv2 is the IPv4 implementation of this protocol). OSPFv3 is a routing protocol which is described in RFC5340 entitled OSPF for IPv6. It is a link-state based IGP (Interior Gateway Protocol) routing protocol. It is widely used with medium to large-sized enterprise networks.

## Overview

OSPFv3 is not supported on the Aruba 6000 and 6100 Switch Series.

The characteristics of OSPFv3 are:

- Provides a loop-free topology using SPF algorithm.
- Allows hierarchical routing using area 0 (backbone area) as the top of the hierarchy.
- Supports load balancing with equal cost routes for same destination.
- OSPFv3 is a classless protocol and allows for a hierarchical design with VLSM (Variable Length Subnet Masking) and route summarization.
- Scales enterprise size network easily with area concept.
- Provides fast convergence with triggered, incremental updates through Link State Advertisements (LSAs).

Some OSPFv3 configuration is done in the configuration context, others in the OSPFv3 router context, or in the interface context. OSPFv3 can be configured on routed ports, VLAN interfaces, LAG interfaces, and loopback interfaces. All such configurations work in the mentioned interfaces context. OSPFv3 mandates the associated interface to be routing interface.

The switch supports congestion control (prioritizing hello packets, inactivity timer reset, and adjacency throttling).

**Supported features**

- OSPFv3 neighbor adjacency, Hello protocol, multiple areas, Inter-area routing
- AS external routes, stub areas, totally stubby areas, NSSA, ASBR
- Designated Router/Backup Designated Router
- Point-to-point interfaces/broadcast interfaces
- Equal-cost multipath
- Null authentication, Simple password authentication, MD5 authentication
- Graceful Restart - un-planned, restart interval, Graceful Restart Helper
- Stub router advertisement
- SPF throttling
- LSA Throttling
- SPF throttling
- Graceful Restart - un-planned, restart interval, Graceful Restart Helper
- Stub router advertisement

- Configuration of interface parameters such as priority, cost, hello-interval, dead-interval, retransmit-interval, transit-delay, etc.
- Configuration of virtual link parameters such as hello-interval, dead-interval, retransmit-interval, transit-delay, etc.
- Route redistribution
- Passive interfaces
- Multi VRF support with each VRF having up to eight OSPF process instances
- Congestion control (prioritizing hello packets, inactivity timer reset, and adjacency throttling)

# How OSPFv3 protocol works

## OSPFv3 protocol

The protocol uses Link State Advertisements (LSAs) transmitted by each router to update neighboring routers regarding its interfaces and the routes available through those interfaces. Each routing switch in an area also maintains a link-state database (LSDB) that describes the area topology. (All routers in a given OSPF area have identical LSDBs.) The routing switches used to connect areas to each other flood summary link LSAs and external link LSAs to neighboring OSPF areas to update them regarding available routes. Through this means, each OSPF router determines the shortest path between itself and a desired destination router in the same OSPF domain (Autonomous System (AS)).

# OSPFv3 concepts

OSPFv3 is a link-state routing protocol applied to IPv6 routers grouped into OSPFv3 areas identified by the IPv6 routing configuration on each routing switch. Each OSPFv3 area includes one or more networks. OSPFv3 routers use hello packets and LSAs to maintain OSPFv3 operation across networks within an area and between areas within an OSPFv3 domain.

OSPFv3 uses hello packets to initiate and preserve relationships between neighboring routers on the same interface.

## OSPFv3 Link-state advertisement (LSA) types

OSPFv3 uses LSAs transmitted by each router to update neighboring routers regarding its interfaces and the routes available through those interfaces.

Each routing switch in an area also maintains an LSDB that describes the area topology. (All routers in a given OSPFv3 area have identical LSDBs, and each router uses the LSDB to build its own shortest-path tree.)

The routing switches used to connect areas to each other flood inter-area-prefix-LSAs, inter-area-router-LSAs, and AS-external-LSAs to backbone area to update it regarding available routes. Each OSPFv3 router determines the shortest path between itself and a desired destination router in the same OSPFv3 domain (AS). Routed traffic in an OSPFv3 AS is classified as one of the following:

- Intra-area traffic
- Inter-area traffic
- External traffic

The routing switches support the LSAs listed in the following table.

| Link-state type | Description | Use | Flood scope |
| --- | --- | --- | --- |
| 0x2001 | Router-LSA | Describes the state of each active interface on a router for Area a given area. (Excludes loopback interfaces and interfaces that have not achieved full adjacency.) | Area |
| 0x2002 | Network-LSA | Describes the OSPFv3 routers in a given network. | Area |
| 0x2003 | Inter-area-prefix-LSA | Describes the route to a prefix in another OSPFv3 area of Area the same AS. (Excludes prefixes for link-local addresses.)<br>Propagated through backbone area to other areas. | Area |
| 0x2004 | Inter-area-router-LSA | Describes the route to an ASBR in another OSPFv3 normal area (including the backbone area) of the same AS.<br>Propagated through backbone area to other areas. (Excludes any ASBR in the same area as the router sending the LSA.) | Area |
| 0x4005 | AS-external-LSA | Describes the route to a destination prefix in another AS (external route). (Excludes prefixes for link-local addresses.) Originated by ASBR in normal or backbone areas of an AS and propagates through backbone area to other normal areas.<br>Does not flood over virtual links and is not summarized in virtual links. For injection into an NSSA, an NSSA ABR generates a type-7-default-LSA advertising the default route (::/0). | AS |
| 0x2007 | NSSA-LSA | Describes the route to a destination in another AS (external route). Originated by ASBR in NSSA. ABR translates type-7 LSAs to AS-external-LSAs for injection into the backbone area. | NSSA |

| Link-state type | Description | Use | Flood scope |
|---|---|---|---|
| 0x0008 | Link-LSA | For other routers on the same VLAN interface (except virtual links), describes the router link-local address and any other IPv6 prefixes reachable on the VLAN. Link LSAs are not flooded over virtual links. | Link-local |
| 0x2009 | Intra-area-prefix-LSA | Generated on transit links within an area by the DR operating on those links. Also, every OSPFv3 router generates this LSA to refer to stub and loopback prefixes on the router. | Area |

## OSPFv3 area types

OSPFv3 is built upon a hierarchy of network areas. All areas for a given OSPFv3 domain reside in the same AS. An AS is defined as a number of contiguous networks that share an interior gateway routing protocol.

An AS can be divided into multiple areas, including the backbone (area 0). Because each area represents a collection of contiguous networks and hosts, the topology of a given area is not known by the internal routers in any other area. Areas define the boundaries to which router-LSAs and network-LSAs are broadcast. This functionality limits the amount of LSA flooding that occurs within the AS. It also helps to control the size of the link-state databases (LSDBs) maintained in OSPFv3 routers.

An area is represented in OSPFv3 by either a 32-bit dotted-decimal address or a number. Area types include: Backbone, Normal, Not-so-stubby (NSSA), and Stub.

### Backbone area

Every AS must have one (and only one) backbone area (identified as area 0 or 0.0.0.0.) The ABRs of all other areas in the same AS connect to the backbone area, either physically through an ABR or through a configured, virtual link. The backbone is a special type of area and serves as a transit area for carrying the inter-area-prefix-LSAs, AS-external-LSAs, and routed traffic between non-backbone areas, as well as the router-LSAs, network-LSAs, and routed traffic internal to the area. ASBRs are allowed in backbone areas.

### Normal area

A normal area allows inter-area-prefix-LSAs and AS-external-LSAs to and from the backbone area. A normal area connects to the AS backbone area through one or more ABRs (physically or through a virtual link) and allows router-LSAs and network LSAs within this area. ASBRs are allowed in normal areas.

### Stub area

Stub area connects to the AS backbone through one or more ABRs. It does not allow an internal ASBR, and does not allow AS-external-LSAs. A stub area supports these actions:

- Advertise the area summary routes to the backbone area.
- Advertise inter-area routes from other areas.
- Use the inter-area-prefix-LSA default route to advertise routes to an ASBR and to other areas.

You can configure the stub area ABR to do the following:

- Suppress advertising on the area's summarized internal routes into the backbone area.
- Suppress LSA traffic from other areas in the AS by replacing inter-area-prefix-LSAs and the default external route from the backbone area with the default summary route (::/0.). This area is called totally stubby area.

Virtual links are not allowed for stub areas.

### Not-so-stubby (NSSA) area

An NSSA area connects to the backbone area through one or more ABRs. NSSAs are used where an ASBR exists in an area where you want to:

- Block injection of external routes from other areas of the AS.
- Advertise type-7-LSA external routes (learned from the ASBR) to the backbone area as AS-external-LSAs.

NSSAs also support the following:

- Advertise inter-area-prefix-LSAs from the backbone area into the NSSA. (If no-summary is enabled, the NSSA ABR suppresses these LSAs from the backbone and, instead, injects the default route into the NSSA.)
- Advertise NSSA inter-area-prefix-LSAs to the backbone area.

Virtual links are not allowed for NSSAs.

## OSPFv3 router types

### Internal routers

Internal OSPFv3 routers belong to only one area. Internal routers flood router-LSAs to all routers in the same area and maintain identical LSDBs.

### Area border routers (ABRs)

Area border routers have membership in multiple areas. ABRs are used to connect the various areas in an AS to the backbone area for that AS. Multiple ABRs can be used to connect a given area to the backbone, and a given ABR can belong to multiple areas other than the backbone.

An ABR maintains a separate LSDB for each area to which it belongs. (All routers within the same area have identical LSDBs.) The ABR is responsible for flooding inter-area-prefix-LSAs and inter-area router LSAs between its border areas.

You can reduce summary LSA flooding by configuring area ranges. An area range enables you to assign an aggregate address to a range of IPv6 addresses. This aggregate address is advertised instead of all the individual addresses it represents. You can assign up to eight ranges in an OSPFv3 area.

### Autonomous system boundary router (ASBR)

Autonomous system boundary routers run one or more interior gateway protocols and serve as a gateway to other autonomous systems operating with interior gateway protocols. The ASBR imports and translates different protocol routes into OSPFv3 through redistribution. ASBRs can be used in backbone areas, normal areas, and NSSAs, but not in stub areas.

### Designated routers (DRs)

In an OSPFv3 network having two or more routers, one router is elected to serve as the designated router (DR) and another router to act as the backup designated router (BDR). All other routers in the same network segment forward their routing information to the DR and BDR, and the DR forwards this information to all routers in the network. This functionality minimizes the amount of repetitive information that is forwarded on the network by eliminating the need for each individual router in the area to forward its routing information to all other routers in the network. If the area includes multiple networks, each network elects its own DR and BDR.

In an OSPFv3 network with no DR and no BDR, the neighboring router with the highest priority is elected the DR and the router with the next highest priority is elected the BDR. If the DR goes off-line, the BDR automatically becomes the DR, and the router with the next highest priority then becomes the new BDR. If multiple routing switches on the same OSPFv3 network are declaring themselves DRs, both priority and router ID are used to select the DR and BDR.

Priority is configurable using the `ipv6 ospfv3 priority` command at the interface level. You can use this command to help bias one router as the DR. If two neighbors share the same priority, the router with the highest router ID is designated the DR. The router with the next highest router ID is designated the BDR. Routers retain DR/BDR states throughout the period for which the adjacency is up and running.

# OSPFv3 configuration task list

## Tasks at a glance

- Configuring OSPFv3 on the routing switch
- Assigning the routing switch to an OSPF area
- Setting OSPFv3 network for the area
- Configuring external route redistribution and control
- Influencing route choice by changing the administrative distance
- Configuring graceful restart
- Configuring OSPFv3 interface settings
- Configuring all OSPFv3 interfaces as passive
- Viewing OSPFv3 information
- Clearing OSPFv3 statistics on a switch

# Configuring OSPFv3 on the routing switch

### Prerequisites

- You must be in the global configuration context, as indicated by the `switch(config)#` prompt to create the OSPFv3 instance and enter the OSPFv3 configuration context.
- To configure a router ID, and other OSPFv3 configuration you must be in the OSPFv3 router configuration context, as indicated by the `switch(config-ospfv3-1 )#` prompt.

Create the OSPFv3 instance and enter the OSPFv3 configuration context. From this context, you can proceed with other OSPFv3 configuration.

### Procedure

1. Create the OSPFv3 instance and enter the OSPFv3 configuration context using the following command. For command details, see router ospfv3.

   ```
   router ospfv3  <PROCESS-ID> [vrf <VRF-NAME>]
   ```

For example, the following command creates OSPF instance 1.

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)#
```

2. Configure a global router ID using the following command. For command details, see router-id.
   `router-id <ROUTER-ADDRESSW>`
   For example, the following command sets the router ID to 1.1.1.1.

   ```
   switch(config-ospfv3-1)# router-id 1.1.1.1
   ```

3. Optionally, if the OSPFv3 process was disabled (it is enabled by default), enable the OSPFv3 process using the following command.
   `enable`
   For command details, see enable. (Refer to disable for disabling the OSPFv3 process).

# Creating an OSPFv3 area

## Prerequisites

You must be in the OSPFv3 router configuration context.

Create a Normal, Stub, or Not So Stubby (NSSA) area.

## Procedure

Create an OSPFv3 area for the routing switch using one of the following commands:

- Create a Normal area using the following command: `area <area-id>`. For command details, see area.
- Create a Stub area using the following command: `area <area-id> stub`. For command details, see area stub.
- Create a Not So Stubby Area using the command: `area <area-id> nssa`.

For example, the following command creates a normal area with an area identifier of 10.1.1.1. Area identifier could alternatively be entered in decimal format such as 1.

```
switch(config-ospfv3-1)# area 1.1.1.1
```

# Setting OSPFv3 network for the area

## Prerequisites

Routing must be enabled on the interface where you are planning to enable OSPFv3.

You must be in the interface configuration context, as indicated by the `switch(config-if)#` prompt.

After you define an OSPFv3 area, you can assign one or more networks to it. The OSPFv3 protocol will run on the interface with the configured link-local ipv6 address. The interfaces that have an IPv6 address configured in this network or in a subset of this network will participate in the OSPFv3 protocol.

## Procedure

1. Set an OSPFv3 network for the area using the following command. For command details, see ipv6 ospfv3 area.

   `ipv6 ospfv3 <PROCESS-ID> area <AREA-ID>`

   For example, use the following command to assign interface 1/1/1 to OSPFv3 area 1. The area can alternatively be entered in IO address format.

```
switch(config)# interface 1/1/1
switch(config-if)# ipv6 ospfv3 1 area 1
```

2. Optionally, you can disable OSPFv3 on the interface using the following command. For command details, see ipv6 ospfv3 area.

   `ipv6 ospfv3 shutdown`

```
switch(config)# interface 1/1/1
switch(config-if)# ipv6 ospfv3 shutdown
```

# Configuring external route redistribution and control

Configuring route redistribution for OSPFv3 establishes the routing switch as an ASBR for importing and translating different protocol routes from other IGP domains into an OSPFv3 domain. When you configure redistribution for OSPFv3, you can specify that routes external to the OSPFv3 domain are imported as OSPFv3 routes.

1. Enable route redistribution using the `redistribute` command.

```
redistribute {bgp | connected | local loopback | static | ripng | ospf <PROCESS-ID>}
             [route-map <ROUTE-MAP-NAME>]
```

   For example, setting redistribution of connected routes as OSPFv3 routes:

```
switch(config-ospfv3-1)# redistribute connected
```

   If a route map is specified, then only the routes that pass the match clause specified in the route map are redistributed to OSPFv3. For example, setting redistribution of local loopback routes, that only match the routes specified by the route map:

```
switch(config-ospf-1)# redistribute local loopback route-map local_routes
```

2. Optionally, modify the default metric for redistribution using the `default-metric` command.

   `default-metric <METRIC-VALUE>`

   For example, setting the default metric for redistribution to 37.

```
switch(config-ospfv3-1)# default-metric 37
```

3. Optionally, set the cost of default-summary LSAs using the area command.

   `area <AREA-ID> default-metric <COST>`

   For example, setting the cost of default summary LSAs to 2.

```
switch (config-ospfv3-1)# area 1 default-metric 2
```

4. Optionally, use the `max-metric router-lsa` command to set the protocol to advertise a maximum metric so that other routers do not prefer this router as an intermediate hop in their shortest path first (SPF) calculations.
   `max-metric router-lsa [on-startup [<ADVERT-TIME>]]`
   For example, setting advertise max-metric router-lsa on startup.

```
switch(config-ospfv3-1)# max-metric router-lsa on-startup 3000
```

# Influencing route choice by changing the administrative distance

## Prerequisites

You must be in the OSPFv3 router configuration context, as indicated by the `switch(config-ospfv3-1)#` prompt.

The administrative distance is used to influence the selection of routes learned by different protocols.

## Procedure

Reconfigure the administrative distance using the following command. For command details, see distance.

`distance <distance>`

For example, use the following command to set administrative distance to 100.

```
switch(config-ospfv3-1)# distance 100
```

# Configuring graceful restart

## Prerequisites

You must be in the OSPFv3 router configuration context, as indicated by the `switch(config-ospfv3-1)#` prompt.

OSPFv3 routing can be gracefully restarted on switches without losing packets that are in transit. There is no effect on the saved switch configuration.

## Procedure

Configure graceful restart of using the following command. For command details, see graceful-restart.

`graceful-restart [restart-interval  <seconds> | helper]`

For example, the following command specifies 50 seconds as the maximum interval another router will wait for this router to gracefully restart.

```
switch(config-ospfv3-1)# graceful-restart restart-interval 50
```

# Configuring OSPFv3 interface settings

## Prerequisites

You must be in the interface configuration context, as indicated by the `switch(config-if)#` prompt.

You can optionally adjust the following OSPFv3 interface settings.

1. Set the interface cost using the following command. For command details, see ipv6 ospfv3 cost.
   ```
   ipv6 ospfv3 cost <interface-cost>
   ```
   For example, the following command sets the cost associated with interface 1/1/1 to 100.
   ```
   switch(config)# interface 1/1/1
   switch(config-if)# ipv6 ospfv3 cost 100
   ```
2. Set the time interval between OSPFv3 hello packets for the OSPFv3 interface using the following command. For command details, see ipv6 ospfv3 hello-interval.
   ```
   ipv6 ospfv3 hello-interval  <seconds>
   ```
3. Set the interval after which a neighbor is declared dead if no hello packet is received on the OSPFv3 interface using the following command. For command details, see ipv6 ospfv3 dead-interval.
   ```
   ipv6 ospfv3 dead-interval <seconds>
   ```
4. Set the time between re-transmitting lost link state advertisements for the OSPFv3 interface using the following command. For command details, see ipv6 ospfv3 retransmit-interval.
   ```
   ipv6 ospfv3 retransmit-interval  <seconds>
   ```
5. Sets the time delay in Link state transmission for the OSPFv3 interface using the following command. For command details, see ipv6 ospfv3 transit-delay.
   ```
   ipv6 ospfv3 transit-delay  <seconds>
   ```
6. Set the OSPFv3 network type for the interface. For command details, see ipv6 ospfv3 network.
   ```
   ipv6 ospfv3 network {broadcast|point-to-point}
   ```
7. Set the OSPFv3 priority on the interface using the following command. For command details, see ipv6 ospfv3 priority.
   ```
   ipv6 ospfv3 priority <number-value>
   ```
8. Set the OSPFv3 interface as OSPFv3 passive interface. With this setting the interface participates in OSPFv3 but does not send or receive packets on that interface. For command details, see ipv6 ospfv3 passive.
   ```
   ipv6 ospfv3 passive
   ```

# Configuring all OSPFv3 interfaces as passive

## Prerequisites

You must be in the OSPFv3 router configuration context, as indicated by the `switch(config-ospfv3-1)#` prompt.

OSPFv3 sends LSAs to all other routers in the same AS. To limit the flooding of LSAs throughout the AS, you can configure OSPFv3 to be passive.

## Procedure

Configure all OSPFv3 interfaces as passive using the following command. For command details, see passive-interface default.

```
passive-interface default
```

```
switch(config-ospfv3-1)# passive-interface default
```

# Configuring SPF throttling timers

SPF calculation is throttled with default timer values (start-time 200ms, hold-time 1000ms, max-wait-time 5000ms). You can throttle SPF calculation by configuring non-default timers to improve performance of a specific network configuration.

### Prerequisites

You must be in the router configuration context, as indicated by the `switch(config-router)#` prompt.

### Procedure

Reconfigure SPF throttling timers using the following command. For command details, see timers throttle spf.

```
timers throttle spf start-time <milliseconds> hold-time <milliseconds> max-wait-time
<milliseconds>
```

For example, use the following command to set start-time to 500ms, hold-time to 3000ms and max-wait-time to 9000ms:

```
switch(config-ospf-1)# timers throttle spf start-time 500 hold-time
3000 max-wait-time 9000
```

# Viewing OSPFv3 information

### Prerequisites

Use these show commands from the Manager context, as indicated by the `switch#` prompt.

### Procedure

To view OSPFv3 information, use the following commands. For command details and examples, click the links.

- To view general OSPFv3, area, state and configuration information use: show ipv6 ospfv3.
- To view information about OSPFv3 interfaces use: show ipv6 ospfv3 interface.
- To view information about OSPFv3 neighbors use: show ipv6 ospfv3 neighbors .
- To view OSPFv3 routing table information use: show ipv6 ospfv3 routes.
- To view OSPFv3 statistics for the OSPFv3 interfaces use show ipv6 ospfv3 statistics interface.

# Clearing OSPFv3 statistics on a switch

Clear OSPFv3 event statics using the following command. For command details, see clear ipv6 ospfv3 statistics

```
clear ipv6 ospfv3 [<PROCESS-ID>] statistics [interface [<INTERFACE-NAME>]] [all-vrfs | vrf
<VRF-NAME>]
```

For example, the following command clears OSPFv3 event statistics from interface 1/1/1.

```
switch# clear ipv6 ospfv3 statistics interface 1/1/1
```

# OSPFv3 commands

# area

```
area <AREA-ID>
no area <AREA-ID>
```

## Description

Creates a normal area with `<AREA-ID>` set if not present. If area is present and is not the normal area, this command changes the area type to normal area.

The `no` form of this command deletes the area with the `<AREA-ID>` specified. The area can be of any type (stub, stub no-summary, and default normal area).

| Parameter | Description |
| --- | --- |
| *<AREA-ID>* | Specifies the area ID is one of the following formats.<br>OSPF area identifier in IPv4 format (`x.x.x.x`), where `x` is a decimal number from 0 to 255.<br>OSPF area identifier in decimal format. Range: 0 to 4294967295. |

## Examples

Creating a normal area:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# area 1
```

Deleting an area:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# no area 1
```

## Command History

| Release | Modification |
| --- | --- |
| 10.07 or earlier | -- |

## Command Information

| Platforms | Command context | Authority |
| --- | --- | --- |
| 6200 | `config-ospfv3-<PROCESS-ID>` | Administrators or local user group members with execution rights for this command. |

# area authentication ipsec

```
area <AREA-ID> authentication ipsec spi <SPI-INDEX> <AUTH-TYPE> [<KEY-TYPE> <AUTH-KEY>]
no area <AREA-ID> authentication
```

## Description

Configures IPsec AH authentication for the specified area. OSPFv3 interfaces which have IPsec configured at the interface context will not use area level IPsec.

The `no` form of this command removes IPsec AH authentication for the specified area.

IPsec is not supported for 6in6 tunnel interfaces.

| Parameter | Description |
|---|---|
| *<AREA-ID>* | Specifies the area ID is one of the following formats.<br>OSPF area identifier in IPv4 format (`x.x.x.x`), where `x` is a decimal number from 0 to 255.<br>OSPF area identifier in decimal format. Range: 0 to 4294967295. |
| `spi` *<SPI-INDEX>* | Specifies the Security Parameters Index (SPI) to use. The SPI is an identification tag carried in the IPsec AH header. It enables the receiving OSPF process to select and use the Security Association (SA) from the SA table. The SPI must be unique on the switch. Range: 256 to 4294967295. |
| *<AUTH-TYPE>* | Specifies the authentication type: `md5` or `sha1`. |
| *<KEY-TYPE>* | Specifies the key type to use: `plaintext` (unencrypted), `hex-string` (encrypted) or `ciphertext` (encrypted). |
| *<AUTH-KEY>* | Specifies the authentication key. |

When the authentication key is not provided on the command line, plaintext key prompting occurs upon pressing Enter. The entered key characters are masked with asterisks.

## Examples

Setting area 0 to use IPsec authentication with a provided plaintext authentication key:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# area 0 authentication ipsec spi 256 sha1 plaintext F82#450
```

Setting area 5 to use IPsec authentication with a prompted plaintext authentication key:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# area 5 authentication ipsec spi 256 sha1
Enter the IPsec authentication key: ********
Re-Enter the IPsec authentication key: ********
```

Removing IPsec authentication from area 1:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# no area 1 authentication
```

## Command History

| Release | Modification |
| --- | --- |
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
| --- | --- | --- |
| 6200 | `config-ospfv3-<PROCESS-ID>` | Administrators or local user group members with execution rights for this command. |

## area encryption ipsec

```
area <AREA-ID> encryption ipsec spi <SPI-INDEX> <AUTH-TYPE>
    [<KEY-TYPE> <AUTH-KEY> <ENCR-TYPE> [<KEY-TYPE> <ENCR-KEY>]]
no area <AREA-ID> encryption
```

### Description

Configures IPsec ESP with the authentication and encryption algorithm types and keys for the specified area. OSPFv3 interfaces with IPsec configured at the interface context will not use area level IPsec ESP configuration.

The `no` form of this command removes IPsec ESP from the specified area.

IPsec is not supported for 6in6 tunnel interfaces.

| Parameter | Description |
| --- | --- |
| *<AREA-ID>* | Specifies the area ID is one of the following formats.<br>OSPF area identifier in IPv4 format (`x.x.x.x`), where `x` is a decimal number from 0 to 255.<br>OSPF area identifier in decimal format. Range: 0 to 4294967295. |
| `spi` *<SPI-INDEX>* | Specifies the Security Parameters Index (SPI) to use. The SPI is an identification tag carried in the IPsec AH header. It enables the receiving OSPF process to select and use the Security Association (SA) from the SA table. The SPI must be unique on the switch. Range: 256 to 4294967295. |
| *<AUTH-TYPE>* | Specifies the authentication type: `md5` or `sha1`. |
| *<KEY-TYPE>* | Specifies the key type to use: `plaintext` (unencrypted), `hex-string` (encrypted) or `ciphertext` (encrypted). |
| *<AUTH-KEY>* | Specifies the authentication key. |
| *<ENCR-TYPE>* | Specifies the encryption type: `des`, `3des`, `aes`, or `null`.<br><br>**NOTE:** Encryption type `aes` is considered to be AES128, AES192 or AES256 based on key length. |
| *<ENCR-KEY>* | Specifies the encryption key. |

When the authentication key is not provided on the command line, plaintext authentication key prompting occurs upon pressing Enter, followed by encryption type prompting, and finally plaintext encryption key prompting. The entered key characters are masked with asterisks.

When the authentication key and encryption type are provided on the command line but the encryption key is not provided, plaintext encryption key prompting occurs upon pressing Enter. The entered key characters are masked with asterisks.

## Examples

Setting area 0 to use IPSec ESP with provided authentication and encryption keys:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# area 0 encryption ipsec spi 256 md5 plaintext F824eva
                                des plaintext F82#450b
```

Setting area 5 to use IPSec ESP with prompted authentication and encryption keys:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# area 5 encryption ipsec spi 256 md5
Enter the IPsec authentication key: ********
Re-Enter the IPsec authentication key: ********

Enter the IPsec encryption type (3des/aes/des/null)? des

Enter the IPsec encryption key: ********
Re-Enter the IPsec encryption key: ********
```

Setting area 2 to use IPsec ESP with provided authentication password and encryption type but a prompted encryption key:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)#  area 2 encryption ipsec spi 256 md5 plaintext F82# des
Enter the IPsec encryption key: ********
Re-Enter the IPsec encryption key: ********
```

Setting area 0 to use IPSec ESP with provided plaintext authentication key and null encryption:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# area 0 encryption ipsec spi 256 md5 plaintext axtw null
```

Removing IPSec ESP from area 0:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# no area 0 encryption
```

## Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | config-ospfv3-<*PROCESS-ID*> | Administrators or local user group members with execution rights for this command. |

## clear ipv6 ospfv3 neighbors

```
clear ipv6 ospfv3 [<PROCESS-ID>] neighbor [<NEIGHBOR>] [interface [<INTERFACE-NAME>]]
[all-vrfs | vrf <VRF-NAME>]
```

### Description

Resets the neighbor and clears the OSPF neighbor information.

| Parameter | Description |
|---|---|
| <*PROCESS-ID*> | Specifies the OSPFv3 process ID to clear the statistics for the particular OSPFv3 process. Range: 1 to 63. |
| <*NEIGHBOR*> | Specifies the router ID of a neighbor. |
| <*INTERFACE-NAME*> | Specifies the OSPFv3 statistics to clear for the specified interface. |
| all-vrfs | Select to clear the OSPFv3 statistics for all VRFs. |
| vrf <*VRF-NAME*> | Specifies the name of a VRF. |

### Example

Clearing the OSPFv3 neighbor information:

```
switch# clear ipv6 ospfv3 1 neighbor 3.3.3.3
switch# clear ipv6 ospfv3 1 neighbor interface 1/1/1
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

# clear ipv6 ospfv3 statistics

```
clear ipv6 ospfv3 [<PROCESS-ID>] statistics [interface [<INTERFACE-NAME>]]
[all-vrfs | vrf <VRF-NAME>]
```

### Description

Clears the OSPFv3 event statistics.

| Parameter | Description |
|---|---|
| *<PROCESS-ID>* | Specifies the OSPFv3 process ID to clear the statistics for the particular OSPFv3 process. Range: 1 to 63. |
| *<INTERFACE-NAME>* | Specifies the OSPFv3 statistics to clear for the specified interface. |
| `all-vrfs` | Select to clear the OSPFv3 statistics for all VRFs. |
| `vrf` *<VRF-NAME>* | Specifies the name of a VRF. |

### Example

Clearing the OSPFv3 event statistics:

```
switch# clear ipv6 ospfv3 statistics
switch# clear ipv6 ospfv3 statistics interface 1/1/1
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

# default-metric

```
default-metric <METRIC-VALUE>
no default-metric
```

### Description

Sets the default metric for redistributed routes in the OSPFv3.

The `no` form of this command sets the default metric to be used for redistributed routes into OSPFv3 to the default of 25.

| Parameter | Description |
| --- | --- |
| *<METRIC-VALUE>* | Specifies the default metric value to use for redistributed routes. Range: 1 to 1677214. Default: 25. |

### Examples

Setting default metric for redistributed routes:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# default-metric 36
```

Setting default metric for redistributed routes to the default:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# no default-metric
```

### Command History

| Release | Modification |
| --- | --- |
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
| --- | --- | --- |
| 6200 | config-ospfv3-*<PROCESS-ID>* | Administrators or local user group members with execution rights for this command. |

## disable

`disable`

### Description

Disables the OSPFv3 process. By default OSPFv3 process is enabled.

This command does not remove the OSPFv3 configurations.

### Example

Disabling OSPFv3 process:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# disable
```

### Command History

| Release | Modification |
| --- | --- |
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
| --- | --- | --- |
| 6200 | `config-ospfv3-<PROCESS-ID>` | Administrators or local user group members with execution rights for this command. |

## distance

```
distance <DISTANCE>
no distance
```

### Description

Defines an administrative distance for OSPFv3. Administrative distance is used as a criteria to select the best route when the same route is learned by multiple routing protocols.

The `no` form of this command sets the OSPFv3 administrative distance to the default of 110.

| Parameter | Description |
| --- | --- |
| `<DISTANCE>` | Specifies OSPFv3 administrative distance. Range: 1 to 255. Default: 110. |

### Examples

Setting OSPF administrative distance:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# distance 100
```

Setting OSPF administrative distance to the default:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# no distance
```

### Command History

| Release | Modification |
| --- | --- |
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
| --- | --- | --- |
| 6200 | `config-ospfv3-<PROCESS-ID>` | Administrators or local user group members with execution rights for this command. |

## enable

```
enable
```

### Description

Enables OSPFv3 process when disabled. By default OSPFv3 process is enabled.

### Example

Enabling OSPFv3 process:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# enable
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config-ospfv3-<PROCESS-ID>` | Administrators or local user group members with execution rights for this command. |

## default-information originate

```
default-information originate
no default-information originate
```

### Description

Configures OSPF to advertise the default route (0.0.0.0/0) to its neighbors if it is present in the routing table.

The `no` form of this command disables advertisement of the default route.

### Examples

Setting advertisement of the default route:

```
switch(config)# router ospf 1
switch(config-ospf-1)# default-information originate
```

Disabling advertisement of the default route:

```
switch(config)# router ospf 1
switch(config-ospf-1)# no default-information originate
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config-ospf-<PROCESS-ID>` | Administrators or local user group members with execution rights for this command. |

# default-information originate always

```
default-information originate always
no default-information originate always
```

## Description

Configures OSPF to advertise the default route (0.0.0.0/0) to its neighbors, regardless if it is present in the routing table or not.

The `no` form of this command sets the OSPF administrative distance to the default of 110.

## Examples

Setting advertisement of the default route:

```
switch(config)# router ospf 1
switch(config-ospf-1)# default-information originate always
```

Disabling advertisement of the default route:

```
switch(config)# router ospf 1
switch(config-ospf-1)# no default-information originate always
```

## Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

## Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config-ospf-<PROCESS-ID>` | Administrators or local user group members with execution rights for this command. |

# graceful-restart

```
graceful-restart {restart-interval <INTERVAL> | helper | strict-lsa-check}
no graceful-restart {restart-interval | helper | strict-lsa-check}
```

## Description

Configures graceful restart for OSPFv3. By default graceful restart is enabled on the OSPFv3 router.

The `no` form of this command sets the restart interval to the default of 120 seconds or disables helper mode depending on the specified parameters.

| Parameter | Description |
|---|---|
| `restart-interval <INTERVAL>` | Specifies the time another router waits for this router to gracefully restart and selects the maximum time to wait in seconds. Range: 5 to 1800. Default: 120. |
| `helper` | Specifies that the router will participate in the graceful restart of a neighbor router. |
| `strict-lsa-check` | Enable strict LSA checking when acting as a restart helper for a restarting peer. |

### Examples

Enabling OSPF graceful restart:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# graceful-restart restart-interval 40
switch(config-ospfv3-1)# graceful-restart helper
switch(config-ospfv3-1)# graceful-restart strict-lsa-check
```

Setting the restart interval to default, disabling helper mode, and strict LSA checking:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# no graceful-restart restart-interval
switch(config-ospfv3-1)# no graceful-restart helper
switch(config-ospfv3-1)# no graceful-restart strict-lsa-check
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config-ospfv3-<PROCESS-ID>` | Administrators or local user group members with execution rights for this command. |

## ipv6 ospfv3 area

```
ipv6 ospfv3 <PROCESS-ID> area <AREA-ID>
no ipv6 ospfv3 <PROCESS-ID> area <area-id>
```

### Description

Runs the OSPFv3 protocol on the interface for the area specified.

To move an interface to a new area, unmap the existing area and then associate a new area with the interface.

The `no` form of this command disables OSPF on the interface and removes the interface from the area. Interfaces which have an IP address configured on the network or in a subset of the network, stop participating in the OSPF protocol

| Parameter | Description |
|---|---|
| *<AREA-ID>* | Specifies the area ID is one of the following formats.<br>Area ID in IPv4 format (`x.x.x.x`), where `x` is a decimal number from 0 to 255.<br>Area ID as a decimal value. Range: 0-4294967295. |
| *<PROCESS-ID>* | Specifies the OSPFv3 process ID. Range: 1 to 63. |

## Examples

Setting OSPFv3 network for the area:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ipv6 ospfv3 1 area 1
switch(config-if-vlan)# ipv6 ospfv3 1 area 0.0.0.1
```

Disabling OSPFv3 network for the area:

```
switch(config)# interface 1/1/1
switch(config-if-vlan)# no ipv6 ospfv3 1 area 1
switch(config-if-vlan)# no ipv6 ospfv3 1 area 0.0.0.1
```

## Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

## Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config-if`<br>`config-if-vlan` | Administrators or local user group members with execution rights for this command. |

# ipv6 ospfv3 authentication null

`ipv6 ospfv3 authentication null`

## Description

Configures null authentication on an interface which disables IPsec authentication.

## Examples

Disabling IPsec on interface VLAN **1**:

```
switch(config)# interface van 1
switch(config-if-vlan)# ipv6 ospfv3 authentication null
```

## Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config-if` `config-if-vlan` | Administrators or local user group members with execution rights for this command. |

## ipv6 ospfv3 authentication ipsec

```
ipv6 ospfv3 authentication ipsec spi <SPI-INDEX> <AUTH-TYPE> [<KEY-TYPE> <AUTH-KEY>]
no ipv6 ospfv3 authentication
```

### Description

Configures IPSec AH authentication. OSPFv3 interfaces that have IPsec configured at the interface context will not use area level IPsec.

The `no` form of this command removes IPsec AH authentication for the specified area.

| Parameter | Description |
|---|---|
| spi *<SPI-INDEX>* | Specifies the Security Parameters Index (SPI) to use. The SPI is an identification tag carried in the IPsec AH header. It enables the receiving OSPF process to select and use the Security Association (SA) from the SA table. The SPI must be unique on the switch. Range: 256 to 4294967295. |
| *<AUTH-TYPE>* | Specifies the authentication type: `md5` or `sha1`. |
| *<KEY-TYPE>* | Specifies the key type to use: `plaintext` (unencrypted), `hex-string` (encrypted) or `ciphertext` (encrypted). |
| *<AUTH-KEY>* | Specifies the authentication key. |

When the authentication key is not provided on the command line, plaintext key prompting occurs upon pressing Enter. The entered key characters are masked with asterisks.

### Examples

Setting interface VLAN **1** to use IPsec authentication with a provided plaintext authentication key:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ipv6 ospfv3 authentication ipsec spi 256 md5 plaintext F82#
```

Setting interface VLAN **4** to use IPsec authentication with a prompted plaintext authentication key:

```
switch(config)# interface vlan 4
switch(config-if-vlan)# ipv6 ospfv3 authentication ipsec spi 256 md5
```

```
Enter the IPsec authentication key: ********
Re-Enter the IPsec authentication key: ********
```

Removing IPsec authentication from interface VLAN **1**:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ipv6 ospfv3 authentication
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config-if`<br>`config-if-vlan` | Administrators or local user group members with execution rights for this command. |

## ipv6 ospfv3 cost

```
ipv6 ospfv3 cost <INTERFACE-COST>
no ipv6 ospfv3 cost
```

### Description

Sets the cost (metric) associated with a particular interface. The interface cost is used as a parameter to calculate the best routes.

The `no` form of this command sets the cost (metric) associated with a particular interface to the default of 1.

| Parameter | Description |
|---|---|
| *<INTERFACE-COST>* | Specifies the interface cost value. Range: 1to 65535. Default: 1. |

### Examples

Setting OSPFv3 interface cost:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ipv6 ospfv3 cost 100
```

Setting the OSPFv3 interface cost to default:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ipv6 ospfv3 cost
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config-if`<br>`config-if-vlan` | Administrators or local user group members with execution rights for this command. |

## ipv6 ospfv3 dead-interval

```
ipv6 ospfv3 dead-interval <INTERVAL>
no ipv6 ospfv3 dead-interval
```

### Description

Sets the interval after a neighbor is declared dead when no hello packet is received on the OSPFv3 interface.

The `no` form of this command sets the interval after which a neighbor is declared dead, to the default for the OSPFv3 interface. The default value is 40 seconds (generally four times the hello packet interval).

| Parameter | Description |
|---|---|
| *<INTERVAL>* | Specifies the time interval for the dead interval, in seconds. Range: 1 to 65535. Default: 40. |

### Examples

Setting OSPFv3 dead interval on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ipv6 ospfv3 dead-interval 30
```

Setting OSPFv3 dead interval to default on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ipv6 ospfv3 dead-interval
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config-if-vlan` | Administrators or local user group members with execution rights for this command. |

# ipv6 ospfv3 encryption ipsec

```
ipv6 ospfv3 encryption ipsec spi <SPI-INDEX> <AUTH-TYPE>
    [<KEY-TYPE> <AUTH-KEY> <ENCR-TYPE> [<KEY-TYPE> <ENCR-KEY>]]
no ipv6 ospfv3 encryption
```

## Description

Configures IPsec ESP authentication. OSPFv3 interfaces that have IPsec configured at the interface context will not use area level IPsec ESP.

The `no` form of this command removes IPsec ESP for the specified area.

| Parameter | Description |
|---|---|
| `spi <SPI-INDEX>` | Specifies the Security Parameters Index (SPI) to use. The SPI is an identification tag carried in the IPsec AH header. It enables the receiving OSPF process to select and use the Security Association (SA) from the SA table. The SPI must be unique on the switch. Range: 256 to 4294967295. |
| `<AUTH-TYPE>` | Specifies the authentication type: `md5` or `sha1`. |
| `<KEY-TYPE>` | Specifies the key type to use: `plaintext` (unencrypted), `hex-string` (encrypted) or `ciphertext` (encrypted). |
| `<AUTH-KEY>` | Specifies the authentication key. |
| `<ENCR-TYPE>` | Specifies the encryption type: `des`, `3des`, `aes`, or `null`.<br><br>**NOTE:** Encryption type `aes` is considered to be AES128, AES192, or AES256 based on key length. |
| `<ENCR-KEY>` | Specifies the encryption key. |

When the authentication key is not provided on the command line, plaintext authentication key prompting occurs upon pressing Enter, followed by encryption type prompting, and finally plaintext encryption key prompting. The entered key characters are masked with asterisks.

When the authentication key and encryption type are provided on the command line but the encryption key is not provided, plaintext encryption key prompting occurs upon pressing Enter. The entered key characters are masked with asterisks.

## Examples

Setting interface VLAN **1** to use IPsec ESP with provided authentication and encryption keys:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ipv6 ospfv3 encryption ipsec spi 256 sha1 plaintext F82
                        des plaintext F82#450b
```

Setting interface VLAN **3** to use IPsec ESP with prompted authentication and encryption keys:

```
switch(config)# interface vlan 3
switch(config-if-vlan)# ipv6 ospfv3 encryption ipsec spi 256 sha1
Enter the IPsec authentication key: ********
Re-Enter the IPsec authentication key: ********

Enter the IPsec encryption type (3des/aes/des/null)? des

Enter the IPsec encryption key: ********
Re-Enter the IPsec encryption key: ********
```

Setting interface VLAN **4** to use IPsec ESP with provided authentication password and encryption type but a prompted encryption key:

```
switch(config)# interface vlan 4
switch(config-if-vlan)# ipv6 ospfv3 encryption ipsec spi 256 sha1 plaintext F82 des
Enter the IPsec encryption key: ********
Re-Enter the IPsec encryption key: ********
```

Setting interface VLAN **1** to use IPSec ESP with provided plaintext authentication key and null encryption:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ipv6 ospfv3 encryption ipsec spi 256 sha1 plaintext F82 null
```

Removing IPsec from interface VLAN **1**:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ipv6 ospfv3 encryption
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config-if`<br>`config-if-vlan` | Administrators or local user group members with execution rights for this command. |

## ipv6 ospfv3 encryption null

`ipv6 ospfv3 encryption null`

### Description

Configures NULL ESP on an interface which disables IPsec ESP.

### Examples

Disable IPsec ESP on interface VLAN *1*:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ipv6 ospfv3 encryption null
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config-if`<br>`config-if-vlan` | Administrators or local user group members with execution rights for this command. |

## ipv6 ospfv3 hello-interval

```
ipv6 ospfv3 hello-interval <INTERVAL>
no ipv6 ospfv3 hello-interval
```

### Description

Sets the time interval between OSPFv3 hello packets for the OSPFv3 interface.

The `no` form of this command sets the time interval between OSPFv3 hello packets to the default for the OSPFv3 interface of 10 seconds.

| Parameter | Description |
|---|---|
| *<INTERVAL>* | Specifies the time interval between hello packets, in seconds. Range: 1 to 65535. Default: 10. |

### Examples

Setting OSPFv3 hello interval on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ipv6 ospfv3 hello-interval 30
```

Setting OSPFv3 hello interval to default on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ipv6 ospfv3 hello-interval
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config-if`<br>`config-if-vlan` | Administrators or local user group members with execution rights for this command. |

## ipv6 ospfv3 network

```
ipv6 ospfv3 network {broadcast|point-to-point}
no ipv6 ospfv3 network
```

### Description

Configures the network type for the interface. By default the network type is broadcast network.

The `no` form of this command sets the network type for the interface to the system default of broadcast network.

| Parameter | Description |
|---|---|
| `broadcast` | Specifies the OSPFv3 network type as a broadcast multiaccess network. |
| `point-to-point` | Specifies the OSPFv3 network type as a point-to-point network. |

### Examples

Setting OSPFv3 network type for the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ipv6 ospfv3 network broadcast
switch(config-if-vlan)# ipv6 ospfv3 network point-to-point
```

Disabling OSPFv3 network type for the interface to system default of broadcast network:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ipv6 ospfv3 network
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config-if`<br>`config-if-vlan` | Administrators or local user group members with execution rights for this command. |

## ipv6 ospfv3 passive

```
ipv6 ospfv3 passive
no ipv6 ospfv3 passive
```

### Description

Configures the interface as an OSPFv3 passive interface. With this setting, the interface participates in the OSPF, but does not send or receive OSPF packets on that interface.

The `no` form of this command resets the interface as active. With this setting, the interface starts sending and receiving OSPF packets.

### Examples

Setting the interface as OSPFv3 passive interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ipv6 ospfv3 passive
```

Setting the interface as OSPFv3 active interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ipv6 ospfv3 passive
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config-if`<br>`config-if-vlan` | Administrators or local user group members with execution rights for this command. |

## ipv6 ospfv3 priority

```
ipv6 ospfv3 priority <number-value>
no ipv6 ospfv3 priority
```

### Description

Sets the OSPFv3 priority for the interface. The larger the numeric value of the priority, the higher the chance it will become the designated router. Setting a priority of 0 makes the router ineligible to become a designated router or back up designated router.

The `no` form of this command sets the OSPFv3 priority for the interface to the default of 1.

| Parameter | Description |
|---|---|
| *<number-value>* | Specifies the OSPFv3 priority value. Default: 1. Range: 0 to 255. |

### Examples

Setting the OSPFv3 priority for the interface:

```
switch(config)# interface vlan /1
switch(config-if-vlan)# ipv6 ospfv3 priority 50
```

Setting the OSPFv3 priority for the interface to the default of 1:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ipv6 ospfv3 priority
```

### Command History

| Release | Modification |
| --- | --- |
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
| --- | --- | --- |
| 6200 | `config-if`<br>`config-if-vlan` | Administrators or local user group members with execution rights for this command. |

## ipv6 ospfv3 retransmit-interval

```
ipv6 ospfv3 retransmit-interval <INTERVAL>
no ipv6 ospfv3 retransmit-interval
```

### Description

Sets the time between retransmitting lost link state advertisements for the OSPFv3 interface.

The `no` form of this command sets the time between retransmitting lost link state advertisements to the default 5 seconds.

| Parameter | Description |
| --- | --- |
| *<INTERVAL>* | Specifies the time interval for the retransmit interval, in seconds. Range: 1 to 3600. Default: 5 |

### Examples

Setting OSPFv3 retransmit interval on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ipv6 ospfv3 retransmit-interval 30
```

Setting OSPFv3 retransmit interval to the default on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ipv6 ospfv3 retransmit-interval
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config-if`<br>`config-if-vlan` | Administrators or local user group members with execution rights for this command. |

## ipv6 ospfv3 shutdown

```
ipv6 ospfv3 shutdown
no ipv6 ospfv3 shutdown
```

### Description

Disables OSPFv3 on the interface. The interface state changes to Down. It does not remove the interface from the OSPF area. To remove the interface, use the command `no ip ospf area`.

The `no` form of this command re-enables OSPFv3 on the interface.

### Examples

Disabling OSPFv3 on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ipv6 ospfv3 shutdown
```

Re-enabling OSPFv3 on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ipv6 ospfv3 shutdown
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config-if`<br>`config-if-vlan` | Administrators or local user group members with execution rights for this command. |

## ipv6 ospfv3 transit-delay

```
ipv6 ospfv3 transit-delay <DELAY>
no ipv6 ospfv3 transit-delay
```

### Description

Sets the time delay in Link state transmission for the OSPFv3 interface.

The `no` form of this command sets the transit delay in Link state transmission to the default of 1 second.

| Parameter | Description |
|---|---|
| *<DELAY>* | Specifies the time delay for the transit delay, in seconds. Range: 1 to 3600. Default: 1. |

### Examples

Setting OSPFv3 transit delay on the interface

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ipv6 ospfv3 transit-delay 30
```

Setting OSPFv3 transit delay to default on the interface

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ipv6 ospfv3 transit-delay
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config-if`<br>`config-if-vlan` | Administrators or local user group members with execution rights for this command. |

## maximum-paths

```
maximum-paths <MAXIMUM>
no maximum-paths
```

### Description

Sets the maximum number of ECMP routes that OSPFv3 can support.

The `no` form of this command sets the maximum number of ECMP routes that OSPFv3 can support to the default value of 4.

| Parameter | Description |
|---|---|
| *<MAXIMUM>* | Specifies the maximum number of ECMP routes. Range: 1 to 8. Default: 4. |

### Examples

Setting maximum number of parallel routes:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# maximum-paths 3
```

Setting maximum number of parallel paths to the default value of 4:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# no maximum-paths
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config-ospfv3-<PROCESS-ID>` | Administrators or local user group members with execution rights for this command. |

## max-metric router-lsa

```
max-metric router-lsa [on-startup <INTERVAL>]
no max-metric router-lsa [on-startup]
```

### Description

Sets the protocol to advertise a maximum metric so that other routers do not prefer the router as an intermediate hop in their shortest path first (SPF) calculations. If the on-startup parameter is used, the router is configured to advertise a maximum metric at startup. That is, for the time specified in seconds, or the default value of 600 seconds.

To disable advertisement of the maximum metric, use the `no` form of the command.

The `no` form of this command advertises the normal cost metrics instead of advertising the maximized cost metric. This setting causes the router to be considered in traffic forwarding.

| Parameter | Description |
|---|---|
| `on-startup <INTERVAL>` | Automatically advertises the stub Router-LSA (or maximize the router-LSA cost metric) for a specified time interval upon OSPFv3 startup.<br>Specifies the time in seconds. Range: 5 to 86400. Default: 600. |

### Examples

Setting to maximize the cost metrics for Router LSA:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# max-metric router-lsa
switch(config-ospfv3-1)# max-metric router-lsa on-startup 3000
```

Setting to advertise the normal cost metrics instead of advertising the maximized cost metric:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# no max-metric router-lsa
switch(config-ospfv3-1)# no max-metric router-lsa on-startup
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config-ospfv3-<PROCESS-ID>` | Administrators or local user group members with execution rights for this command. |

## passive-interface default

```
passive-interface default
no passive-interface default
```

### Description

Sets all OSPFv3 interfaces as passive.

The `no` form of this command sets all the OSPFv3 interfaces as active.

### Examples

Setting OSPFv3-enabled interfaces as passive:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# passive-interface default
```

Setting OSPFv3-enabled interfaces as active:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# no passive-interface default
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config-ospfv3-<PROCESS-ID>` | Administrators or local user group members with execution rights for this command. |

## redistribute

```
redistribute {bgp | connected | local loopback | static | ripng | ospf <PROCESS-ID>}
             [route-map <ROUTE-MAP-NAME>]
no redistribute {bgp | connected | local loopback | static | ripng | ospf <PROCESS-ID>}
                [route-map <ROUTE-MAP-NAME>]
```

### Description

Redistributes routes originating from other protocols, or from another OSPFv3 process, to the current OSPFv3 process.

if a route map is specified, then only the routes that pass the match clause specified in the route map are redistributed to OSPFv3. Configuration is not allowed if the referenced route map has not yet been configured.

If you try to redistribute routes from an OSPFv3 process which is not created, you are prompted to allow the OSPFv3 process to be auto-created before proceeding with redistribution. If you confirm at the prompt, the OSPFv3 process is created with defaults and redistribution configuration applied. If you deny at the prompt, redistribution configuration is skipped.

If command `route-redistribute active-routes-only` has been issued, only the routes from other protocols which are selected for forwarding are considered for redistribution into OSPFv3.

The `no` form of this command disables redistribution of routes to the current OSPFv3 process.

| Parameter | Description |
|---|---|
| `bgp` | Specifies redistributing BGP (Border Gateway Protocol) routes. |
| `connected` | Specifies redistributing connected (directly attached subnet or host). |
| `local loopback` | Specifies redistributing local routes of the loopback interface. |
| `static` | Specifies redistributing static routes. |
| `ripng` | Specifies redistributing RIPng routes. |
| `ospf <PROCESS-ID>` | Specifies redistributing routes from the specified OSPFv3 process ID. Range: 1 to 63. |
| `route-map <ROUTE-MAP-NAME>` | Specifies redistribution filtering by route map. To create a route map, use command `route-map`. |

### Examples

Redistributing routes to OSPFv3:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# redistribute connected
switch(config-ospfv3-1)# redistribute connected route-map connected_routes
switch(config-ospfv3-1)# redistribute local loopback
switch(config-ospfv3-1)# redistribute local loopback route-map local_routes
switch(config-ospfv3-1)# redistribute static
switch(config-ospfv3-1)# redistribute static route-map static_networks
switch(config-ospfv3-1)# redistribute ripng
switch(config-ospfv3-1)# redistribute ripng route-map rip-routes
switch(config-ospfv3-1)# redistribute ospf 2
```

Disabling redistributing routes to OSPFv3:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# no redistribute connected
switch(config-ospfv3-1)# no redistribute connected route-map connected_routes
switch(config-ospfv3-1)# no redistribute local loopback
switch(config-ospfv3-1)# no redistribute local loopback route-map local_routes
switch(config-ospfv3-1)# no redistribute static
switch(config-ospfv3-1)# no redistribute static route-map static_networks
switch(config-ospfv3-1)# no redistribute ripng
switch(config-ospfv3-1)# no redistribute ripng route-map rip-routes
switch(config-ospfv3-1)# no redistribute ospf 2
```

### Command History

| Release | Modification |
|---|---|
| 10.08 | Added `route-map` support for supported redistribute source-protocols. Updated information and examples. |
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config-ospfv3-<PROCESS-ID>` | Administrators or local user group members with execution rights for this command. |

## reference-bandwidth

```
reference-bandwidth <BANDWIDTH>
no reference-bandwidth
```

### Description

Sets the reference bandwidth for OSPFv3. If the OSPFv3 interface cost is not explicitly set, then the cost of all the OSPFv3 interfaces is recalculated based on the reference bandwidth and link speed of the interface.

For VLAN interfaces the calculated link speed value is 1 Gbps (if the OSPFv3 interface cost is not explicitly set).

The `no` form of this command sets the reference bandwidth for OSPF to the default of 100000 Mbps.

| Parameter | Description |
|---|---|
| *<BANDWIDTH>* | Specifies the reference bandwidth used to calculate the cost of an interface in Mbps. Range: 1 to 4000000. Default: 100000. |

### Examples

Setting the reference bandwidth:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# reference-bandwidth 40000
```

Setting the reference bandwidth to the default value:

```
switch(config)# routerv3 ospf 1
switch(config-ospfv3-1)# no reference-bandwidth
```

### Command History

| Release | Modification |
| --- | --- |
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
| --- | --- | --- |
| 6200 | `config-ospfv3-<PROCESS-ID>` | Administrators or local user group members with execution rights for this command. |

## retransmit-interval

```
retransmit-interval <INTERVAL>
no retransmit-interval
```

### Description

Sets the time between retransmitting lost link state advertisements for virtual links.

The `no` form of this command sets the time between retransmitting lost link state advertisements to the default of 5 seconds for virtual links.

| Parameter | Description |
| --- | --- |
| *<INTERVAL>* | Specifies the time interval for the retransmit interval, in seconds. Range: 1 to 3600. Default: 5. |

### Examples

Setting OSPFv3 virtual links retransmit interval:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# area 100 virtual-link 100.0.1.1
switch(config-router-vlink6)# retransmit-interval 30
```

Setting OSPFv3 virtual links retransmit interval to default:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# area 100 virtual-link 100.0.1.1
switch(config-router-vlink6)# no retransmit-interval
```

### Command History

| Release | Modification |
| --- | --- |
| 10.07 or earlier | -- |

## Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config-router-vlink` | Administrators or local user group members with execution rights for this command. |

# router-id

```
router-id <ROUTER-ADDRESS>
no router-id
```

## Description

Sets an ID for the router in an IPv4 address format.

The `no` form of this command unconfigures the router-id for the instance and sets the router-id to the default. The router-id is changed to the dynamically selected router-id. The default router-id 0.0.0.0 updates to the routing stack that triggers to auto-elect a router-id based on the highest IP address of loopback interface, or the highest IP address of interfaces.

| Parameter | Description |
|---|---|
| *<ROUTER-ADDRESS>* | Specifies the router address in IPv4 format (`x.x.x.x`), where `x` is a decimal number from 0 to 255. |

## Examples

Setting router-id in the OSPFv3 context:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1) # router-id 1.1.1.1
```

Unconfiguring router-id:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# no router-id
```

## Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

## Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config-ospfv3-<PROCESS-ID>` | Administrators or local user group members with execution rights for this command. |

# router ospfv3

```
router ospfv3 <PROCESS-ID> [vrf <VRF-NAME>]
```

```
no router ospfv3 <PROCESS-ID> [vrf <VRF-NAME>]
```

### Description

Creates the OSPFv3 process (if not created already) and enters the router OSPFv3 instance context. Optionally if specified, you can specify a named VRF, or the default VRF if the <*vrf-name*> is not specified. Only one OSPFv3 process is allowed per VRF.

The `no` form of this command removes the OSPFv3 instance. If a VRF is specified, it removes the OSPF instance from the named VRF, or the default VRF if the <*var-name*> is not specified.

| Parameter | Description |
|---|---|
| `<PROCESS-ID>` | Specifies an OSPFv3 process ID. Length: 1 to 63. |
| `vrf <VRF-NAME>` | Specifies the name of a VRF. Default: default. |

### Examples

Entering the router OSPFv3 instance:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)#
```

Setting the router OSPFv3 VRF instance:

```
switch(config)# router ospfv3 1 vrf vrf_red
```

Removing the router OSPFv3 instance:

```
switch#(config)# no router ospfv3 1
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config` | Administrators or local user group members with execution rights for this command. |

## show ipv6 ospfv3

```
show ipv6 ospfv3 [<PROCESS-ID>] [all-vrfs | vrf <VRF-NAME>]
```

### Description

Shows OSPFv3 information including area, state, and configuration information.

| Parameter | Description |
|---|---|
| *<PROCESS-ID>* | Specifies an OSPFv3 process ID optionally to show OSPFv3 information for a particular OSPFv3 process. Range: 1 to 63. |
| all-vrfs | Select to show OSPFv3 information for all VRFs. |
| vrf *<VRF-NAME>* | Specifies the name of a VRF. Default: default. |

## Example

Showing general OSPFv3 configurations:

```
switch# show ipv6 ospfv3 1
Routing Process 1 with ID: 1.1.1.1 VRF default
-----------------------------------------------

OSPFv3 Protocol is enabled
Graceful-restart is configured
Restart Interval: 120, State: inactive
Last Graceful Restart Exit Status: none
Area Border: false
AS Border: false
SPF: Start Time: 200ms, Hold Time: 1000ms, Max Wait Time: 5000ms
LSA: Start Time: 5000ms, Hold Time: 0ms, Max Wait Time: 0ms
LSA Arrival: 1000ms
Maximum Paths to Destination: 4
Number of External LSAs 0, checksum sum 0
Summary address:
  prefix fd00::0/64 advertise tag 10
Number of areas is: 2, 2 normal, 0 stub, 0 NSSA
Number of active areas is: 1, 1 normal, 0 stub, 0 NSSA
Reference Bandwidth: 100000 Mbps
Area (0.0.0.0) (Active)
  Interfaces in this Area: 1 Active Interfaces: 1
  Passive Interfaces: 0 Loopback Interfaces: 0
  SPF calculation has run 9 times
  Number of LSAs: 4, checksum sum 195745
Area (0.0.0.1) (Inactive)
  Interfaces in this Area: 0 Active Interfaces: 0
  Passive Interfaces: 0 Loopback Interfaces: 0
  SPF calculation has run 8 times
  Area ranges:
    fd00::1/64, inter-area, advertise
  Number of LSAs: 0, checksum sum 0
```

## Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

## Command Information

| Platforms | Command context | Authority |
| --- | --- | --- |
| 6200 | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show ipv6 ospfv3 border-routers

`show ipv6 ospfv3 [<PROCESS-ID>] border-routers [all-vrfs | vrf <VRF-NAME>]`

### Description

Shows the OSPFv3 routing table entries for Area Border Router (ABR) and Autonomous System Border Router (ASBR).

| Parameter | Description |
| --- | --- |
| *<PROCESS-ID>* | Specifies an OSPFv3 process ID to show the OSPFv3 routing table entries for ABR and ASBR for the particular OSPFv3 process. Range: 1-63. |
| `all-vrfs` | Select to show OSPFv3 border router information for all VRFs. |
| `vrf <VRF-NAME>` | Specifies the name of a VRF. Default: default. |

### Example

Showing OSPFv3 border routers information for default VRF:

```
switch# show ipv6 ospfv3 border-routers
OSPFv3 Process ID 1 VRF default, Internal Routing Table
--------------------------------------------------------
Codes: i - Intra-area route, I - Inter-area route
i 2.2.2.2 [10], ABR, Area 0.0.0.0, SPF 17 via
        fe80::98f2:b301:2c68:1d48, Interface 1/1/1
i 2.2.2.2 [10], ABR, Area 0.0.0.1, SPF 17 via
        fe80::98f2:b301:3068:1d48, Interface 1/1/2
```

Showing OSPFv3 border routers information for default VRF:

```
switch# show ipv6 ospfv3 border-routers
OSPFv3 Process ID 1 VRF default, Internal Routing Table
--------------------------------------------------------
Codes: i - Intra-area route, I - Inter-area route
i 2.2.2.2 [10], ABR, Area 0.0.0.0, SPF 17 via
        fe80::98f2:b301:2c68:1d48, Interface 1/1/1
i 2.2.2.2 [10], ABR, Area 0.0.0.1, SPF 17 via
        fe80::98f2:b301:3068:1d48, Interface 1/1/2
```

### Command History

| Release | Modification |
| --- | --- |
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
| --- | --- | --- |
| 6200 | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show ipv6 ospfv3 interface

```
show ipv6 ospfv3 [<PROCESS-ID>] interface [<INTERFACE-NAME>] [brief]
[all-vrfs | vrf <VRF-NAME>]
```

### Description

Shows information about OSPFv3 enabled interfaces.

| Parameter | Description |
| --- | --- |
| *<PROCESS-ID>* | Specifies an OSPFv3 process ID optionally to show the OSPFv3 enabled interfaces for the particular OSPFv3 process. Range: 1-63. |
| *<INTERFACE-NAME>* | Selects to show information only for the specified OSPFv3 enabled interface. |
| brief | Select to show the overview information for the OSPFv3 enabled interface. |
| all-vrfs | Select to show OSPF-enabled interface information for all VRFs. |
| vrf *<VRF-NAME>* | Specifies the name of a VRF. Default: default. |

### Examples

Showing OSPFv3 information for all interfaces in default VRF:

```
switch# show ipv6 ospfv3 interface
Interface 1/1/1 is up, Line Protocol is up
-------------------------------------------
IPv6 address fe80::98f2:b301:468:be02
Process ID 1 VRF default, Area 0.0.0.0
    State Dr, Status up, Network type Broadcast
    Link Speed: 10000 Mbps
    Cost Configured NA, Calculated 10
    Transit Delay 1 sec, Router Priority 1
    Designated Router ID: 1.1.1.1
    Backup Designated Router ID: 2.2.2.2
    Timer Intervals: Hello 10, Dead 40, Retransmit 5
    Number of Link LSAs: 2, checksum sum 67473

Interface 1/1/2 is up, Line Protocol is up
-------------------------------------------

IPv6 address fe80::98f2:b301:868:be02
Process ID 1 VRF default, Area 0.0.0.1
    State Waiting, Status up, Network type Broadcast
    Link Speed: 10000 Mbps
    Cost Configured NA, Calculated 10
    Transit Delay 1 sec, Router Priority 1
```

```
    No designated router on this network
    No backup designated router on this network
    Timer Intervals: Hello 10, Dead 40, Retransmit 5
    Number of Link LSAs: 1, checksum sum 24592
```

Showing overview information for OSPFv3 enabled interfaces for all VRFs:

```
switch# show ipv6 ospfv3 interface brief all-vrfs
OSPFv3 Process ID 1 VRF default
================================

Total Number of Interfaces: 2

Interface     Area                Cost      State              Status
----------------------------------------------------------------------
1/1/1         0.0.0.0             10        Dr                 up
1/1/2         0.0.0.1             10        Backup-dr          up
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show ipv6 ospfv3 neighbors

```
show ipv6 ospfv3 [<PROCESS-ID>] neighbors [<NEIGHBOR-ID>]
     [interface <INTERFACE-NAME>] [detail | summary]
     [all-vrfs | vrf <VRF-NAME>]
```

### Description

Shows information about OSPFv3 neighbors.

| Parameter | Description |
|---|---|
| *<PROCESS-ID>* | Specifies an OSPFv3 process ID to show OSPFv3 neighbor information for the particular OSPFv3 process. Range: 1-63. |
| `neighbors` *<NEIGHBOR-ID>* | Shows information about a particular neighbor, specified in IPv4 format (A.B.C.D). |
| `interface` *<INTERFACE-NAME>* | Shows neighbor information only for the specified interface. |
| `detail` | Shows detailed information for the neighbors. |

| Parameter | Description |
|---|---|
| `summary` | Shows summary information for the neighbors. |
| `all-vrfs` | Shows neighbor information for all VRFs. |
| `vrf <VRF-NAME>` | Specifies the name of a VRF. Default: default. |

## Examples

Showing OSPFv3 neighbors information:

```
switch# show ipv6 ospfv3 neighbors
OSPFv3 Process ID 1 VRF default
================================

Total Number of Neighbors: 1

Neighbor ID      Priority  State             Interface
-------------------------------------------------------
2.2.2.2          1         FULL/BDR          1/1/1
  Neighbor address fe80::98f2:b301:2c68:1d48
```

Showing OSPFv3 neighbors information for a specific neighbor:

```
switch# show ipv6 ospfv3 neighbors 2.2.2.2
Neighbor 2.2.2.2, Interface address fe80::98f2:b301:2c68:1d48
--------------------------------------------------------------
Process ID 1 VRF default, in Area 0.0.0.0 via Interface 1/1/1
    Neighbor priority is 1, State is FULL
    Options is 0x13
    Dead Timer due in 00:00:38
    Time since last state change 00h:20m:26s
```

Showing detail OSPFv3 neighbors information for a specific neighbor:

```
switch# show ipv6 ospfv3 neighbors 2.2.2.2 detail
Neighbor 2.2.2.2, Interface address fe80::98f2:b301:2c68:1d48
--------------------------------------------------------------
Process ID 1 VRF default, in Area 0.0.0.0 via Interface 1/1/1
    Neighbor priority is 1, State is FULL
    DR is 1.1.1.1, BDR is 2.2.2.2
    Options is 0x13
    Dead Timer due in 00:00:36
    Retransmission Queue Length 0
    Time since last state change 00h:20m:38s
```

Showing OSPFv3 neighbors information for interface **1/1/1**:

```
switch# show ipv6 ospfv3 neighbors 2.2.2.2 interface 1/1/1
Neighbor 2.2.2.2, Interface address fe80::98f2:b301:2c68:1d48
--------------------------------------------------------------
Process ID 1 VRF default, in Area 0.0.0.0 via Interface 1/1/1
    Neighbor priority is 1, State is FULL
    Options is 0x13
```

```
  Dead Timer due in 00:00:32
  Time since last state change 00h:20m:52s
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show ipv6 ospfv3 routes

```
show ipv6 ospfv3 [<PROCESS-ID>] routes [<PREFIX/LENGTH>]
     [all-vrfs | vrf <VRF-NAME>]
```

### Description

Shows the OSPFv3 routing table information.

| Parameter | Description |
|---|---|
| *<PROCESS-ID>* | Specifies an OSPFv3 process ID that shows information from the OSPFv3 routing table for the particular OSPFv3 process. Range: 1-63. |
| *<PREFIX/LENGTH>* | Specifies the IPv6 destination prefix showing information about a particular destination prefix. For example, 2010:bd9::/32. |
| all-vrfs | Select to show OSPFv3 routing table information for all VRFs. |
| vrf *<VRF-NAME>* | Specifies the name of a VRF. Default: default. |

### Examples

Showing OSPFv3 routing table information:

```
switch# show ipv6 ospfv3 routes
Codes: i - Intra-area route, I - Inter-area route
       E1 - External type-1, E2 - External type-2

OSPFv3 Process ID 1 VRF default, Routing Table
-----------------------------------------------

Total Number of OSPFv3 Routes : 2

111::/64           (i) area:0.0.0.0
     directly attached to interface 1/1/1, cost 1 distance 110
```

```
fd00::/64          (i) area:0.0.0.1
     directly attached to interface vlan10, cost 1 distance 110
```

Showing OSPFv3 routing table information for VRF red:

```
switch# show ipv6 ospfv3 routes vrf red

Codes: i - Intra-area route, I - Inter-area route
       E1 - External type-1, E2 - External type-2

OSPFv3 Process ID 2 VRF red, Routing Table
-------------------------------------------

Total Number of OSPFv3 Routes : 1

222::/64           (i) area:0.0.0.1
     directly attached to interface 1/1/2, cost 1 distance 110
```

Showing OSPFv3 routing table information for destination prefix fd00::/64:

```
switch# show ipv6 ospfv3 1 routes fd00::/64
Codes: i - Intra-area route, I - Inter-area route
       E1 - External type-1, E2 - External type-2

OSPFv3 Process ID 1 VRF default, Routing Table for prefixes fd00::/64
----------------------------------------------------------------------

Total Number of OSPFv3 Routes : 1

fd00::/64          (i) area:0.0.0.1
     directly attached to interface vlan10, cost 1 distance 110
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show ipv6 ospfv3 statistics

```
show ipv6 ospfv3 [<PROCESS-ID>] statistics
     [all-vrfs | vrf <VRF-NAME>]
```

### Description

Shows OSPFv3 statistics.

| Parameter | Description |
|---|---|
| *<PROCESS-ID>* | Specifies an OSPFv3 process ID that shows information on the OSPFv3 SPF statistics for the particular OSPFv3 process. Range: 1-63. |
| `all-vrfs` | Select to show OSPFv3 SPF statistics information for all VRFs. |
| `vrf` *<VRF-NAME>* | Specifies the name of a VRF. Default: default. |

### Examples

Showing OSPFv3 statistics information:

```
switch# show ipv6 ospfv3 statistics
OSPFv3 Process ID 1 VRF default, Statistics (cleared 3h 2m 21s ago)
--------------------------------------------------------------------

Unknown Interface Drops          : 0
Unknown Virtual Interface Drops : 0
Bad IPv6 Header Length Drops    : 0
Wrong OSPFv3 Version Drops      : 0
Bad Source IPv6 Drops           : 0
Resource Failure Drops          : 0
Bad Header Length Drops         : 0
Total Drops                     : 0
```

Showing OSPFv3 statistics information for VRF red:

```
switch# show ipv6 ospfv3 2 statistics vrf red

OSPFv3 Process ID 2 VRF red, Statistics (cleared 3h 2m 30s ago)
----------------------------------------------------------------

Unknown Interface Drops          : 0
Unknown Virtual Interface Drops : 0
Bad IPv6 Header Length Drops    : 0
Wrong OSPFv3 Version Drops      : 0
Bad Source IPv6 Drops           : 0
Resource Failure Drops          : 0
Bad Header Length Drops         : 0
Total Drops                     : 0
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show ipv6 ospfv3 statistics interface

```
show ipv6 ospfv3 [<PROCESS-ID>] statistics interface [<INTERFACE-NAME>]
     [all-vrfs | vrf <VRF-NAME>]
```

### Description

Shows the OSPFv3 statistics for the OSPFv3-enabled interfaces.

| Parameter | Description |
|---|---|
| *<PROCESS-ID>* | Specifies an OSPFv3 process ID to show OSPF-enabled interface statistics information on the specified OSPFv3 process. Range: 1 to 63. |
| *<INTERFACE-NAME>* | Selects to show information only for the specified interface. |
| `all-vrfs` | Select to show OSPF-enabled interface statistics information for all VRFs. |
| `vrf` *<VRF-NAME>* | Specifies the name of a VRF. Default: default. |

### Example

Showing OSPFv3-enabled interfaces information for interface 1/1/1:

```
switch# show ipv6 ospfv3 statistics interface all-vrfs
OSPFv3 Process ID 1 VRF default, Interface vlan2000 Statistics  (cleared 0h 2m 52s
ago)

=====================================================================================
==

Tx Hello packets      : 0                Rx Hello packets      : 0
Tx Hello bytes        : 0                Rx Hello bytes        : 0
Tx DD packets         : 0                Rx DD packets         : 0
Tx DD bytes           : 0                Rx DD bytes           : 0
Tx LS request packets : 0                Rx LS request packets : 0
Tx LS request bytes   : 0                Rx LS request bytes   : 0
Tx LS update packets  : 0                Rx LS update packets  : 0
Tx LS update bytes    : 0                Rx LS update bytes    : 0
Tx LS ack packets     : 0                Rx LS ack packets     : 0
Tx LS ack bytes       : 0                Rx LS ack bytes       : 0

Total IPsec packets processed : 0
Total IPsec bytes processed   : 0
Total Number of State Changes : 0
Number of LSAs                : 0
LSA Checksum Sum              : 0

Total OSPFv3 Packets Discarded: 0
----------------------------------

Reason                           Packets Dropped
--------------------------------------------------
Invalid Type                     0
Invalid Length                   0
Invalid Version                  0
Bad or Unknown Source            0
Area Mismatch                    0
Self-originated                  0
```

```
Duplicate Router ID              0
Interface Standby                0
Total Hello Packets Dropped      0
  Hello Interval Mismatch        0
  Dead Interval Mismatch         0
  Options Mismatch               0
  MTU Mismatch                   0
  Neighbor Ignored               0
Resource Failures                0
Bad LSA Length                   0
Others                           0
IPsec Authentication Errors      0
IPsec ESP Errors                 0

Total LSAs Ignored : 0
-----------------------
Bad Type             : 0
Bad Length           : 0
Invalid Data         : 0
Invalid Checksum     : 0
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## summary-address

```
summary-address  <IPV6-ADDR>/<MASK> [no-advertise | tag <TAG-VALUE>]
no summary-address <prefix/length> [no-advertise | tag <tag-value>
```

### Description

Summarizes the external routes with the matching address and mask. When advertising this route, its metric is set to the lowest cost path from among the routes that were summarized.

The `no` form of this command disables route summarization.

This command only works for an ASBR (Autonomous System Boundary Router).

| Parameter | Description |
|---|---|
| *<IPV6-ADDR>* | Specifies an IP address in IPv6 format (`xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx`), where `x` is a hexadecimal number from 0 to F. |

| Parameter | Description |
|---|---|
| *<MASK>* | Specifies the number of bits in the address mask in CIDR format (`x`), where `x` is a decimal number from 0 to 128. |
| `no-advertise` | Do not advertise the aggregate route. Suppress routes that match the specified prefix/mask pair. |
| `tag` *<TAG-VALUE>* | Specify the tag for the aggregate route. The summary prefix will be advertised along with the tag value in External LSAs. Range: 0 to 4294967295 |

### Examples

Setting OSPF route summarization:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# summary-address 2001:DB8::1/32
```

Disabling route summarization:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# no summary-address 2001:DB8::1/32
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config-ospfv3-`*<PROCESS-ID>* | Administrators or local user group members with execution rights for this command. |

## timers lsa-arrival

```
timers lsa-arrival <DELAY>
no timers lsa-arrival
```

### Description

Configures the minimum delay between receiving the same LSA from a peer. The same LSA is an LSA that contains the same LSA ID number, LSA type, and advertising router ID. If an instance of the same LSA arrives sooner before the delay expires, the LSA is dropped. Generally, the LSA arrival timer should be set to a value less than or equal to the start-time value for the command `timers throttle lsa start` on the neighbor.

The `no` form of this command sets the LSA timers to default values.

| Parameter | Description |
|---|---|
| *<DELAY>* | Specifies the delay in milliseconds. Range: 0 to 600000. Default: 1000. |

### Examples

Setting the LSA arrival timer:

```
switch(config)# router ospf 1
switch(config-ospfv3-1)# timers lsa-arrival 10
```

Setting the LSA arrival timer to default:

```
switch(config)# router ospf 1
switch(config-ospfv3-1)# no timers lsa-arrival
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | config-ospfv3-*<PROCESS-ID>* | Administrators or local user group members with execution rights for this command. |

## timers throttle lsa

```
timers throttle lsa start-time <START-TIME> hold-time <HOLD-TIME> max-wait-time <WAIT-TIME>
no timers throttle lsa
```

### Description

Configures the timers for LSA generation.

The `no` form of this command sets the LSA timers to default values.

| Parameter | Description |
|---|---|
| start-time *<START-TIME>* | Specifies the initial wait time in milliseconds after which LSAs are generated. When set to 0, the LSAs are generated without any delay. Range: 0 to 600000. Default: 5000. |
| hold-time *<HOLD-TIME>* | Specifies the amount of time, in milliseconds, between regeneration of an LSA. The hold time doubles each time the same LSA must be regenerated, until `max-wait-time` is reached. When set to 0, LSA regeneration time is not increased. Range: 0 to 600000. Default: 0. |

| Parameter | Description |
|---|---|
| `max-wait-time <WAIT-TIME>` | Specifies the maximum wait time, in milliseconds, for regeneration of the same LSA. When set to 0, LSA regeneration time is not increased. Range: 0 to 600000. Default: 0. |

### Examples

Setting the LSA timers:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# timers throttle lsa start-time 100 hold-time 1000 max-wait-
time 10000
```

Setting LSA timers to default values:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# no timers throttle lsa
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config-ospfv3-<PROCESS-ID>` | Administrators or local user group members with execution rights for this command. |

## timers throttle spf

```
timers throttle spf start-time <START-TIME> hold-time <HOLD-TINME>
max-wait-time <WAIT-TIME>
no timers throttle spf
```

### Description

Configures timers for SPF calculation. There are three timers:

- `start-time` Is the initial delay before an SPF calculation is started. Default is 200 milliseconds.
- `hold-time` Is the progressive backoff time to wait before next scheduled SPF calculation. Default is 1000 milliseconds. If a route change event occurs during this period, the value doubles until it reaches the *max-wait-time*.
- `max-wait-time` Is the maximum time to wait before the next scheduled SPF calculation. Default is 5000 milliseconds. This is used to limit the SPF hold timer and also defines the time to be considered for which the OSPF LSDB has to be stable, after which the SPF throttle mechanism is reset.

The `no` form of this command sets all the configured non-default timers to default value.

| Parameter | Description |
|---|---|
| *<START-TIME>* | Time in milliseconds to set timer for initial SPF delay. Default: 200. |
| *<HOLD-TINME>* | Time in milliseconds to set the minimum hold time between two consecutive SPF calculations. Default: 1000. |
| *<WAIT-TIME>* | Time in milliseconds to set the maximum wait time between two consecutive SPF calculations. Default: 5000. |

### Examples

Setting non-default timer values for SPF throttling:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# timers throttling spf start-time 500 hold-time 3000 max-
wait-time 9000
Switch(config-ospfv3-1)# show running-config current-context
router ospfv3 1
               area 0.0.0.0
```

Setting default timer values for SPF throttling after configuring non-default values:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# no timers throttling spf
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config-ospfv3-<PROCESS-ID>` | Administrators or local user group members with execution rights for this command. |

## transit-delay

```
transit-delay <DELAY>
no transit-delay
```

### Description

Sets the time delay in Link state transmission for virtual links.

The `no` form of this command sets the delay in Link state transmission to the default of 1 second for virtual links.

| Parameter | Description |
|---|---|
| *<DELAY>* | Specifies the time delay for the transit delay, in seconds. Range: 1 to 3600. Default: 1. |

### Examples

Setting OSPFv3 virtual links transit delay:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# area 100 virtual-link 100.0.1.1
switch(config-router-vlink6)# transit-delay 30
```

Setting OSPFv3 virtual links transit delay to default:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# area 100 virtual-link 100.0.1.1
switch(config-router-vlink6)# no transit-delay
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config-router-vlink` | Administrators or local user group members with execution rights for this command. |

## trap-enable

```
trap-enable
no trap-enable
```

### Description

Enables the notification of the events to be sent as traps to the SNMP management stations for OSPFv3.

The `no` form of this command disables the notification of the events to be sent as traps to the SNMP management stations for OSPFv3.

### Examples

Enabling sending notification of events as traps:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# trap-enable
```

Disabling sending notification of events as traps:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# no trap-enable
```

## Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

## Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config-ospfv3-<PROCESS-ID>` | Administrators or local user group members with execution rights for this command. |

# Chapter 7
# Equal Cost Multipath (ECMP)

ECMP is not supported on the Aruba 4100i Switch Series.

ECMP is not supported on the Aruba 6000 and 6100 Switch Series.

Equal Cost Multipath (ECMP) allows a router to load balance traffic destined to some network that is reachable through multiple equal cost route nexthops. ECMP functionality is dependent on L3 routing being enabled on the router.

## Overview

The nexthop chosen out of the ECMP group of nexthops for a given packet is typically determined by its header information, usually by hashing source and destination IP addresses as well as potentially source and destination L4 ports. The hashing algorithm used is dependent on the type of switch in use as well as whether the packet is flowing through the slow-path or fast-path.

By default all of the following fields are enabled and utilized for the load balancing decision.

- L3 Destination IP address
- L3 Source IP address
- L4 Destination port
- L4 Source port

ECMP support is NOT dependent on address family. The ECMP feature applies to both IPv4 and IPv6 traffic flowing through the router.

If two different routing protocols are (mis)configured to use the same Administrative Distance, then they have the opportunity to create ECMP groups together.

## ECMP commands

### show ip ecmp

```
show ip ecmp
```

#### Description

Displays the Equal Cost Multipath (ECMP) configuration.

#### Examples

```
switch# show ip ecmp
```

```
ECMP Configuration
---------------------

 ECMP Status        : Enabled

ECMP Load Balancing by
------------------------
 Source IP          : Enabled
 Destination IP     : Enabled
 Source Port        : Enabled
 Destination Port   : Enabled
```

### Command History

| Release | Modification |
| --- | --- |
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
| --- | --- | --- |
| 6200 | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

# Virtual Router Redundancy Protocol (VRRP)

VRRP is not supported on the Aruba 4100i Switch Series.

VRRP is not supported on the Aruba 6000 and 6100 Switch Series.

This section describes how to configure and verify VRRP configuration and operational states.

- All VRRP configurations work in VRRP context.
- VRRP can be configured on physical ports, VLAN interfaces, and LAG interfaces.
- VRRP mandates the associated interface to be routing interface.

## Overview

In many networks, edge devices are configured to send packets to a statically configured default router. If this router becomes unavailable, devices that use it as a first-hop router become isolated from the network.

VRRP uses dynamic failover to ensure the availability of an end node default router. The IP address used as the default route is assigned to a virtual router (VR). The VR includes:

- A Master router assigned to forward traffic designated for the virtual route
- One or more prioritized Backup routers (If a Backup is forwarding traffic for the VR, it has replaced the Owner as the Master router.)

This redundancy provides a backup for gateway IP addresses (first-hop routers). If a VR Master router becomes unavailable, the traffic it supports will be transferred to a Backup router without major delays or operator intervention. This operation can eliminate single-point-of-failure problems and provide dynamic failover (and failback) support. As long as one physical router in a VR configuration is available, IP addresses assigned to the VR are always available. Edge devices can send packets to these IP addresses without interruption.

Advantages to using VRRP include:

- Minimizing failover time and bandwidth overhead if a primary router becomes unavailable
- Minimizing service disruptions during a failover
- Providing backup for a load-balanced routing solution
- Avoiding the need to make configuration changes in the end nodes if a gateway router fails
- Eliminating the need for router discovery protocols to support failover operation

## Terminology

**Owner router**: The Owner router is automatically configured with the highest VRRP router priority in the VR (255). It operates as the Master router for the VR unless it becomes unavailable to the network. All Virtual Router members can be configured so virtual IP is not the same as physical (or real) IP. Such virtual address can be called pure virtual IP address.

**Master router**: The Owner or Backup router that is the physical forwarding agent for routed traffic using the VR as a gateway. There can be only one router operating as the Master for a network. If the router configured as the Owner for a VR is available to the network, it will also be the Master. If the Owner fails or loses availability to the network, the highest-priority Backup becomes the Master.

**Backup**: A router configured in a VR as a Backup to the Owner/Master for the same VR. There must be a minimum of one Backup in a VR to support VRRP operation if the Owner/Master fails. Every backup is created with a configurable priority (default: 100). This priority determines the precedence for becoming the Master of the VR if the Owner or another Backup operating as Master becomes unavailable.

When the current Master router is no longer available, the Backup router with highest priority will become current master. When a router with higher priority becomes available, it is switched to master. The user can override this behavior by disabling preemption mode.

**Virtual Router (VR)**: Consists of one Owner/Master router and one or more Backup routers that belong to the same network. The Owner is the router that owns any IP addresses associated with the VR. The VR has one virtual IP address (multiple virtual IP addresses for multinetted interfaces) that correspond to a real IP address on the Owner. A VR includes the following:

- A VR identification (VRID) configured on all VRRP routers in the same network. For a multinetted interface, the VRID is configured on all routers in the same subnet.
- The same VIP configured on each instance of the same VR.
- The status of either Owner or Backup configured on each instance of the same VR. There can be one Owner and one or more Backups configured on each VR.
- The priority level configured on each instance of the VR. On the Owner router, the highest priority setting of 255 is automatically fixed. On Backups, the default priority setting is 100 and configurable.
- A VR MAC address that is not configurable.

**Virtual IP address (VIP)**: The VIP associated with a VR must be a real IP address already configured in the associated interface on the Owner router.

- If the VIP is an IPv6 address, a link-local address must be configured before adding a global IPv6 address.
- The Owner and all Backup routers belonging to the VR have this IP address configured in their VRID contexts as the VIP.
- A subnetted interface allows multiple VIPs. If there are fewer IP addresses in an interface and a user wants VRRP support on multiple subnets, configure a separate VR instance for each IP address in the interface.

**VRID**: The identifier for a VR configured on an interface. A VRID can be used for only one VR in any interface on a router. It can be used again for a different VR in a different interface.

# VRRP operation

VRRP supports router redundancy through a prioritized election process among routers configured as members of the same virtual router (VR).

A VR includes two or more member routers configured with:

- A virtual IP address that is also configured as a real IP address on one of the routers
- A virtual router MAC address

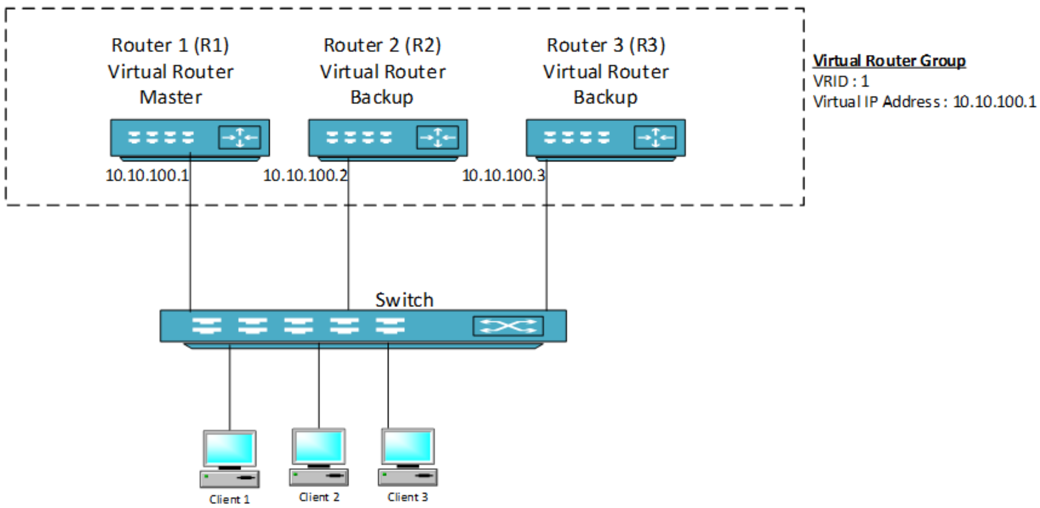
The router that owns the IP address is configured to operate as the Owner of the VR for traffic-forwarding purposes. By default, this router has the highest VRRP priority in the VR. Any other routers in the VR have a lower priority and are configured to operate as Backups if the Owner router becomes unavailable.

The Owner normally operates as the Master for a VR. If the Owner becomes unavailable, a failover to a Backup router belonging to the same VR occurs. This Backup becomes the Master. If the Owner recovers, a failback occurs, and Master status reverts to the Owner.

- Using more than one Backup provides additional redundancy. If the Owner and the highest-priority Backup fail, another lower-priority Backup can take over as Master.
- In a VRRP owner scenario, DAD should be disabled for VRRP to work.

The image shown here depicts a basic VLAN topology. In this example, Routers 1, 2 and 3 form a VRRP group. The IP address of the group is the same address configured for the SVI interface of R1 (10.10.100.1).

**Figure 1** *Basic VLAN Topology*




Because the virtual IP address uses the IP address of the SVI interface of Router R1, Router R1 is the master (also known as IP address Owner).

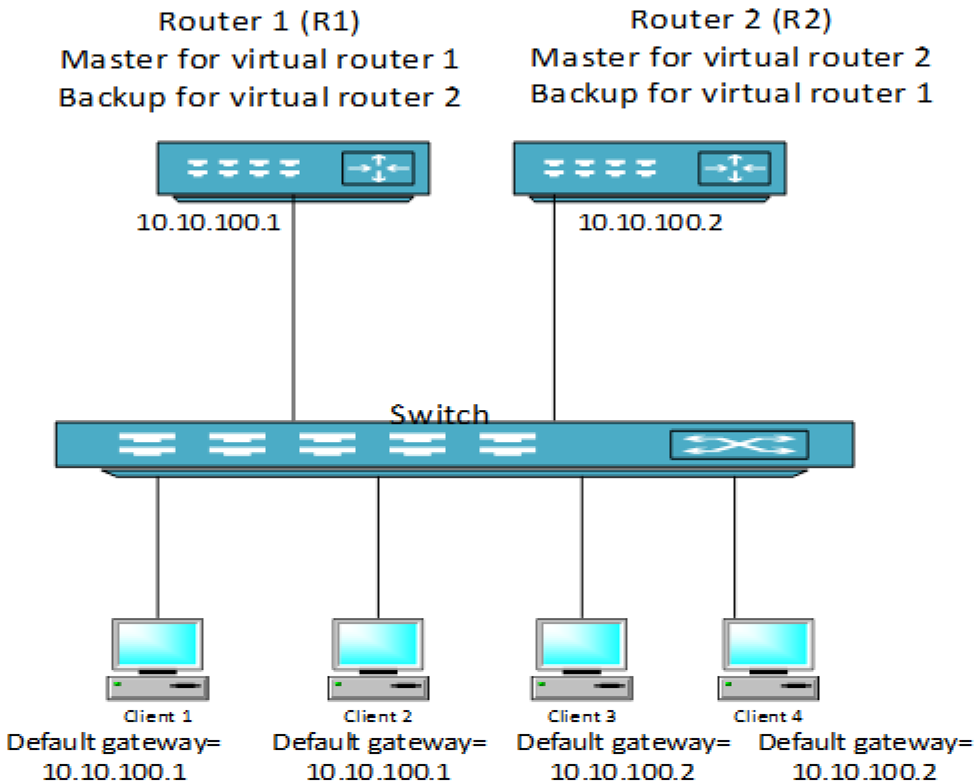
- As the master, R1 owns the virtual IP address of the VRRP group and forwards packets sent to this IP address.
- Clients 1 through 3 are configured with the default gateway IP address of 10.10.100.1.
- Routers R2 and R3 function as backups. If the master fails, the backup router with highest priority becomes the master and takes over the virtual IP address to provide uninterrupted service for the clients connected to L2 Switch. When Router R1 recovers, it becomes master again.

# Multiple VRRP groups

A user can configure multiple VRRP groups on a router (or L3) interface (Physical, LAG, or SVI interfaces). In a topology in which multiple VRRP groups are configured on a router interface, the interface can act as a master for one VRRP group and a backup for one or more other VRRP groups.

This image displays a topology in which VRRP is configured so that Routers R1 and R2 share the traffic to and from clients 1 through 4. Routers R1 and R2 act as backups to each other if either router fails.

**Figure 1** *Routers share the traffic to and from clients 1 through 4.*




This topology contains two virtual IP addresses for two VRRP groups that overlap. For VRRP group 1, Router R1 is the owner of IP address 10.10.100.1 and is the Master. Router R2 is the Backup to Router R1. Clients 1 and 2 are configured with the default gateway IP address of 10.10.100.1.

For VRRP group 2, Router R2 is the owner of IP address 10.10.100.2 and is the Master. Router R1 is the Backup to Router R2. Clients 3 and 4 are configured with the default gateway IP address of 10.0.0.2.

# VRRP priority

In a Backup router VR configuration, the virtual router priority defaults to 100. The priority for the configured Owner is automatically set to the highest value of 255.

In a VR in which there are two or more Backup routers, priority settings can be reconfigured to define the order in which Backups will be reassigned as Master if a failover occurs.

# VRRP preemption

In a Backup router VR configuration, the virtual router priority defaults to 100. The priority for the configured Owner is automatically set to the highest value of 255.

When multiple Backup routers exist in a VR:

- If the current Master fails and the highest-priority Backup is not available, VRRP selects the next-highest priority Backup to operate as Master.
- If the highest-priority Backup later becomes available, it preempts the lower-priority Backup and takes over the Master function.

To remove this preemptive ability on a VR, disable it with the `no preempt` command.

Preemption applies only to VRRP routers configured as Backups.

# Virtual Router MAC address

When a Virtual Router (VR) instance is configured, the protocol automatically assigns a MAC address based on the standard MAC prefix for VRRP packets plus the VRID number (as described in RFC 3768). The first five octets form the standard MAC prefix for VRRP and the last octet is the configured VRID. For example: `00-00-5E-00-01-<VRid>`

# VRRP and ARP

The current Master for a VR responds to ARP requests for the virtual IP addresses with the VR-assigned MAC address. The virtual MAC address is also used as source MAC address for periodic advertisements sent by the current Master.

The VRRP router responds to ARP requests for non-virtual IP addresses with the system MAC address. Non-virtual IP addresses are not configured as virtual IP addresses for any VR on the interface.

# VRRP tracking

VRRP supports object tracking. This functionality allows monitoring of the state of a configured object. The state determines the priority of the VRRP router in a VRRP group. A track object can be created using the `track <object-id>` command and the object can be associated with an interface. A change in interface state will then affect the priority of a VRRP group.

# High availability

VRRP supports high availability through stateful restarts and stateful switchovers.

- A stateful restart occurs when the VRRP process (or daemon) fails and is restarted.
- A stateful switchover occurs when the active management module (AMM) switches to the standby management module (SMM).

# VRRP and Neighbor Discovery for IPv6

Neighbor Discovery (ND) is the IPv6 equivalent of the IPv4 ARP for layer 2 address resolution. It uses IPv6 ICMP messages to do the following:

- Determine the link-layer address of neighbors
- Verify that a neighbor is reachable
- Track neighbor (local) routers

Neighbor Discovery enables functions such as:

- Router and neighbor solicitations and discovery
- Detecting address changes for devices
- Identifying a replacement for a router or router path that has become unavailable
- Duplicate address detection (DAD)

- Router Advertisement processing
- Neighbor reachability
- Resolution of destination addresses
- Changes to link-layer addresses

An instance of Neighbor Discovery is triggered on a device when a new or changed IPv6 address is detected. VRRPv3 provides a faster failover to a backup router by not using standard ND procedures. A failover to a backup router can occur in approximately three seconds without any interaction with hosts and a minimum of VRRPv3 traffic.

# Duplicate address detection (DAD)

Duplicate Address Detection verifies that a configured unicast IPv6 address is unique before it is assigned to an interface. When the Owner router fails, the backup VRRP router assumes the Master role. When the Owner router becomes operational, DAD will fail because a backup VRRP router is in the Master role that responds to the DAD request. To avoid this, on virtual routers that are in owner mode (priority = 255) DAD needs to be disabled for the interface on which the owner VR is configured.

# Guidelines and limitations

- VRRP must be enabled at the global level using the `router vrrp enable` command. It must also be enabled at the interface level at which you configure VRRP.
- A maximum of 8 IPv4 and 8 IPv6 VRRP groups are supported on an interface.
- A maximum of 256 VRRP groups is supported on a router. The groups can be IPv4 or IPv6.
- Supported virtual router identification (VRID) range is 1-255. A maximum of 8 VRIDs can be used to achieve a scale of 256 VRRP groups by using the same VRID across different interfaces.
- Proxy-ARP and Active-Gateway are not supported in conjunction with VRRP.

# VRRP commands

## address

```
address <IP-ADDR> [ primary | secondary ]
no address <IP-ADDR> [ primary | secondary ]
```

### Description

Configures a primary or secondary IPv4 or IPv6 address for the VRRP group. To use secondary IP addresses in a VRRP group, you must first configure a primary IP address on the same group. A maximum of 16 IP addresses per IPv4 VRRP group and 8 IPv6 addresses per IPv6 VRRP group are supported.

Do not configure an IPv4 VRRP group using addresses from the /30, /31, and /32 subnets of the interface IP address.

16 Virtual IP addresses per IPv4 VR and 8 Virtual IP addresses per IPv6 VR are supported.

The total number of VIPs supported by a switch is:

- 1024 VIPs for IPv4 VRs
- 512 VIPs for IPv6 VRs

The `no` form of this command deletes a primary or secondary IPv4 or IPv6 address from the VRRP group.

| Parameter | Description |
|---|---|
| *<IP-ADDR>* | Configures the IPv4 or IPv6 address. |
| `primary` | Configures a primary address. |
| `secondary` | Configures a secondary address. |

## Examples

```
switch(config)# system internal-vlan-range 4041-4050
switch(config)# router vrrp enable
```

```
switch(config)# interface 1/1/1
switch(config-if)# vrrp 1 address-family ipv4
switch(config-if-vrrp)# address 10.0.0.1 primary


switch(config)# interface 1/1/1
switch(config-if)# vrrp 1 address-family ipv6
switch(config-if-vrrp)# address fe80::1 primary


switch(config)# interface 1/1/1
switch(config-if)# vrrp 1 address-family ipv4
switch(config-if-vrrp)# no address 10.0.0.1 primary


switch(config)# interface 1/1/1
switch(config-if)# vrrp 1 address-family ipv6
switch(config-if-vrrp)# no address fe80::1 primary
```

## Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

## Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config` | Administrators or local user group members with execution rights for this command. |

# authentication

```
authentication {text | md5} [{plaintext | ciphertext} <KEY>]
no authentication
```

## Description

This command enables authentication mode and the authentication key for VRRP groups on an interface. VRRP members or routers of the same VRRP group must use the same authentication mode and authentication key.

The no form of this command disables authentication mode and the authentication key for VRRP groups on an interface.

IPv4 VRRPv3 and IPv6 VRRPv3 do not support VRRP packet authentication. Authentication mode and key configuration take effect only in VRRPv2 (IPv4 only - RFC2338).

In VRRPv3, authentication mode and authentication key settings do not take effect because VRRP Authentication was removed from RFC5798.

VRRP provides the following authentication modes as described in RFC2338:

### Simple authentication

The sender fills an authentication key into the VRRP packet and the receiver compares the received authentication key with its local authentication key.

If the two authentication keys match, the received VRRP packet is legitimate. Otherwise, the received packet is illegitimate and is discarded.

Authentication key text is sent in the clear and can be seen in a packet trace. This makes MD5 authentication more secure than text.

### MD5 authentication

The sender computes a digest for the packet that will be sent using the authentication key and MD5 algorithm, and saves the result in the VRRP packet.

The receiver performs the same operation with the authentication key and MD5 algorithm, and compares the result with the content in the authentication header. If the results match, the received VRRP packet is legitimate. Otherwise, the received packet is illegitimate and is discarded.

| Parameter | Description |
|---|---|
| `text` | Configures the simple authentication type. |
| `md5` | Configures the MD5 (message-digest) authentication type. |
| `plaintext` | Specifies that the key is provided as plaintext. |
| `ciphertext` | Specifies that the key is provided as ciphertext. |
| *<KEY>* | Specifies the key in the chosen format. |

When the key is not provided on the command line, plaintext key prompting occurs upon pressing Enter. The entered key characters are masked with asterisks.

## Examples

Enabling VRRP authentication using MD5 with a provided plaintext key:

```
switch(config)# interface 1/1/1
switch(config-if)# vrrp 1 address-family ipv4
switch(config-if-vrrp)# version 2
switch(config-if-vrrp)# authentication md5 plaintext testvrrpkey
```

Enabling VRRP authentication using MD5 with a prompted plaintext key:

```
switch(config)# interface 1/1/1
switch(config-if)# vrrp 1 address-family ipv4
switch(config-if-vrrp)# version 2
switch(config-if-vrrp)# authentication md5
Enter the authentication key: *************
Re-Enter the authentication key: *************
```

Enabling VRRP authentication using MD5 with a provided ciphertext key:

```
switch(config)# interface 1/1/1
switch(config-if)# vrrp 1 address-family ipv4
switch(config-if-vrrp)# version 2
switch(config-if-vrrp)# authentication md5 ciphertext AQBapfciFZ/P...biBAAAAOjc0a8=
```

Disabling VRRP authentication:

```
switch(config)# interface 1/1/1
switch(config-if)# vrrp 1 address-family ipv4
switch(config-if-vrrp)# version 2
switch(config-if-vrrp)# no authentication
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config-if-vrrp` | Administrators or local user group members with execution rights for this command. |

## preempt

```
preempt
no preempt
```

### Description

Enables the preempt option. The default value is enabled. In default mode, a backup router with a higher priority than another backup that is operating as master will take over the master function.

Applies to VRRP backup routers only and is used to minimize network disruption caused by unnecessary preemption of the master operation among backup routers.

The `no` form of this command disables the preempt option, thus preventing the higher-priority backup from taking over the master operation from a lower-priority backup. This command does not prevent an owner router from resuming the master function after recovering from being unavailable.

### Examples

```
switch(config)# interface 1/1/1
switch(config-if)# vrrp 1 address-family ipv4
switch(config-if-vrrp)# preempt
```

```
switch(config)# interface 1/1/1
switch(config-if)# vrrp 1 address-family ipv4
switch(config-if-vrrp)# no preempt
```

### Command History

| Release | Modification |
| --- | --- |
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
| --- | --- | --- |
| 6200 | `config` | Administrators or local user group members with execution rights for this command. |

## preempt delay minimum

```
preempt delay minimum <DELAY-IN-SECONDS>
no preempt delay minimum <DELAY-IN-SECONDS>
```

### Description

Sets the time in seconds (1-3600) that the router will wait before taking control of the virtual IP and starting to route packets.

The `no` form of this command sets the preempt delay for the VRRP group to the default preempt delay of 0 seconds.

The VRRP Preempt Delay Timer (PDT) allows admin users to configure a period of time before the VR takes control of the virtual IP address. It does not transition to the Master state until the timer period expires.

The timer value configured should be long enough to allow upper layer protocol to converge. The PDT is applied only during initialization of the router.

| Parameter | Description |
| --- | --- |
| *<DELAY-IN-SECONDS>* | Selects the time in seconds (1-3600). Default is 0 seconds. |

### Examples

```
switch(config)# interface 1/1/1
switch(config-if)# vrrp 1 address-family ipv4
switch(config-if-vrrp)# preempt delay minimum 30

switch(config)# interface 1/1/1
switch(config-if)# vrrp 1 address-family ipv4
switch(config-if-vrrp)# no preempt delay
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config` | Administrators or local user group members with execution rights for this command. |

## priority

```
priority <1-254>
no priority
```

### Description

Sets the priority for the VRRP group.

The `no` form of this command sets the priority for the VRRP group as default priority.

- The default value for non-Owner virtual routers is 100.
- The default value for Owner virtual router is 255, which cannot be changed.

### Examples

```
switch(config)# interface 1/1/1
switch(config-if)# vrrp 1 address-family ipv4
switch(config-if-vrrp)# priority 150
```

```
switch(config)# interface 1/1/1
switch(config-if)# vrrp 1 address-family ipv4
switch(config-if-vrrp)# no priority
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config` | Administrators or local user group members with execution rights for this command. |

## router vrrp {enable | disable}

```
router vrrp {enable | disable}
no router vrrp {enable | disable}
```

### Description

Enables or disables VRRP protocol globally. You must globally enable the VRRP feature for VRRP virtual router.

`no router vrrp enable` disables VRRP protocol globally but does not remove all VRRP configurations.

`no router vrrp disable` enables VRRP protocol globally.

### Example

Enabling VRRP protocol globally:

```
switch(config)# router vrrp enable
```

Disable VRRP protocol globally:

```
switch(config)# router vrrp disable
```

Disable VRRP protocol globally:

```
switch(config)# no router vrrp enable
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config` | Administrators or local user group members with execution rights for this command. |

## no router vrrp

```
no router vrrp
```

### Description

Removes VRRP configuration and VRRP global protocol. If `auto-confirm` is enabled or VRRP is not configured on any interface, this command will not ask for user confirmation.

## Examples

Removing VRRP configuration:

```
switch(config)# no router vrrp
All VRRP configuration will be deleted.
Do you want to continue (y/n)?
```

## Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

## Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config` | Administrators or local user group members with execution rights for this command. |

# show track

`show track [brief | <OBJECT-ID>]`

## Description

Shows all or specific track object information.

| Parameter | Description |
|---|---|
| `brief` | Displays brief information about all or specific track objects |
| `<OBJECT-ID>` | Displays information about a specified track object (1-128) |

## Examples

```
switch# show track
Track 1
  interface 1/1/1
  Interface is DOWN
```

```
switch# show track brief
Track  Interface        State

1      1/1/1            Down
2      None             Down
3      1/1/2            Up
```

## Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show track brief

`show track brief`

### Description

Shows brief information for all track objects.

### Examples

Showing brief information for all track objects:

```
switch# show track brief
Track  Interface        State

1      1/1/1            Down
2      None             Down
3      1/1/2            Up
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show vrrp

`show vrrp [brief | detail | interface <INTERFACE-NAME> | interface <LAG-NAME> | interface <VLAN-NAME> | ipv4 | ipv6 | statistics | statistics interface <INTERFACE-NAME> | statistics interface <LAG-NAME> | statistics interface <VLAN-NAME>]`

### Description

Shows all VRRP virtual routers information.

| Parameter | Description |
|---|---|
| `brief` | Displays brief output of all VRRP virtual routers<br>Keywords used in displayed information:<br>Grp: VRRP virtual router group ID.<br>A-F: Address Family.<br>Pri: Priority.<br>Time: Uptime of VRRP virtual router since it moved out of INIT state.<br>Pre: Preempt mode (Y is enabled, N if not enabled). |
| `detail` | Displays detailed output of all VRRP virtual routers |
| `interface <INTERFACE-NAME>` | Displays VRRP information for a specific interface |
| `interface <LAG-NAME>` | Displays VRRP information for a specific LAG interface |
| `interface <VLAN-NAME>` | Displays VRRP information for a specific VLAN interface |
| `ipv4` | Displays IPv4 address family |
| `ipv6` | Displays IPv6 address family |
| `statistics` | Displays VRRP statistics information for all interfaces |
| `statistics interface <INTERFACE-NAME>` | Displays VRRP statistics information for a specific interface |
| `statistics interface <LAG-NAME>` | Displays VRRP statistics information for a specific LAG interface |
| `statistics interface <VLAN-NAME>` | Displays VRRP statistics information for a specific VLAN interface |

## Examples

```
switch# show vrrp

VRRP is enabled

Interface 1/1/1 - Group 1 - Address-Family IPv4
  State is MASTER
  State duration 56 mins 57.826 secs
  Virtual IP address is 10.0.0.1
  Virtual MAC address is 00:00:5e:00:01:01
  Advertisement interval is 1000 msec
  Preemption enabled
  Priority is 100
  BFD is enabled
  Master Router is 10.0.0.2 (local), priority is 100
  Master Advertisement interval is 1000 msec
  Master Down interval is unknown
  Tracked object ID is 1, and state Down

Interface 1/1/2 - Group 1 - Address-Family IPv4
  State is INIT (Interface Down)
  State duration 45 mins 28.313 secs
  Virtual IP address is no address
  Virtual MAC address is 00:00:5e:00:01:01
```

```
  Advertisement interval is 1000 msec
  Preemption enabled
  Priority is 100
  BFD is disabled
  Master Router is unknown, priority is unknown
  Master Advertisement interval is unknown
  Master Down interval is unknown
```

```
switch# show vrrp brief
 Interface   Grp  A-F   Pri   Time Owner Pre State   Master addr/Group addr
  1/1/1       1   IPv4  100    0    N    Y   MASTER  10.0.0.2(local) 10.0.0.1
  1/1/2       1   IPv4  100    0    N    Y   INIT    AF-UNDEFINED no address
  1/1/2       1   IPv6  100    0    N    Y   INIT    AF-UNDEFINED no address
```

```
switch# show vrrp detail

VRRP is enabled

Interface 1/1/1 - VRRPv2 Statistics
   Invalid group ID packets received : 0
   Invalid version packets received : 0
   Invalid checksum packets received : 0

Interface 1/1/1 - VRRPv3 Statistics
   Invalid group ID packets received : 0
   Invalid version packets received : 0
   Invalid checksum packets received : 0

Interface 1/1/1 - Group 1 - Address-Family IPv4
  State is MASTER
  State duration 1 mins 35.486 secs
  Virtual IP address is 10.0.0.1
  Virtual MAC address is 00:00:5e:00:01:01
  Advertisement interval is 1000 msec
  Version 3
  Preemption enabled
  Priority is 100
  BFD is enabled
  Master Router is 10.0.0.2 (local), priority is 100
  Master Advertisement interval is 1000 msec
  Master Down interval is unknown
  Tracked object ID is 1, and state Down
  VRRPv3 Advertisements: sent 3931 (errors 0) - rcvd 0
  VRRPv2 Advertisements: sent 0 (errors 0) - rcvd 0
  Group Discarded Packets: 3537
    IP address Owner conflicts: 0
    IP address configuration mismatch : 3537
    Advert Interval errors : 0
    Adverts received in Init state: 0
    Invalid group other reason: 0
  Group State transition:
    Init to master: 0
    Init to backup: 2 (Last change Mon Jun 16 11:19:36.316 UTC)
    Backup to master: 2 (Last change Mon Jun 16 11:19:39.926 UTC)
    Master to backup: 0
    Master to init: 1 (Last change Mon Jun 16 11:17:49.978 UTC)
    Backup to init: 0
```

```
switch# show vrrp interface 1/1/1

VRRP is enabled

Interface 1/1/1 - Group 1 - Address-Family IPv4
  State is MASTER
  State duration 11 mins 21.617 secs
  Virtual IP address is 10.0.0.1
  Virtual MAC address is 00:00:5e:00:01:01
  Advertisement interval is 1000 msec
  Version 3
  Preemption enabled
  Priority is 100
  BFD is enabled
  Master Router is 10.0.0.2 (local), priority is 100
  Master Advertisement interval is 1000 msec
  Master Down interval is unknown
```

```
switch# show vrrp interface lag10

VRRP is enabled

Interface lag10 - Group 1 - Address-Family IPv4
  State is MASTER
  State duration 11 mins 21.617 secs
  Virtual IP address is 10.0.0.1
  Virtual MAC address is 00:00:5e:00:01:01
  Advertisement interval is 1000 msec
  Version 3
  Preemption enabled
  Priority is 100
  BFD is enabled
  Master Router is 10.0.0.2 (local), priority is 100
  Master Advertisement interval is 1000 msec
  Master Down interval is unknown
```

```
switch# show vrrp interface vlan100

VRRP is enabled

Interface vlan100 - Group 1 - Address-Family IPv4
  State is MASTER
  State duration 11 mins 21.617 secs
  Virtual IP address is 10.0.0.1
  Virtual MAC address is 00:00:5e:00:01:01
  Advertisement interval is 1000 msec
  Version 3
  Preemption enabled
  Priority is 100
  BFD is enabled
  Master Router is 10.0.0.2 (local), priority is 100
  Master Advertisement interval is 1000 msec
  Master Down interval is unknown
```

```
switch# show vrrp statistics

VRRP is enabled
```

```
Interface 1/1/1 - VRRPv2 Statistics
   Invalid group ID packets received : 0
   Invalid version packets received : 0
   Invalid checksum packets received : 0

Interface 1/1/1 - VRRPv3 Statistics
   Invalid group ID packets received : 0
   Invalid version packets received : 0
   Invalid checksum packets received : 0

VRRP Statistics for interface 1/1/1 - Group 1 - Address-Family IPv4
  State is MASTER
  State duration 6 mins 55.006 secs
  VRRPv3 Advertisements: sent 4288 (errors 0) - rcvd 0
  VRRPv2 Advertisements: sent 0 (errors 0) - rcvd 0
  Group Discarded Packets: 3856
    IP address Owner conflicts: 0
    IP address configuration mismatch : 0
    Advert Interval errors : 0
    Adverts received in Init state: 0
    Invalid group other reason: 0
  Group State transition:
    Init to master: 0
    Init to backup: 2 (Last change Mon Jun 16 11:19:36.316 UTC)
    Backup to master: 2 (Last change Mon Jun 16 11:19:39.926 UTC)
    Master to backup: 0
    Master to init: 1 (Last change Mon Jun 16 11:17:49.978 UTC)
    Backup to init: 0

Interface 1/1/2 - VRRPv2 Statistics
   Invalid group ID packets received : 0
   Invalid version packets received : 0
   Invalid checksum packets received : 0

Interface 1/1/2 - VRRPv3 Statistics
   Invalid group ID packets received : 0
   Invalid version packets received : 0
   Invalid checksum packets received : 0

VRRP Statistics for Interface 1/1/2 - Group 1 - Address-Family IPv4
  State is INIT (No Primary Group Address)
  State duration 54 mins 43.027 secs
  VRRPv3 Advertisements: sent 0 (errors 0) - rcvd 0
  VRRPv2 Advertisements: sent 0 (errors 0) - rcvd 0
  Group Discarded Packets: 0
    IP address Owner conflicts: 0
    Invalid address count: 0
    IP address configuration mismatch : 0
    Advert Interval errors : 0
    Adverts received in Init state: 0
    Invalid group other reason: 0
  Group State transition:
    Init to master: 0
    Init to backup: 0
    Backup to master: 0
    Master to backup: 0
    Master to init: 0
    Backup to init: 0

VRRP Statistics for Interface 1/1/2 - Group 1 - Address-Family IPv6
  State is INIT (Interface Down)
  State duration 29 mins 34.508 secs
```

```
  VRRPv3 Advertisements: sent 0 (errors 0) - rcvd 0
  VRRPv2 Advertisements: sent 0 (errors 0) - rcvd 0
  Group Discarded Packets: 0
    IP address Owner conflicts: 0
    IP address configuration mismatch : 0
    Advert Interval errors: 0
    Adverts received in Init state: 0
    Invalid group other reason: 0
  Group State transition:
    Init to master: 0
    Init to backup: 0
    Backup to master: 0
    Master to backup: 0
    Master to init: 0
    Backup to init: 0
```

```
switch# show vrrp statistics interface 1/1/1

VRRP is enabled

Interface 1/1/1 - VRRPv2 Statistics
   Invalid group ID packets received : 0
   Invalid version packets received : 0
   Invalid checksum packets received : 0

Interface 1/1/1 - VRRPv3 Statistics
   Invalid group ID packets received : 0
   Invalid version packets received : 0
   Invalid checksum packets received : 0

VRRP Statistics for interface 1/1/1 - Group 1 - Address-Family IPv4
  State is MASTER
  State duration 6 mins 55.006 secs
  VRRPv3 Advertisements: sent 4288 (errors 0) - rcvd 0
  VRRPv2 Advertisements: sent 0 (errors 0) - rcvd 0
  Group Discarded Packets: 3856
    IP address Owner conflicts: 0
    IP address configuration mismatch : 0
    Advert Interval errors : 0
    Adverts received in Init state: 0
    Invalid group other reason: 0
  Group State transition:
    Init to master: 0
    Init to backup: 2 (Last change Mon Jun 16 11:19:36.316 UTC)
    Backup to master: 2 (Last change Mon Jun 16 11:19:39.926 UTC)
    Master to backup: 0
    Master to init: 1 (Last change Mon Jun 16 11:17:49.978 UTC)
    Backup to init: 0
```

```
switch# show vrrp statistics interface lag10

VRRP is enabled

Interface lag10 - VRRPv2 Statistics
   Invalid group ID packets received : 0
   Invalid version packets received : 0
   Invalid checksum packets received : 0

Interface lag10 - VRRPv3 Statistics
   Invalid group ID packets received : 0
```

```
   Invalid version packets received : 0
   Invalid checksum packets received : 0

VRRP Statistics for interface lag10 - Group 1 - Address-Family IPv4
  State is MASTER
  State duration 6 mins 55.006 secs
  VRRPv3 Advertisements: sent 4288 (errors 0) - rcvd 0
  VRRPv2 Advertisements: sent 0 (errors 0) - rcvd 0
  Group Discarded Packets: 3856
    IP address Owner conflicts: 0
    IP address configuration mismatch : 0
    Advert Interval errors : 0
    Adverts received in Init state: 0
    Invalid group other reason: 0
  Group State transition:
    Init to master: 0
    Init to backup: 2 (Last change Mon Jun 16 11:19:36.316 UTC)
    Backup to master: 2 (Last change Mon Jun 16 11:19:39.926 UTC)
    Master to backup: 0
    Master to init: 1 (Last change Mon Jun 16 11:17:49.978 UTC)
    Backup to init: 0
```

```
switch# show vrrp statistics interface vlan100

VRRP is enabled

Interface vlan100 - VRRPv2 Statistics
   Invalid group ID packets received : 0
   Invalid version packets received : 0
   Invalid checksum packets received : 0

Interface vlan100 - VRRPv3 Statistics
   Invalid group ID packets received : 0
   Invalid version packets received : 0
   Invalid checksum packets received : 0

VRRP Statistics for interface vlan100 - Group 1 - Address-Family IPv4
  State is MASTER
  State duration 6 mins 55.006 secs
  VRRPv3 Advertisements: sent 4288 (errors 0) - rcvd 0
  VRRPv2 Advertisements: sent 0 (errors 0) - rcvd 0
  Group Discarded Packets: 3856
    IP address Owner conflicts: 0
    IP address configuration mismatch : 0
    Advert Interval errors : 0
    Adverts received in Init state: 0
    Invalid group other reason: 0
  Group State transition:
    Init to master: 0
    Init to backup: 2 (Last change Mon Jun 16 11:19:36.316 UTC)
    Backup to master: 2 (Last change Mon Jun 16 11:19:39.926 UTC)
    Master to backup: 0
    Master to init: 1 (Last change Mon Jun 16 11:17:49.978 UTC)
    Backup to init: 0
```

## Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

## Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

# shutdown

```
shutdown
no shutdown
```

## Description

Disables VRRP group operation.

The `no` form of this command enables VRRP group operation.

## Examples

Disabling VRRP group operation:

```
switch(config)# interface 1/1/1
switch(config-if)# vrrp 1 address-family ipv4
switch(config-if-vrrp)# shutdown

switch(config)# interface 1/1/1
switch(config-if)# vrrp 1 address-family ipv4
switch(config-if-vrrp)# no shutdown
```

## Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

## Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config` | Administrators or local user group members with execution rights for this command. |

# timers advertise

```
timers advertise <ADVERTISE-IN-MILLISECONDS>
no timers advertise
```

## Description

Sets the advertisement interval in ms (100-40950). The default value is 1000. Advertisement interval can be configured in multiples of 1,000 ms.

The `no` form of this command sets the advertisement interval in ms to the default value of 1000.

This release does not support sub-second timer for VRRPv3.

### Examples

Setting the advertisement interval in ms to 2000:

```
switch(config)# interface 1/1/1
switch(config-if)# vrrp 1 address-family ipv4
switch(config-if-vrrp)# timers advertise 2000
```

Setting the advertisement interval in ms to the default value of 1000:

```
switch(config)# interface 1/1/1
switch(config-if)# vrrp 1 address-family ipv4
switch(config-if-vrrp)# no timers advertise
```

### Command History

| Release | Modification |
| --- | --- |
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
| --- | --- | --- |
| 6200 | `config` | Administrators or local user group members with execution rights for this command. |

## track (VRRP group)

```
track <OBJECT-ID>
no track <OBJECT-ID>
```

### Description

Sets the track object ID (1-128) for the group. The track object is first configured globally for the interface and then attached to the VRRP virtual router.

The track object must not track the same interface for which a VRRP group is configured.

The `no` form of this command removes the track object ID from the group.

### Examples

Setting the track object ID for the group:

```
switch(config)# interface 1/1/1
switch(config-if)# vrrp 1 address-family ipv4
switch(config-if-vrrp)# track 1
```

Removing the track object ID from the group:

```
switch(config)# interface 1/1/1
switch(config-if)# vrrp 1 address-family ipv4
switch(config-if-vrrp)# no track 1
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config` | Administrators or local user group members with execution rights for this command. |

## track (VRRP virtual router)

`track <OBJECT-ID>`
`no track <OBJECT-ID>`

### Description

Configures a track object that can be associated with an interface. A change in interface state will then affect the priority of a VRRP group. By default, no interface is associated to a track object, so state is down.

The `no` form of this command deletes a tracked object for an interface. If it is not associated with a VRRP virtual router, a track object cannot be deleted.

Track cannot be configured by using port with no routing.

| Parameter | Description |
|---|---|
| *<OBJECT-ID>* | Specify the track object ID value. Range: 1 to 128. |

### Examples

Configuring a tracked object:

```
switch(config)# track 1
```

Deleting a tracked object:

```
switch(config)# no track 1
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

## Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config` | Administrators or local user group members with execution rights for this command. |

# track by

```
track by <OBJECT-ID>
no track by <OBJECT-ID>
```

## Description

Specifies an interface to be tracked when changes in the state of the interface affect the priority of a VRRP group. Once track is associated with an interface, the track state reflects the interface forwarding state.

The `no` form of this command removes an interface from tracking, affecting VRRP states of any interfaces associated with VRRP groups.

The VLAN interface 1 is always tracked.

| Parameter | Description |
|---|---|
| *<OBJECT-ID>* | Specifies the track object ID value. Range: 1 to 128. |

## Example

Specifying an interface to be tracked:

```
switch (config)# interface 1/1/1
switch (config-if)# track by 1
```

## Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

## Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config` | Administrators or local user group members with execution rights for this command. |

# version

```
version <VERSION-NUMBER>
```

## Description

Sets the protocol version for the VRRP group. Version change is allowed only for the IPv4 address-family. The default value is 2, which supports IPv4 with minimum 1 second advertisement interval. Value 3 supports IPv4 and IPv6 with minimum 1 second advertisement interval.

| Parameter | Description |
|---|---|
| *<VERSION-NUMBER>* | Specifies the VRRP protocol version. Possible values: 2 or 3. The default value is 2, which supports IPv4 with a minimum 1 second advertisement interval. |

### Example

Setting the protocol version for the VRRP group to 3:

```
switch(config)# interface 1/1/1
switch(config-if)# vrrp 1 address-family ipv4
switch(config-if-vrrp)# version 3
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config` | Administrators or local user group members with execution rights for this command. |

## vrrp

```
vrrp <VRID> address-family {ipv4  | ipv6}
no vrrp <VRID> address-family {ipv4  | ipv6}
```

### Description

Creates a VRRP group and establishes VRRP group configuration context.

- A maximum of 16 VRRP groups, including both IPv4 and IPv6, are supported on an interface.
- A maximum of 511 VRRP groups is supported on a router. The groups can be IPv4 or IPv6 as per first come first serve basis.

The `no` form of this command deletes a VRRP group.

| Parameter | Description |
|---|---|
| *<VRID>* | Selects the VRRP router ID value. Range: 1 to 255. |
| `address-family [IPv4 | IPv6]` | Specifies which address family to use, IPv4 or IPv6. |

### Examples

```
switch(config)# interface 1/1/1
switch(config-if)# routing
switch(config-if-vrrp)# vrrp 1 address-family-ipv4
switch(config-if-vrrp)#

switch(config-if)# no vrrp 1 address-family-ipv4
```

## Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

## Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config` | Administrators or local user group members with execution rights for this command. |

# Chapter 9 Policy Based Routing (PBR)

PBR is not supported on the 4100i Switch Series.

PBR is not supported on the 6000 and 6100 Switch Series.

Policy Based routing (PBR) lets you manipulate the path of a packet based on the various attributes of the packet. Packets that can be manipulated by PBR are packets that are already routing through the system at Layer 3, with a destination IP address that is on a network other than the packet ingress Layer 3 interface. PBR is an extension of the existing classifier policy system where traffic to be manipulated is matched by a classifier **class**, and policy **actions** to be executed on the matching traffic. Matching traffic with the same destination can be routed over different paths so that different types of traffic, such as VoIP or traffic with special security requirements, can be better managed.

PBR's ability to influence a packet path is limited to the current router in that the next router to which the traffic is redirected makes an independent decision on where to forward traffic to next, if anywhere. Packets not matched by a PBR policy entry are not affected and take routes specified in the system route table.

## PBR actions

- **interface null:** equivalent to the policy **drop** policing action. Any packets matching the class criteria for that policy entry will be dropped and not routed any further.
- **nexthop:** allows for overriding the routing table's longest prefix match next-hop router for matching packets. If no such routing table entry exists for matching packets, (default or not) this action still affects matching packets.
- **default-nexthop:** allows for specifying a next-hop router for matching packets when there is no longest prefix match for those packets in the routing table. Such a default-nexthop overrides a system default route if already configured and also applies if there is no system default route.

- Next-hop and default-nexthop facilitate routing matching packets where they otherwise might not be, due to the absence of routing table entries.
- Unlike next-hop, default-nexthop only applies if there is no destination lookup match in the main routing table for matching packets.

## PBR policy action and action list

Entries in a policy use a class to specify the criteria on what packets to match and a set of one or more policy actions to take on the matching packets such as dscp or mirror.

PBR is configured for use as another such policy action in a policy entry by using the `pbr` keyword followed by the name of a user configured PBR action list. This list may contain up to eight PBR action entries which are listed according to priority by sequence number. The PBR actions section lists the different action entry choices.

In a given action list applied to an interface, zero entries may be available (that is, reachable). Of the entries that are available, only one can be active for that action list. Due to real-time changes in the network operating environment, list entry availability can change at any moment. At any given time, the highest priority available entry in the list is selected as the active PBR action for that list. If the active entry becomes unavailable, the next highest available entry is automatically selected and replaces that now-unavailable entry as the active entry. If a higher priority entry (other than the current active entry) becomes available, this entry is selected and replaces the current active entry, as the new active entry (even though that current entry is still available). If no entries are available or the only available entry becomes unavailable, routing occurs based on the routing table and is not influenced by PBR.

The **interface null** action is always considered available. If there are no higher priority PBR actions available, it will be selected as the active entry when configured in a PBR action list, and applied to an interface.

## PBR action list maximum entries

There is a limit of eight entries per action list and a maximum of 16 action lists in the system. Only one entry from an action list can be 'active' at any given moment, so the maximum number of unique active entries in the system is also 16.

For example, a given next-hop specified in an action list applied to an interface counts against the limit once. If that same next-hop address is entered in a different action list and applied to another interface in the same VRF, this next-hop is not unique in the VRF and does not affect the limit further. If one of those action lists (that specify the address) is applied to an interface in a different VRF, the entry will affect the limit since addresses are unique to each VRF.

Another example is if two PBR action lists with eight unique next-hop entries (for a total of 16 unique next-hops) are configured as PBR actions in a policy. If that policy is applied to 32 route-only ports (each of which is in a different VRF), the entry supply will be exhausted.

- PBR action list maximum capacity is 16.

## IP versions in an action list

Configuring PBR actions of different IP versions in the same PBR action list is not supported. This limitation applies to the next-hop and default-nexthop actions specifically. If an action of one IP version is already configured, configuring an action with a different IP version in the same action list will be blocked. This limitation applies regardless of whether the action list is part of a policy, applied to an interface or whether any of the entries are selected or not.

Furthermore, use cases similar to the following are not supported:

- Applying a PBR action list with entries of one IP version (for example IPv4 next-hop) to an interface of the other IP version (for example a route-only port with an IPv6 address only).
- Creating a policy entry with a class of one IP version and a PBR action list with entries of the other IP version.

Configuration mismatches while not prevented are not supported and behavior is not defined. At best, traffic will not reach its intended destination.

# Specifying valid next-hop and default-nexthop addresses

PBR does not support remote routers as next-hop or default-next-hop routers (a.k.a recursive). PBR only supports routers that are on a directly connected network. During configuration you may specify any IPv4 or IPv6 address as a next-hop or default-nexthop, however only those addresses reachable through a directly connected network will be installed as the active PBR entry.

If a specified address is reachable from the router, it must be on a directly connected subnet or it will not become active.

# Hardware path PBR versus software path PBR

Traffic to be routed that cannot be handled by the switching ASIC (for example IPv4 packets with IP options), is handled by the switch operating system kernel routing software.

The characteristics of network usage determine the proportion of traffic handled by software path PBR relative to the hardware path. Under normal conditions, software path PBR is likely to range from a fraction of a percent of all traffic to almost none.

Packets with IP options are sent to the hardware ASIC as well as the CPU. Since the software path PBR actions are applied to the packets in the CPU, a default entry is programmed in the hardware to take no action on those packets. This hardware entry count can be seen in diagnostics with the command `diag-dump policy basic`.

# Hardware versus software path for default-nexthop action

When a policy containing an entry with PBR is applied to an interface and the active action for that entry action list is 'default-nexthop', packets that match the class criteria and have no explicit route table hit (that is, no destination address longest prefix match, a.k.a 'route-miss'), they are forwarded to the specified active PBR default-nexthop. This is true for packets that follow the hardware and software paths through the switch.

There is a difference between PBR hardware and software path behavior when a route table hit occurs for class-matched packets (when the policy entry is a PBR default-nexthop).

Hardware path match criteria for a PBR policy entry with default-nexthop is extended to include the route table miss along with the class qualifiers, resulting in a **policy entry hit** for that policy entry. Conversely when there is a route table hit, the result is a policy entry miss in hardware path. When a policy entry miss occurs, policy processing moves on to the next entry in the policy and takes whatever action is specified, if any exist. This can include a different PBR routing action including interface null or no PBR action at all, for example.

In software path, the class match criteria is the only criteria required to achieve a policy entry hit. When that occurs, policy processing will stop. When there is a class match and a route table hit (with PBR action default-nexthop), the packet is forwarded according to that route table entry, not the PBR default-nexthop entry, nor is it influenced by any subsequent policy entries.

This difference in behavior is due to a limitation in the software path matching and routing engine, relative to hardware.

**Table 1:** *default-nexthop behaviors*

| Routing packet | System route miss | System route hit |
|---|---|---|
| Hardware | Take PBR default-nexthop route. | Policy entry miss, continue policy processing with next entry, if present. No further matches result in system default route being used, if present. |
| Software | Take PBR default-nexthop route. | Policy entry hit is overridden by system route hit. System route entry used, policy processing stops. |

## Software path and system default route

AOS-CX default behavior (for packets with destination networks that lack entries in the system route table with a reachable next-hop address), sends packets to the system CPU for special handling, if any are specified (for example, sending an ICMP unreachable reply message).

A side effect of this CPU routing is that it has higher priority than an applied PBR next-hop action. Even though a policy is applied with an entry that matches the traffic and specifies a PBR next-hop action which is reachable, the traffic will still be routed to the system CPU (due to the absence of a reachable next-hop in the system route table), and not through the desired PBR hardware path. With traffic routing to the system CPU, it will be properly routed by PBR software path, but it will also be rate limited by the control plane policing feature. Loss can occur at higher traffic rates.

The workaround for this issue is to create a default next-hop route in the system with a reachable next-hop router/host. This will result in a route hit, or reachable next-hop detected for the traffic with no further need to route traffic to the CPU. The PBR hardware path next-hop action will then occur, as desired.

## PBR, ECMP, and routing protocols

When a policy with PBR actions is applied to an interface, the highest priority action from the action list is applied to matching traffic on that interface and overrides any static, ECMP or dynamically installed routes in the system.

## PBR, VSX, and VLAN ACLs

Multichassis Link Aggregation Group (MCLAG) with VSX is a high-availability feature where a switch containing LAG members is connected to multiple switches to allow for node-level redundancy on that link. If one of the other switches goes down the LAG remains up and can continue to carry all the LAG traffic, bandwidth permitting.

A LAG (and an MCLAG) can be a member of a VLAN and a Layer 3 VSI (virtual switch interface), which can be created for that VLAN for the purposes of routing a policy with Layer 3 specific actions (that is, PBR), which can therefore be applied to that interface to influence routing decisions for matching Layer 3 packets on the MCLAG; like any other route-only port or VSI. Under such a configuration, it is likely that it is desirable to use VLAN ACLs applied to the VLAN the MCLAG is a member of.

There is a limit with this particular combination when the VLAN ACL specifies both IPv4 and IPv6 entries and the PBR policy has entries with IPv4 and IPv6 classes. This configuration will exhaust the switching ASIC resources and will fail to apply. To achieve a similar configuration, you may use port ACLs for IPv4 traffic (applying the ACL to all ports individually in the VLAN) while preserving the rest of the configuration.

# PBR software path, VSX, and VRRP

Due to the dynamic nature of the VSX and VRRP protocols, applying a policy with PBR to VSX or VRRP Layer 3 kernel interfaces and then making further changes to those protocols may not result in expected behavior. Apply policies with PBR to VSX and VRRP Layer 3 interfaces after all changes have been made. If further changes are necessary the policy should be removed first and reapplied when configuration updates are complete.

# PBR and next-hop router reachability

When a policy with PBR actions is applied to an interface, all next-hop and default-nexthop entries in the associated action list are continuously monitored for reachability. Probing occurs every 5 seconds for each such entry and guarantees that the reaction to the loss of reachability, or detection of the new reachability of any such entry will be approximately 5 seconds.

For example, if there are two reachable next-hops in an applied action list and the active next-hop (highest priority by lowest sequence number) entry becomes unreachable (loss of link or next-hop power event for example), the switch over to the lower priority next-hop will occur within approximately 5 seconds. If there is only one nexthop/default-nexthop entry and no entries of any other type in the action list, routing decisions will return to the system routing table.

Conversely if an applied action list with a single unreachable next-hop becomes reachable, the switch back to routing to that next-hop will occur when reachability of the next-hop on the network is achieved. On an action list with multiple next-hops (where the active next-hop is not the highest priority entry); once the next-hop with the higher priority becomes available, the now reachable next-hop will promote to active once new reachability on the network is achieved.

# CLI errors

**Table 1:** *Error messages and triggering events*

| Message | Event |
|---|---|
| `Failed to create action list` | A PBR action list create operation was unable to create the OVSDB row. |
| `PBR action list '%s' doesn't exist` | A PBR action list update or delete operation was unable to locate the list OVSDB row.<br>OR<br>A PBR action list entry create, update, or delete operation was unable to locate its parent action list OVSDB row. |
| `Unable to automatically set sequence number` | The automatically generated sequence number for a new PBR action list entry is higher than the maximum permitted sequence number. |

| Message | Event |
|---|---|
| `Unable to add PBR action list entry` | A PBR action list entry create failed to create an entry row in OVSDB. |
| `Configuration of IPv4 and IPv6 addresses in the same PBR action list is not supported` | An attempt to create both IPv4 and IPv6 addresses in the same action list was made. |
| `Invalid sequence number` | A PBR action list entry delete did not specify the entry sequence number. |
| `Cannot add entry because its automatic sequence number would exceed the maximum` | The automatic sequence number exceeded the maximum. |
| `Action list entry does not exist` | A PBR action list entry delete could not locate the OVSDB entry row. |
| `Action list name required` | A PBR action list delete was passed an invalid list name. |
| `Invalid action list name length` | A PBR action list name was longer than the permitted length. |

# PBR commands

## apply policy

```
apply policy <POLICY-NAME> routed-in
no apply policy <POLICY-NAME> routed-in
```

### Description

Applies a classifier policy containing a PBR action to an interface. A policy with PBR actions is only applicable to L3/routing interfaces.

The `no` form of this command removes a classifier policy containing a PBR action from an interface.
`config-if`

| Parameter | Description |
|---|---|
| *<POLICY-NAME>* | Specifies name of the policy. |

### Restrictions

- Layer 3 LAGs are not supported on Aruba 6200 series switches. PBR poliices cannot be applied to Layer 3 LAGs.

### Usage

To use route-only ports (ROPs) as Layer 3 interfaces, an internal VLAN range must be configured first. A policy with PBR actions can be applied to ROPs.

### Example

Apply policy to an interface:

```
switch(config)# interface 1/1/10
switch(config-if)# routing
switch(config-if)# apply policy pbr_policy routed-in
switch(config-if)# exit
```

### Command History

| Release | Modification |
|---|---|
| 10.08 | PBR support added for 6200 Switch Series. |
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config-if` | Administrators or local user group members with execution rights for this command. |

## pbr-action-list

```
pbr-action-list <ACTION-LIST-NAME>

    [<SEQUENCE-NUMBER>]
      {nexthop | default-nexthop} <NEXT-HOP-IP-ADDR>
       interface null

    no [<SEQUENCE-NUMBER>]
      {nexthop | default-nexthop} <IP-ADDR>
       interface null

no pbr-action-list <ACTION-LIST-NAME>
```

### Description

Creates a PBR action list or modifies its entries.

The `no` form of this command can be used to delete an action list or an individual action list entry.

| Parameter | Description |
|---|---|
| *<ACTION-LIST-NAME>* | Specifies the action list name. An action list name can be 1 to 64 alphanumeric characters. |
| *<SEQUENCE-NUMBER>* | Specifies list entry sequence number. Range: 1-4294967295<br>{nexthop \| default-nexthop}<br>Selects a regular next-hop (`nexthop`) or a default next-hop (`default-nexthop`). These parameters specify the address of a next-hop router to forward traffic matched by a class under different conditions. |
| `nexthop` | Sets the next hop for routing the packet. |

| Parameter | Description |
|---|---|
| `default-nexthop` | Sets the next hop for routing the packet when there is no explicit route for its destination. |
| `<NEXTHOP-IP-ADDR>` | Specifies IPv4 or IPv6 address of the next-hop router. |
| `interface null` | Selects keyword interface: `null`. |
| `null` | Specifies to drop matching traffic. |

### Restrictions

Reachability of next-hop routers in the list is not guaranteed. Such reachability can change at any time due to the dynamic nature of the network environment.

### Usage

Each action list may contain up to four entries of three different entry types:

- `interface null`
- `nexthop`
- `default-nexthop`

List entries have a unique sequence number which, if not user specified, are automatically assigned beginning at 10 and continuing at intervals of 10 for each subsequent new list entry, for example 20, 30, and 40. Sequence numbers of any value can be specified manually, a different interval may be set, and new entries can be added to (or removed from) any location in the list at any time.

Specifying an existing sequence number causes the existing list entry to be replaced by the new details. The list entry with the lowest sequence number has the highest priority entry in the list. The sequence numbers may be renumbered with the pbr-action-list resequence command.

Only one next-hop router or interface from the list is used per packet matched. This router or interface is defined as the highest priority list entry that is reachable or available at the time of the traffic match. If the highest priority list entry next-hop router is reachable - that list entry is chosen, the search is stopped, and the traffic is forwarded to the next-hop router or interface for the entry. If the highest priority list entry next-hop router is not reachable, the next highest priority list entry reachability is determined and used if reachable, otherwise the process continues down the list. If none of the routers in the list are reachable, the packet may be dropped (through the null interface entry if configured) or forwarded according to a system route table entry.

An action list that contains a next-hop of one IP version cannot also contain an entry of another IP version. For example, an action list must contain only IPv4 or IPv6 next-hop addresses.

### Examples

The list name is included in the context prompt for easy current-list identification. Any list name over 10 characters will be truncated at 10 characters and terminated with the tilde character (~) to indicate a reduced list name display. This reduction affects the prompt display of the list name only:

```
switch(config)# pbr-action-list eighteenchars
switch(config-pbr-action-list-eighteench~)#
```

The following example creates an action list with two IPv4 next-hops, a default IPv4 next-hop, and a null interface. The example uses default sequence numbering for its list entries.

```
switch(config)# pbr-action-list test1
switch(config-pbr-action-list-test1)# nexthop 1.1.1.1
switch(config-pbr-action-list-test1)# nexthop 2.2.2.2
switch(config-pbr-action-list-test1)# default-nexthop 9.9.9.9
switch(config-pbr-action-list-test1)# interface null
switch(config-pbr-action-list-test1)# end

switch(config)# show pbr-action-list test1


           Name
  Sequence Type                            Address/Interface
-------------------------------------------------------------------
           test1
        10 nexthop                         1.1.1.1
        20 nexthop                         2.2.2.2
        30 default-nexthop                 9.9.9.9
        40 interface                       null
```

The following example creates an action list with an IPv4 next-hop and a null interface with manual sequence numbers for its entries.

```
switch(config)# pbr-action-list test2
switch(config-pbr-action-list-test2)# 6 default-nexthop 4.4.4.4
switch(config-pbr-action-list-test2)# 1 interface null
switch(config-pbr-action-list-test2)# exit

switch(config)# show pbr-action-list test2

           Name
  Sequence Type                            Address/Interface
-------------------------------------------------------------------
           test2
         1 interface                       null
         6 default-nexthop                 4.4.4.4
```

The following example creates an action list with two IPv6 next-hops and the null interface, with manual sequence numbers.

```
switch(config)# pbr-action-list test4
switch(config-pbr-action-list-test4)# 5 nexthop 2000:abcd::cccc:dddd
switch(config-pbr-action-list-test4)# 6 nexthop 1000:abcd::1234:5678
switch(config-pbr-action-list-test4)# 7 interface null
switch(config-pbr-action-list-test4)# end

switch(config)# show pbr-action-list test4

           Name
  Sequence Type                            Address/Interface
-------------------------------------------------------------------
           test4
         5 nexthop                         2000:abcd::cccc:dddd
         6 nexthop                         1000:abcd::1234:5678
         7 interface                       null
```

## Command History

| Release | Modification |
|---|---|
| 10.08 | PBR support added for 6200 Switch Series. |
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config`<br>The `pbr-action-list` *`<ACTION-LIST-NAME>`* command takes you into the `config-pbr-action-list-`*`<ACTION-LIST-NAME>`* context where you modify entries for a PBR action list. | Administrators or local user group members with execution rights for this command. |

## pbr-action-list copy

`pbr-action-list <ACTION-LIST-NAME> copy <DESTINATION-ACTION-LIST-NAME>`

### Description

Copies an existing PBR action list.

| Parameter | Description |
|---|---|
| *`<ACTION-LIST-NAME>`* | Specifies the action list name to be copied. |
| *`<DESTINATION-ACTION-LIST-NAME>`* | Specifies the name of the copied action list. A destination action list name can be 1 to 64 alphanumeric characters. |

### Examples

The following example copies test4 action list to test 5.

```
switch(config)# show pbr-action-list test4

           Name
  Sequence Type                            Address/Interface
-------------------------------------------------------------------
           test4
        5  nexthop                         2000:abcd::cccc:dddd
        6  nexthop                         1000:abcd::1234:5678
        7  interface                       null

switch(config)# pbr-action-list test4 copy test5
switch(config-pbr-action-list-test4)# show pbr-action-list test5

           Name
  Sequence Type                            Address/Interface
-------------------------------------------------------------------
           test4
        1  nexthop                         2000.abcd::cccc.dddd
```

```
        11  nexthop                         1000.abcd::1234.5678
        21  interface                       null
```

### Command History

| Release | Modification |
|---|---|
| 10.08 | PBR support added for 6200 Switch Series. |
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config` | Administrators or local user group members with execution rights for this command. |

## pbr-action-list resequence

`pbr-action-list <ACTION-LIST-NAME> resequence <STARTING-SEQUENCE-NUMBER> <INCREMENT>`

### Description

Renumbers the entries in an action list. The list entry with the lowest sequence number has the highest priority entry in the list.

| Parameter | Description |
|---|---|
| *<ACTION-LIST-NAME>* | Specifies the action list name to have its entries resequenced. |
| *<STARTING-SEQUENCE-NUMBER>* | Specifies the starting sequence number. Range: 1-4294967295 |
| *<INCREMENT>* | Specifies the increment of the resequencing. Range: 1-4294967295 |

### Examples

The following command shows how a PBR action list is resequenced. In the following example, an action list named `test4` is resequenced so that instead of its entries starting at 5 and being numbered sequentially, its entries start now at 1 and they are numbered in increments of 10:

```
switch(config)# show pbr-action-list test4

           Name
  Sequence Type                             Address/Interface
-------------------------------------------------------------------
           test4
         5  nexthop                         2000.abcd::cccc.dddd
         6  nexthop                         1000.abcd::1234.5678
         7  interface                       null

switch(config)# pbr-action-list test4 resequence 1 10
```

```
          Name
 Sequence Type                            Address/Interface
-----------------------------------------------------------------
          test4
        1 nexthop                         2000.abcd::cccc.dddd
       11 nexthop                         1000.abcd::1234.5678
       21 interface                       null
```

## Command History

| Release | Modification |
| --- | --- |
| 10.08 | PBR support added for 6200 Switch Series. |
| 10.07 or earlier | -- |

## Command Information

| Platforms | Command context | Authority |
| --- | --- | --- |
| 6200 | `config` | Administrators or local user group members with execution rights for this command. |

# pbr-action-list reset

`pbr-action-list <ACTION-LIST-NAME> reset`

## Description

Resets a specified PBR action list to its last successful configuration.

| Parameter | Description |
| --- | --- |
| *<ACTION-LIST-NAME>* | Specifies the action list name to be reset. |

## Examples

```
switch(config)# pbr-action-list test reset
```

## Command History

| Release | Modification |
| --- | --- |
| 10.08 | PBR support added for 6200 Switch Series. |
| 10.07 or earlier | -- |

## Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config` | Administrators or local user group members with execution rights for this command. |

## policy

```
policy <POLICY-NAME>

    [<SEQUENCE-NUMBER>]
      class {ip|ipv6|mac} <CLASS-NAME>
      action {<REMARK-ACTIONS> | <POLICE-ACTIONS> | <OTHER-ACTIONS>}
      [{<REMARK-ACTIONS> | <POLICE-ACTIONS> | <OTHER-ACTIONS>}]

    [<SEQUENCE-NUMBER>]
    comment ...

no policy <POLICY-NAME>
```

### Description

Creates, modifies, or deletes a classifier policy. A policy contains one or more policy entries ordered and prioritized by sequence numbers. Each entry has an IPv4/IPv6/MAC class and one or more policy actions associated with it. An applied policy processes a packet sequentially against policy entries in the list until the last entry in the list has been evaluated or the packet matches an entry. If a match occurs the related entry, actions are taken.

The `no` form of this command is used to delete a policy or an individual policy entry.

| Parameter | Description |
|---|---|
| *<POLICY-NAME>* | Specifies the name of the policy. |
| *<SEQUENCE-NUMBER>* | Specifies a sequence number for the policy entry. Optional. Range: 1 to 4294967295. |
| `comment` | Stores the remaining entered text as a policy entry comment. |
| `class {ip|ipv6|mac}` *<CLASS-NAME>* | Specifies a type of class, `ip` for IPv4, `ipv6` for IPv6 and `mac` for a MAC policy. And specifies a class name. |
| *<REMARK-ACTIONS>* | Remark actions can be any of the following options: `{pbr <ACTION-LIST> | pcp <PRIORITY> | ip-precedence <IP-PRECEDENCE-VALUE> | dscp <DSCP-VALUE> | local-priority <LOCAL-PRIORITY-VALUE>}` where:<br>`pbr` *<ACTION-LIST>*<br>Specifies the PBR action list to be used.<br>`pcp` *<PCP-VALUE>*<br>Specifies Priority Code Point (PCP) value. Range: 0 to 7.<br>`ip-precedence` *<IP-PRECEDENCE-VALUE>*<br>Specifies the numeric IP precedence value. Range: 0 to 7.<br>`dscp` *<DSCP-VALUE>*<br>Specifies a Differentiated Services Code Point (DSCP) value. Enter either a numeric value (0 to 63) or a keyword as follows:<br>`AF11` - DSCP 10 (Assured Forwarding Class 1, low drop probability)<br>`AF12` - DSCP 12 (Assured Forwarding Class 1, medium |

| Parameter | Description |
|---|---|
| | drop probability)<br>`AF13` - DSCP 14 (Assured Forwarding Class 1, high drop probability)<br>`AF21` - DSCP 18 (Assured Forwarding Class 2, low drop probability)<br>`AF22` - DSCP 20 (Assured Forwarding Class 2, medium drop probability)<br>`AF23` - DSCP 22 (Assured Forwarding Class 2, high drop probability)<br>`AF31` - DSCP 26 (Assured Forwarding Class 3, low drop probability)<br>`AF32` - DSCP 28 (Assured Forwarding Class 3, medium drop probability)<br>`AF33` - DSCP 30 (Assured Forwarding Class 3, high drop probability)<br>`AF41` - DSCP 34 (Assured Forwarding Class 4, low drop probability)<br>`AF42` - DSCP 36 (Assured Forwarding Class 4, medium drop probability)<br>`AF43` - DSCP 38 (Assured Forwarding Class 4, high drop probability)<br>`CS0` - DSCP 0 (Class Selector 0: Default)<br>`CS1` - DSCP 8 (Class Selector 1: Scavenger)<br>`CS2` - DSCP 16 (Class Selector 2: OAM)<br>`CS3` - DSCP 24 (Class Selector 3: Signaling)<br>`CS4` - DSCP 32 (Class Selector 4: Real time)<br>`CS5` - DSCP 40 (Class Selector 5: Broadcast video)<br>`CS6` - DSCP 48 (Class Selector 6: Network control)<br>`CS7` - DSCP 56 (Class Selector 7)<br>`EF` - DSCP 46 (Expedited Forwarding)<br>`local-priority` *<LOCAL-PRIORITY-VALUE>*<br>Specifies a local priority value. Range: 0 to 7. |
| *<POLICE-ACTIONS>* | Police actions can be the following `{cir` *<RATE-BPS>* `cbs` *<BYTES>* `exceed}` where:<br>`cir` *<RATE-BPS>*<br>Specifies a Committed Information Rate value in Kilobits per second. Range: 1 to 4294967295.<br>`cbs` *<BYTES>*<br>Specifies a Committed Burst Size value in bytes. Range: 1 to 4294967295.<br>`exceed`<br>Specifies action to take on packets that exceed the rate limit. |
| *<OTHER-ACTIONS>* | Other actions can be the following:<br>`drop`<br>Specifies drop traffic. |

## Restrictions

MAC classes are not applicable to policies containing PBR actions. Applying such policies to an interface are blocked.

## Usage

- For Policy Based Routing, the policy action keyword is `pbr` which itself takes the name of a PBR action list as a parameter.
- A policy entry that contains a PBR action can contain other action types as well.
- An applied policy processes a packet sequentially against policy entries in the list until the last policy entry in the list has been evaluated or the packet matches an entry.

Entering an existing `<POLICY-NAME>` value will cause the existing policy to be modified, with any new *`<SEQUENCE-NUMBER>`* value creating an additional policy entry, and any existing *`<SEQUENCE-NUMBER>`* value replacing the existing policy entry with the same sequence number.

- If no sequence number is specified, a new policy entry is appended to the end of the entry list with a sequence number equal to the highest policy entry currently in the list plus 10.

### Examples

Create a policy with two PBR actions:

```
switch(config)# policy pbr_policy
switch (config-policy)# 10 class ip v4_class action pbr action_list1
switch (config-policy)# 20 class ipv6 v6_class action pbr action_list2
switch (config-policy)# exit
```

### Command History

| Release | Modification |
|---|---|
| 10.08 | PBR support added for 6200 Switch Series. |
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config`<br>The `policy` command takes you into the `config-policy` context where you enter the policy entries. | Administrators or local user group members with execution rights for this command. |

## show pbr

`show pbr {interface <INTERFACE-NAME>|vrf <VRF-NAME>|summary}`

### Description

Shows a detailed view of Policy Based Routing (PBR) in the system.

| Parameter | Description |
|---|---|
| *`<VRF-NAME>`* | Specifies name of a VRF. |
| *`<INTERFACE-NAME>`* | Specifies an interface. Format: `member/slot/port`. |

## Usage

Show commands can only reference the default VRF.

## Examples

Showing PBR summary information when there is no active next-hop in the system:

```
switch# show pbr summary
VRF      Port   Policy    PBR      Seq  Type       Nexthop
--------------------------------------------------------------------
No active PBR nexthop found
--------------------------------------------------------------------
```

Showing PBR summary information when there are active next-hops in the system:

```
switch# show pbr summary

VRF      Port   Policy    PBR      Seq  Type       Nexthop
--------------------------------------------------------------------
default  1/1/1  policy_1  pbr_1    10   nexthop    1.1.1.1 (active)
         1/1/2  policy_2  pbr_2    20   nexthop    5.5.5.5 (active)
--------------------------------------------------------------------
```

## Command History

| Release | Modification |
|---|---|
| 10.08 | PBR support added for 6200 Switch Series. |
| 10.07 or earlier | -- |

## Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

# show pbr-action-list

`show pbr-action-list [<ACTION-LIST-NAME>] [commands] [configuration] [vsx-peer]`

## Description

Shows the current PBR action list configuration. Action list entries are displayed in ascending order of their sequence number.

| Parameter | Description |
|---|---|
| *<ACTION-LIST-NAME>* | Specifies the PBR action list name. |
| commands | Formats output as CLI commands. |

| Parameter | Description |
|---|---|
| `configuration` | Displays user-specified configuration. |
| `vsx-peer` | Displays VSX peer switch information. |

### Restrictions

If an action list entry is modified to an invalid value (for example through the REST interface), this command will indicate a mismatch for that action entry when run. In this event, use the `pbr-action-list <NAME> reset` command to restore it to the previous valid value.

### Usage

- This command does not indicate whether the action list is configured in a policy or applied to an interface. Use the `show pbr` command for PBR status involving action lists.
- A single action list is shown by specifying its name or you can show all action lists by omitting a name argument.
- Using the additional commands keyword, you can change the tabulated output to a configuration style output for single or all list display.

### Examples

Create two PBR action lists then run `show pbr-action-list` to display all configured action lists in the default configuration mode:

```
switch(config)# pbr-action-list v4_pbr
switch(config-pbr-action-list-v4_pbr)# 1 nexthop 1.1.1.1
switch(config-pbr-action-list-v4_pbr)# 5 default-nexthop 2.2.2.2
switch(config-pbr-action-list-v4_pbr)# 10 interface null
switch(config-pbr-action-list-v4_pbr)# exit
switch(config)#
switch(config)# pbr-action-list v6_pbr
switch(config-pbr-action-list-v6_pbr)# 20 nexthop 2000:abcd::cccc:dddd
switch(config-pbr-action-list-v6_pbr)# 40 default-nexthop 1000:abcd::1234:5678
switch(config-pbr-action-list-v6_pbr)# 60 interface null
switch(config-pbr-action-list-v6_pbr)# exit
switch#

switch# show pbr-action-list
           Name
           Additional PBR-Action-List Parameters
Sequence     Type                 Nexthop
------------------------------------------------------------------------------------
---
             v4_pbr
1            nexthop              1.1.1.1
5            default-nexthop      2.2.2.2
10           interface            null

             v6_pbr
20           nexthop              2000:abcd::cccc:dddd
40           default-nexthop      1000:abcd::1234:5678
60           interface            null
```

### Command History

| Release | Modification |
| --- | --- |
| 10.08 | PBR support added for 6200 Switch Series. |
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
| --- | --- | --- |
| 6200 | `config` | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show running-config current-context

`show running-config current-context`

### Description

Displays the configuration of the PBR action list in the current configuration context, in commands mode.

| Parameter | Description |
| --- | --- |
| `running-config` | Shows configuration currently running on switch. |
| `current-context` | Limits display to current config context only, in commands mode. |

### Usage

Useful for reexamining entries previously entered into the action list after its entries have scrolled off the terminal due to other output or upon reentering the context of an existing action list.

### Examples

Creating two PBR action lists and running `show running-configuration curent-context` to display the action list configuration in commands mode:

```
switch(config)# pbr-action-list v4_pbr
switch(config-pbr-action-list-v4_pbr)# 1 nexthop 1.1.1.1
switch(config-pbr-action-list-v4_pbr)# 5 default-nexthop 2.2.2.2
switch(config-pbr-action-list-v4_pbr)# 10 interface null
switch(config-pbr-action-list-v4_pbr)# exit
switch(config)#
switch(config)# pbr-action-list v6_pbr
switch(config-pbr-action-list-v6_pbr)# 20 nexthop 2000:abcd::cccc:dddd
switch(config-pbr-action-list-v6_pbr)# 40 default-nexthop 1000:abcd::1234:5678
switch(config-pbr-action-list-v6_pbr)# 60 interface null
switch(config-pbr-action-list-v6_pbr)#
switch(config-pbr-action-list-v6_pbr)# show running-config current-context
pbr-action-list v6_pbr
    20 nexthop 2000:abcd::cccc:dddd
    40 default-nexthop 1000:abcd::1234:5678
    60 interface null
```

Switching context back to the first actionl ist and running the same command:

```
switch(config-pbr-action-list-v6_pbr)# pbr-action-list v4_pbr
switch(config-pbr-action-list-v4_pbr)#
switch(config-pbr-action-list-v4_pbr)# show running-config current-context
pbr-action-list v4_pbr
    1 nexthop 1.1.1.1
    5 default-nexthop 2.2.2.2
    10 interface null
```

Removing action list entry number 5 and running the command again:

```
switch(config-pbr-action-list-v4_pbr)# no 5
switch(config-pbr-action-list-v4_pbr)# show running-config current-context
pbr-action-list v4_pbr
    1 nexthop 1.1.1.1
    10 interface null
```

### Command History

| Release | Modification |
|---|---|
| 10.08 | PBR support added for 6200 Switch Series. |
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config` | Administrators or local user group members with execution rights for this command. |

Key chain is not supported on the Aruba 6000 and 6100 Switch Series.

This chapter provides details for configuring and verifying the functions of a key chain. Key chain configurations are spread across `config` context, `keychain` context, and `keychain-key` context.

A key chain provides seamless authentication-key rollover. When authentication method is enabled, using key chain allows OSPF to overcome static key configuration to change the key periodically. Using key chain also reduces risks of keys being guessed by configuring rotating keys.

A key chain must be used by an application like OSPF that communicates by using the keys with its peers.

# Key chain commands

## accept-lifetime

```
accept-lifetime [start-time <time> <month>/<day>/<year>] {duration {<seconds> | infinite} |
end-time <time> <month>/<day>/<year>}
```

### Description

Configures the duration for which the key is valid for receiving packets.

The `no` form of this command configures the key packet receiving duration to the default value of an infinite time.

| Parameter | Description |
|---|---|
| `start-time` | Time at which the key chain lifetime starts. Required. Format: HH:MM:SS |
| `end-time` | Time at which the key chain lifetime expires. Required. Format: HH:MM:SS |
| `day` | Day of the month. Required. Range: 1-31. |
| `month` | Month of the year. Required. |
| `year` | Year. Required. Range: 2020-2050 |
| `duration` | Time in seconds. Optional. Range: 1-2147483646. |
| `infinite` | Specifies infinite time for the key. Optional. |

### Examples

Configuring the duration for which the key is valid for receiving packets:

```
switch# configure terminal
switch(config)# keychain ospf_keys
switch(config-keychain)# key 1
switch(config-keychain-key)# accept-lifetime start-time 10:10:10 10/25/2020 end-time
10:10:10 11/25/2020
switch(config-keychain-key)# accept-lifetime start-time 10:10:10 10/25/2020 duration
1000
switch(config-keychain-key)# accept-lifetime start-time 10:10:10 10/25/2020 duration
infinite
switch(config-keychain-key)# accept-lifetime end-time 10:10:10 11/25/2020
switch(config-keychain-key)# accept-lifetime duration 1000
switch(config-keychain-key)# accept-lifetime duration infinite
```

Configuring the key packet receiving duration to the default value of an infinite time:

```
switch# configure terminal
switch(config)# keychain ospf_keys
switch(config-keychain)# key 1
switch(config-keychain-key)# no accept-lifetime
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| All platforms | `config-keychain-key` | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## cryptographic-algorithm

*Not applicable to the 6000 or 6100 Switch Series.*

```
cryptographic-algorithm {hmac-sha-1 | hmac-sha-256 | hmac-sha-384 | hmac-sha-512 | md5}
no cryptographic-algorithm
```

### Description

Configures the cryptographic algorithm for the key. Choose one of the authentication algorithms from the following parameters.

The `no` form of this command configures the default cryptographic algorithm for a key, `md5`.

| Parameter | Description |
|---|---|
| `hmac-sha-1` | Sets the authentication algorithm for the key to SHA-1. |
| `hmac-sha-256` | Sets the authentication algorithm for the key to SHA-256. |
| `hmac-sha-384` | Sets the authentication algorithm for the key to SHA-384. |

| Parameter | Description |
|---|---|
| hmac-sha-512 | Sets the authentication algorithm for the key to SHA-512. |
| md5 | Sets the authentication algorithm for the key to md5. Maximum length of the key string supported: 16 bytes (config-if context), 64 bytes (config-keychain-key context). |

### Examples

Set the authentication algorithm for the key to SHA-384:

```
switch(config)# keychain ospf_keys
switch(config-keychain)# key 1
switch(config-keychain-key)# cryptographic-algorithm hmac-sha-384
```

Set the authentication algorithm to the default, md5:

```
switch(config)# keychain ospf_keys
switch(config-keychain)# key 1
switch(config-keychain-key)# no cryptographic-algorithm
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | config-keychain-key | Administrators or local user group members with execution rights for this command. |

## key

key *<KEY-ID>*

### Description

Creates the key for a key chain and enters the key chain key context. A maximum of 64 keys can be configured per key chain.

The no form of this command deletes the key from the key chain.

| Parameter | Description |
|---|---|
| *<KEY-ID>* | ID of the key. Required. Range: 1-255. |

### Examples

Creating a key for a key chain:

```
switch# configure terminal
switch(config)# keychain ospf_keys
switch(config-keychain)# key 1
```

Deleting a key from a key chain:

```
switch# configure terminal
switch(config)# keychain ospf_keys
switch(config-keychain)# no key 1
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| All platforms | `config-keychain` | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## keychain

`keychain <KEYCHAIN-NAME>`

### Description

Creates the key chain and enters the key chain context. A maximum of 64 key chains can be configured in the system.

The `no` form of this command removes the key chain if it is not used by any subscribers.

| Parameter | Description |
|---|---|
| *<KEYCHAIN-NAME>* | Name of the key chain. Required. |

### Examples

Creating a key chain:

```
switch# configure terminal
switch(config)# keychain ospf_keys
```

Removing a key chain:

```
switch# configure terminal
switch(config)# keychain ospf_keys
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| All platforms | `config` | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## key-string

`key-string [{ciphertext | plaintext} <PASSWORD>]`

### Description

Sets the key password. The password is internally stored in encrypted form. The key is not valid until its password has been set.

The `no` form of this command deletes the password used for the key.

| Parameter | Description |
|---|---|
| `ciphertext` | Specifies that the key password is provided as ciphertext. |
| `plaintext` | Specifies that the key password is provided as plaintext. |
| `<PASSWORD>` | Specifies the key password. |

When the key password is not provided on the command line, plaintext password prompting occurs upon pressing Enter. The entered password characters are masked with asterisks.

### Examples

Setting the key password with plaintext:

```
switch(config)# keychain ospf_keys
switch(config-keychain)# key 1
switch(config-keychain-key)# key-string plaintext F82#450bHP
```

Setting the key password with plaintext prompting:

```
switch(config)# keychain ospf_keys
switch(config-keychain)# key 1
switch(config-keychain-key)# key-string
Enter the key password: *************
Re-Enter the key password: *************
```

Setting the key password with ciphertext:

```
switch(config)# keychain ospf_keys
switch(config-keychain)# key 1
switch(config-keychain-key)# key-string ciphertext AQBpfciFZ/P...biAAAOjc0a8=
```

Deleting the password for the key:

```
switch(config)# keychain ospf_keys
switch(config-keychain)# key 1
switch(config-keychain-key)# no key-string
```

### Command History

| Release | Modification |
| --- | --- |
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
| --- | --- | --- |
| All platforms | `config-keychain-key` | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## send-lifetime

`send-lifetime [start-time <time> <month>/<day>/<year>] {duration {<seconds> | infinite} | end-time <time> <month>/<day>/<year>}`

### Description

Configures the duration for which the key is valid for sending packets.

The `no` form of this command configures the key packet sending duration to the default value of an infinite time.

| Parameter | Description |
| --- | --- |
| `start-time` | Time at which the key chain lifetime starts. Required. Format: HH:MM:SS |
| `end-time` | Time at which the key chain lifetime expires. Required. Format: HH:MM:SS |
| `day` | Day of the month. Required. Range: 1-31. |
| `month` | Month of the year. Required. |
| `year` | Year. Required. Range: 2020-2050 |
| `duration` | Time in seconds. Optional. Range: 1-2147483646. |
| `infinite` | Specifies infinite time for the key. Optional. |

### Examples

Configuring the duration for which the key is valid for sending packets:

```
switch# configure terminal
switch(config)# keychain ospf_keys
switch(config-keychain)# key 1
switch(config-keychain-key)# send-lifetime start-time 10:10:10 10/25/2020 end-time
10:10:10 11/25/2020
switch(config-keychain-key)# send-lifetime start-time 10:10:10 10/25/2020 duration
1000
switch(config-keychain-key)# send-lifetime start-time 10:10:10 10/25/2020 duration
infinite
switch(config-keychain-key)# send-lifetime end-time 10:10:10 11/25/2020
switch(config-keychain-key)# send-lifetime duration 1000
switch(config-keychain-key)# send-lifetime duration infinite
```

Configuring the key packet sending duration to the default value of an infinite time:

```
switch# configure terminal
switch(config)# keychain ospf_keys
switch(config-keychain)# key 1
switch(config-keychain-key)# no send-lifetime
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| All platforms | `config-keychain-key` | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show capacities keychain

`show capacities keychain`

### Description

Shows the maximum number of key chains and keys configurable in a key chain.

### Example

```
switch# show capacities keychain

System Capacities: Filter Keychain
Capacities Name
                 Value
------------------------------------------------------------------------------------
-----------------------
Maximum number of keychains supported in the system
                  64
```

```
Maximum number of Keys supported in a single Keychain
                  64
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| All platforms | Operator (>) | Administrators or local user group members with execution rights for this command. |

## show keychain

`show keychain [<KEYCHAIN-NAME>]`

### Description

Shows information about configured and active keys of a named key chain or (if `keychain-name` is not specified) all configured key chains.

| Parameter | Description |
|---|---|
| *<KEYCHAIN-NAME>* | Name of the key chain. Optional. |

### Example

```
switch# show keychain
Keychain Name : ospf_keys
  Number of Keys      : 2
  Active Send Key ID  : 7
  Active Recv Key IDs : 7, 200

  Key ID              : 7
    Key string        : AQBapZ1OHiO9W3JwRqnjtLfbV73BPLS1S6TGVg+Lzl7N4e5eBAAAAPWaPBE=
    Send Key Validity : 00:00:01 10/1/2020 to 23:59:01 10/1/2021
    Recv Key Validity : 00:00:01 10/1/2020 to infinite
  Key ID              : 200
    Key string        : AQBapZ1OHiO9W3JwRqnjtLfbV73BPLS1S6TGVg+Lzl7N4e5eBAAAAPWaPBE=
    Send Key Validity : 00:00:01 10/1/2020 to 23:59:01 10/1/2021
    Recv Key Validity : 00:00:01 10/1/2020 to 23:59:01 10/1/2021

Keychain Name : bgp_keys
  Number of Keys      : 2
  Active Send Key ID  : 7
  Active Recv Key IDs : 7

  Key ID              : 7
    Key string        : AQBapZ1OHiO9W3JwRqnjtLfbV73BPLS1S6TGVg+Lzl7N4e5eBAAAAPWaPBE=
    Send Key Validity : 00:00:01 10/26/2020 to 23:59:01 10/1/2021
    Recv Key Validity : 00:00:01 10/22/2020 to infinite
```

```
  Key ID              : 8
    Key string        : AQBapZ1OHiO9W3JwRqnjtLfbV73BPLS1S6TGVg+Lzl7N4e5eBAAAAPWaPBE=
    Send Key Validity : 00:00:01 10/1/2021 to 23:59:01 10/1/2021
    Recv Key Validity : 00:00:01 10/1/2021 to 23:59:01 10/1/2021


Keychain Name : ospf_keys
  Number of Keys      : 2
  Active Send Key ID  : 7
  Active Recv Key IDs : 7, 200

  Key ID              : 7
    Key string        : AQBapZ1OHiO9W3JwRqnjtLfbV73BPLS1S6TGVg+Lzl7N4e5eBAAAAPWaPBE=
    Send Key Validity : 00:00:01 10/1/2020 to 23:59:01 10/1/2021
    Recv Key Validity : 00:00:01 10/1/2020 to infinite
  Key ID              : 200
    Key string        : AQBapZ1OHiO9W3JwRqnjtLfbV73BPLS1S6TGVg+Lzl7N4e5eBAAAAPWaPBE=
    Send Key Validity : 00:00:01 10/1/2020 to 23:59:01 10/1/2021
    Recv Key Validity : 00:00:01 10/1/2020 to 23:59:01 10/1/2021
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| All platforms | Operator (>) | Administrators or local user group members with execution rights for this command. |

## show running-config keychain

`show runnning-config keychain`

### Description

Shows the configurations for key chain protocol.

### Example

```
switch# show running-config keychain
keychain ospf_keys
  key 1
    key-string ciphertext
AQBapZ1OHiO9W3JwRqnjtLfbV73BPLS1S6TGVg+Lzl7N4e5eBAAAAPWaPBE=
    accept-lifetime start-time 10:10:10 10/25/2020 end-time 10:10:10 11/25/2020
    send-lifetime start-time 10:10:10 10/25/2020 end-time 10:10:10 11/25/2020
  key 45
    key-string ciphertext
AQBapZ1OHiO9W3JwRqnjtLfbV73BPLS1S6TGVg+Lzl7N4e5eBAAAAPWaPBE=
    accept-lifetime start-time 10:10:10 10/25/2020 end-time 10:10:10 11/25/2020
  key 33
switch#
```

## Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

## Command Information

| Platforms | Command context | Authority |
|---|---|---|
| All platforms | Operator (>) | Administrators or local user group members with execution rights for this command. |

# Chapter 11 IP Client Tracker

The client IP address tracking feature will learn and update the IP addresses of the access devices and clients connected to the switch. It can track addresses of directly connected clients, as well as clients connected to a downstream device such as a wireless access point.

IP Client tracker can be enabled for user-based tunneling (UBT) clients to learn the IP addresses on the required VLANs and interfaces.

Enabling IP Client Tracker based on UBT modes:

- If the UBT mode is Local VLAN, enable IP Client tracker for the UBT client VLAN defined using the `ubt-client-vlan` command.
- If the UBT mode is VLAN extend, enable IP Client tracker for the UBT client VLAN defined under role.

## IP Client Tracker commands

### client track ip

`client track ip`

#### Description

Enables client IP address tracking on the switch. The default is disabled on global and VLAN levels.

Admin users can enable client IP address tracking at the VLAN level.

> Tracking enabling will take effect only if the client IP address tracking is enabled at system and VLAN level.

The `no` form of the command disables client IP address tracking. If tracking is disabled at switch level, it will be stopped even if it is enabled at VLAN or port level.

#### Example

Enable client IP address tracking at switch level:

```
switch(config)# client track ip
```

Enable client IP address tracking on VLAN 100:

```
switch(config)# vlan 100
switch(config-vlan-100)# client track ip
Enable client IP address tracking on VLANs 10 to 100:
switch(config)# vlan 10-100
switch(config-vlan-<10-100>)# client track ip
```

#### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 4100i<br>6000<br>6100<br>6200 | `config` | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## client track ip { enable | disable | auto }

`client track ip { enable | disable | auto }`

### Description

Enables client IP address tracking on the specified set of interfaces. Tracking will take effect only if client IP address tracking is enabled at both the system level and for the VLAN to which the port belongs. Default: auto.

The `no` form of the command disables client IP address tracking on the specified set of interfaces.

| Parameter | Description |
|---|---|
| `enable` | Specifies that all client IP addresses will be tracked in the port. |
| `disable` | Specifies that client IP addresses will not be tracked in the port. |
| `auto` | Specifies the following:<br>For LLDP devices: Only the specified client IP address will be tracked in the port and other client IP addresses will not be tracked.<br>For non-LLDP devices: All client IP addresses will be tracked in the port. |

### Example

Enable client IP address tracking on interface 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# client track ip enable
```

Enable client IP address tracking on interfaces 1/1/1 to 1/1/5:

```
switch(config)# interface 1/1/1-1/1/5
switch(config-if-<1/1/1-1/1/5>)# client track ip enable
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 4100i<br>6000<br>6100<br>6200 | `config-if` | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## client track ip client-limit

`client track ip client-limit <CLIENT-LIMIT>`

### Description

Configures the maximum number of clients to be tracked on the specified set of interfaces.

For the 6000 and 6100 Switch Series the `no` form of the command resets the client limit to the default value of 128.

For the 6200 Switch Series the `no` form of the command resets the client limit to the default value of 2048.

| Parameter | Description |
|---|---|
| *CLIENT-LIMIT* | Specifies the maximum number of clients tracked on a port. Required. For the 6000 and 6100 Switch Series, Range: 1-128. Default: 128.For the 6200 Switch Series, Range: 1-2048. Default: 2048. |

### Example

Configure the maximum number of clients to be tracked on interface 1/1/5:

```
switch(config)# interface 1/1/5
switch(config-if)# client track ip client-limit 32
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 4100i<br>6000<br>6100<br>6200 | `config-if` | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

# client track ip update-interval

`client track ip update-interval <INTERVAL>`

## Description

Configures how often client IP addresses are updated.

The `no` form of the command resets the update interval to the default of 1800 seconds.

| Parameter | Description |
|---|---|
| *INTERVAL* | Specifies the update interval in seconds. Required. Range: 60-28000. Default: 1800. |

## Example

Configure the update interval for an interface:

```
switch(config)# interface 1/1/1
switch(config-if)# client track ip update-interval 600
```

## Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

## Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 4100i<br>6000<br>6100<br>6200 | `config-if` | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

# client track ip update-method probe

`client track ip update-method probe`

## Description

Enables probing the client to update the IP address.

The probe is sent to all clients on the tracking list that have an IP address in the following scenarios:

1. IP packets are not received from the clients during the IP address update cycle.
2. There is no IP packet from a learned IP address. In this case, a probe will be sent for the IP address to confirm if it is still owned by that client.

The `no` form of the command disables probing.

## Example

Disable probing to update the client IP address:

```
switch(config)# no client track ip update-method probe
```

## Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

## Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 4100i<br>6000<br>6100<br>6200 | `config` | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

# show capacities

```
show capacities
```

## Description

Shows the capacities configured on the switch.

## Example

```
switch# show capacities

System Capacities:
Capacities Name                                                               Value
-----------------------------------------------------------------------------------
Maximum number of Access Control Entries configurable in a system            14336
Maximum number of Access Control Lists configurable in a system               1024
Maximum number of class entries configurable in a system                      1024
Maximum number of classes configurable in a system                             512
Maximum number of entries in an Access Control List                           1024
Maximum number of entries in a class                                          1024
Maximum number of entries in a policy                                         1024
Maximum number of classifier policies configurable in a system                 512
Maximum number of policy entries configurable in a system                     1024
Maximum number of clients supported for tracking the IP address in the system  128

switch# show capacities client-track-ip-client-limit

System Capacities: Filter Client Track IP Client Limit
Capacities Name                                                                Value
------------------------------------------------------------------------------------
Maximum number of clients supported for tracking the IP address in the system
```

## Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

## Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 4100i<br>6000<br>6100<br>6200 | Operator (>) | Administrators or local user group members with execution rights for this command. |

# show client ip { count | port | vlan }

```
show client ip { count | port | vlan }
```

## Description

Shows number of client IP addresses or information about client IP addresses tracked on ports and VLANs.

| Parameter | Description |
|---|---|
| *count* | Displays number of clients tracked. |
| *port* | Displays client IP addresses tracked on the ports. |
| *vlan* | Displays client IP addresses tracked on the VLANs. |

## Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

## Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 4100i<br>6000<br>6100<br>6200 | Operator (>) | Administrators or local user group members with execution rights for this command. |

Routing Information Protocol (RIP, RIPv2 and RIPng) is not supported on the Aruba 6000 and 6100 Switch Series.

Routing Information protocol (RIP, RIPv2, RIPng) is a distance-vector routing protocol that uses hop count as a routing metric. It is useful in small networks. RIP can be configured on L3 ports, LAG, VLAN and Loopback interfaces. All configurations work in the interface context and require the associated interface to be routing interface.

- **RIP/RIPv2:** IPv4 implementation of Routing Information protocol defined in RFC 2453.
- **RIPng:** IPv6 implementation of Routing Information protocol defined in RFC 2080.

# Overview

RIP characteristics:

- Prevents routing loops by adding a limit to the number of hops allowed in a path from source to destination.
- Allows a maximum of 15 hops which limits the size of the network that RIP can support.
- Uses broadcast UDP data packets to exchange routing information.
- RIPv2 is a classless protocol that supports variable-length subnet mask (VLSM), classless inter-domain routing (CIDR) and route summarization.
- RIPng uses ipsec for message authentication.

RIP limitations:

- RIPv1 is not supported.
- Only poison reverse is supported for loop prevention.

# RIPv2 (IPv4) commands

## Configuration commands

### router rip

```
router rip <PROCESS-ID> [vrf <VRF-NAME>]
no router rip <PROCESS-ID> [vrf <VRF-NAME>]
```

#### Description

Creates RIP process if not already created and enters the `router rip <PROCESS-ID>` context for the VRF mentioned. If no VRF is mentioned, a default is used. Only one RIP process is allowed per VRF.

The `no` form of this command deletes the RIP instance for the VRF. If no VRF is mentioned the default is deleted.

| Parameter | Description |
|---|---|
| *<PROCESS-ID>* | Specifies name of the RIP process ID. Range: 1-63. |
| vrf *<VRF-NAME>* | Specifies VRF name. |

### Examples

Creating RIP process and naming the VRF:

```
switch(config)# router rip 2 vrf red
```

Deleting RIP process:

```
switch(config)# no router rip 2 vrf red
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | config | Administrators or local user group members with execution rights for this command. |

# Interface commands

## ip rip

```
ip rip <PROCESS-ID> {all-ip | ip-address}
no ip rip <PROCESS-ID> {all-ip | ip-address}
```

### Description

Enables RIP process on an interface.

The `no` form of this command deletes the RIP process from an interface.

| Parameter | Description |
|---|---|
| ip rip *<PROCESS-ID>* | Specifies RIP process ID. Range: 1-63. |
| all-ip | Specifies RIP for all IP addresses configured on the interface. |
| ip-address | Specifies IP address for RIP on the interface. |

### Usage

- If an IP address is removed from an interface configured with RIP, all RIP configurations will be removed from the interface.
- If `ip rip 1 all-ip` is configured and a new IP address is added to the interface, RIP configurations will not be applicable for the newly added IP address.

### Examples

Configuring RIP for all IP addresses configured on the interface:

```
switch(config)# interface 1/1/1
switch(config-if)# ip rip 1 all-ip
```

Deleting RIP for all IP addresses configured on the interface:

```
switch(config)# interface 1/1/1
switch(config-if)# no ip rip 1 all-ip
```

### Command History

| Release | Modification |
| --- | --- |
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
| --- | --- | --- |
| 6200 | `config-if` | Administrators or local user group members with execution rights for this command. |

## ip rip all-ip enable

```
ip rip all-ip enable
no ip rip all-ip enable
```

### Description

Enables RIP process for all RIP enabled IP addresses configured on interface.

The `no` form of this command disables RIP process on the interface.

### Usage

- Default settings allow an interface to receive RIP packets.
- If an IP address is removed from an interface configured with RIP, all RIP configurations will be removed from the interface.

### Examples

Enabling RIP process for all RIP enabled IP addresses on interface:

```
switch(config)# interface 1/1/1
switch(config-if)# ip rip all-ip enable
```

Disabling RIP process on interface:

```
switch(config)# interface 1/1/1
switch(config-if)# no ip rip all-ip enable
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config-if` | Administrators or local user group members with execution rights for this command. |

## ip rip all-ip disable

```
ip rip all-ip disable
no ip rip all-ip disable
```

### Description

Disables RIP process for all RIP enabled IP addresses configured on the interface.

The `no` form of this command enables RIP process for all RIP enabled IP addresses configured on the interface.

### Usage

- Default settings allow an interface to receive RIP packets.
- If an IP address is removed from an interface configured with RIP, all RIP configurations will be removed from the interface.

### Examples

Disabling RIP process for all RIP enabled IP addresses on interface:

```
switch(config)# interface 1/1/1
switch(config-if)# ip rip all-ip enable
```

Enabling RIP process for all RIP enabled IP addresses on interface:

```
switch(config)# interface 1/1/1
switch(config-if)# no ip rip all-ip enable
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

## Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config-if` | Administrators or local user group members with execution rights for this command. |

# ip rip all-ip send disable

```
ip rip all-ip send disable
no ip rip all-ip send disable
```

## Description

Disables interface from sending RIP packets for all RIP enabled IP addresses.

The `no` form of this command enables interface to send RIP packets for all RIP enabled IP addresses.

## Usage

- Default settings allow an interface to send RIP packets.
- If an IP address is removed from an interface configured with RIP, all RIP configurations will be removed from the interface.

## Examples

Disabling interface from sending RIP packets for all RIP enabled IP addresses on interface:

```
switch(config)# interface 1/1/1
switch(config-if)# ip rip all-ip send disable
```

Enabling interface to send RIP packets for all RIP enabled IP addresses on interface:

```
switch(config)# interface 1/1/1
switch(config-if)# no ip rip all-ip send disable
```

## Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

## Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config-if` | Administrators or local user group members with execution rights for this command. |

# ip rip all-ip receive disable

```
ip rip all-ip receive disable
no ip rip all-ip receive disable
```

### Description

Disables interface from receiving RIP packets for all enabled IP addresses.

The `no` form of this command enables interface to receive RIP packets for all RIP enabled IP addresses.

### Usage

- Default settings allow an interface to receive RIP packets.
- If an IP address is removed from an interface configured with RIP, all RIP configurations will be removed from the interface.

### Examples

Disabling interface from receiving RIP packets for all RIP enabled IP addresses on interface:

```
switch(config)# interface 1/1/1
switch(config-if)# ip rip all-ip receive disable
```

Enabling interface to receive RIP packets for all RIP enabled IP addresses on interface:

```
switch(config)# interface 1/1/1
switch(config-if)# no ip rip all-ip receive disable
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config-if` | Administrators or local user group members with execution rights for this command. |

## Routing commands

### enable

```
enable
no enable
```

### Description

Enables RIP process if disabled. By default RIP process is enabled.

The `no` form of this command disables the RIP process.

### Examples

Enabling RIP process when disabled:

```
switch(config)# router rip 1
switch(config-rip-1)# enable
```

Disabling RIP process when enabled:

```
switch(config)# router rip 1
switch(config-rip-1)# no enable
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config` | Administrators or local user group members with execution rights for this command. |

## disable

```
disable
no disable
```

### Description

Disables RIP process.

The `no` form of this command enables the RIP process.

### Examples

Disabling RIP process:

```
switch(config)# router rip 1
switch(config-rip-1)# disable
```

Enabling RIP process:

```
switch(config)# router rip 1
switch(config-rip-1)# no disable
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
| --- | --- | --- |
| 6200 | `config` | Administrators or local user group members with execution rights for this command. |

## distance

```
distance <DISTANCE>
no distance
```

### Description

Configures administrative distance for RIP. Administrative distance is used as criteria to select the best route when multiple protocols have the same route.

The `no` form of this command sets the RIP administrative distance to the default. Default: 120.

| Parameter | Description |
| --- | --- |
| *<DISTANCE>* | Specifies RIP administrative distance. Range: 1 to 255. |

### Examples

Configuring administrative distance for RIP:

```
switch(config)# router rip 1
switch(config-rip-1)# distance 100
```

Setting administrative distance for RIP to default values:

```
switch(config)# router rip 1
switch(config-rip-1)# no distance
```

### Command History

| Release | Modification |
| --- | --- |
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
| --- | --- | --- |
| 6200 | `config` | Administrators or local user group members with execution rights for this command. |

## maximum-paths

```
maximum-paths <MAX-VALUE>
no maximum-paths
```

### Description

Sets the maximum number of ECMP routes that RIP can support.

The `no` form of this command sets the maximum number of ECMP routes to the default value of 4.

| Parameter | Description |
| --- | --- |
| *<MAX-VALUE>* | Sets the number of RIP ECMP routes. Range: 1-8. |

### Examples

Setting maximum number of RIP ECMP routes:

```
switch(config)# router rip 1
switch (config-rip-1)# maximum-paths 8
```

Setting maximum number of RIP ECMP routes to default:

```
switch(config)# router rip 1
switch (config-rip-1)# no maximum-paths
```

### Command History

| Release | Modification |
| --- | --- |
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
| --- | --- | --- |
| 6200 | config | Administrators or local user group members with execution rights for this command. |

## redistribute

```
redistribute {bgp | connected | ospf <PROCESS-ID> | static}
no redistribute {bgp | connected | ospf <PROCESS-ID> | static}
```

### Description

Redistributes routes originating from other protocols into RIP.

The `no` form of this command disables redistribution of routes originating from other protocols into RIP.

| Parameter | Description |
| --- | --- |
| bgp | Specifies BGP routes to redistribute into RIP. |
| connected | Specifies connected routes (directly attached subnet or host) to redistribute into RIP. |
| ospf *<PROCESS-ID>* | Specifies OSPF route to redistribute into RIP. |
| static | Specifies static route to redistribute into RIP. |

### Examples

Redistributing BGP routes into RIP:

```
switch(config)# router rip 1
switch(config-rip-1)# redistribute bgp
```

Disabling BGP routes that originate from other protocols and redistribute into RIP:

```
switch(config)# router rip 1
switch(config-rip-1)# no redistribute bgp
```

### Command History

| Release | Modification |
| --- | --- |
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
| --- | --- | --- |
| 6200 | `config` | Administrators or local user group members with execution rights for this command. |

## timers update

```
timers update <INTERVAL> timeout <DURATION> garbage-collection <PERIOD>
no timers
```

### Description

Configures RIP timers with specific values.

The `no` form of this command sets all RIP timers to default values.

| Parameter | Description |
| --- | --- |
| `timers update <INTERVAL>` | Specifies frequency at which RIP sends updates to all of its peers. Range: 1 to 2147484. Default: 30. |
| `timeout <DURATION>` | Specifies timeout duration from the point of the last refresh after a route is received from a peer timeout and is marked as expired. Range: 1 to 255. Default: 180. |
| `garbage-collection <PERIOD>` | Specifies amount of time route remains in routing table after route expiration. Range: 1 to 255. Default: 120. |

### Examples

Configuring RIP timers with specific values:

```
switch(config)# router rip 1
switch(config-rip-1)# timers update 40 timeout 200 garbage-collection 150
```

Configuring RIP timers with default values:

```
switch(config)# router rip 1
switch(config-rip-1)# no timers
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config` | Administrators or local user group members with execution rights for this command. |

# RIPv2 clear commands

## clear ip rip statistics

`clear ip rip [<PROCESS-ID>] statistics [all-vrfs | vrf <VRF-NAME>]`

### Description

Clears RIP event statistics.

| Parameter | Description |
|---|---|
| *<PROCESS-ID>* | Specifies RIP process ID. Range: 1-63 |
| `all-vrfs` | Clears statistics for all VRFs. |
| `vrf` | Selects VRF to clear statistics for. |
| *<VRF-NAME>* | Specifies VRF name. |

### Examples

Clearing RIP event statistics:

```
switch# clear ip rip statistics
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | Operator (>) or Manager (#) | Administrators or local user group members with execution rights for this command. |

## RIPv2 interface commands

### enable

```
enable
no enable
```

#### Description

Enables RIP process for RIP enabled IP address configured on interface.

The `no` form of this command disables RIP process on interface.

#### Usage

- Default settings allow an interface to receive RIP packets.
- If an IP address is removed from an interface configured with RIP, all RIP configurations will be removed from the interface.

#### Examples

Enabling RIP process for RIP enabled IP address:

```
switch(config)# interface 1/1/1
switch(config-if)# ip rip 1 10.1.1.1
switch(config-if-rip)# enable
```

Disabling RIP process on interface:

```
switch(config)# interface 1/1/1
switch(config-if)# ip rip 1 10.1.1.1
switch(config-if-rip)# no enable
```

#### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

#### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config-if` | Administrators or local user group members with execution rights for this command. |

### disable

```
disable
no disable
```

## Description

Disables RIP process for RIP enabled IP addresses configured on interface.

The `no` form of this command enables RIP process on interface.

## Examples

Disabling RIP process for RIP enabled IP addresses configured on interface:

```
switch(config)# interface 1/1/1
switch(config-if)# ip rip 1 10.1.1.1
switch(config-if-rip)# disable
```

Enabling RIP process on interface:

```
switch(config)# interface 1/1/1
switch(config-if)# ip rip 1 10.1.1.1
switch(config-if-rip)# no disable
```

## Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

## Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config-if` | Administrators or local user group members with execution rights for this command. |

# send disable

```
send disable
no send disable
```

## Description

Disables an interface from sending RIP packets for a specific IP address.

The `no` form of this command enables interface for sending RIP packets for a specific IP address.

## Usage

- Default settings allow an interface to send and receive RIP packets.
- If an IP address is removed from an interface configured with RIP, all RIP configurations will be removed from the interface.

## Examples

Disabling interface from sending RIP packets for a specific IP address :

```
switch(config)# interface 1/1/1
switch(config-if)# ip rip 1 10.1.1.1
switch(config-if-rip)# send disable
```

Enabling interface to send RIP packets for a specific IP address:

```
switch(config)# interface 1/1/1
switch(config-if)# ip rip 1 10.1.1.1
switch(config-if-rip)# no send disable
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config-if` | Administrators or local user group members with execution rights for this command. |

## receive disable

```
receive disable
no receive disable
```

### Description

Disables interface from receiving RIP packets for a specific IP address.

The `no` form of this command enables interface for receiving RIP packets for a specific IP address.

### Usage

- Default settings allow an interface to receive RIP packets.
- If an IP address is removed from an interface configured with RIP, all RIP configurations will be removed from the interface.

### Examples

Disabling interface from receiving RIP packets for a specific IP address:

```
switch(config)# interface 1/1/1
switch(config-if)# ip rip 1 10.1.1.1
switch(config-if-rip)# receive disable
```

Enabling interface for receiving RIP packets for a specific IP address:

```
switch(config)# interface 1/1/1
switch(config-if)# ip rip 1 10.1.1.1
switch(config-if-rip)# no receive disable
```

## Command History

| Release | Modification |
| --- | --- |
| 10.07 or earlier | -- |

## Command Information

| Platforms | Command context | Authority |
| --- | --- | --- |
| 6200 | `config-if` | Administrators or local user group members with execution rights for this command. |

# RIPv2 show commands

## show capacities rip

```
show capacities rip
```

## Description

Displays maximum number of RIP interfaces, routes and process.

## Examples

Displaying maximum number of RIP interfaces, routes and process:

```
switch# show capacities rip

System Capacities: Filter RIP
Capacities Name                                                              Value
---------------------------------------------------------------------------------
Maximum number of RIP interfaces configurable in the system                  32
Maximum number of RIP processes supported across each VRF                    1
Maximum number of routes in RIP supported across all VRFs                    2540
```

## Command History

| Release | Modification |
| --- | --- |
| 10.07 or earlier | -- |

## Command Information

| Platforms | Command context | Authority |
| --- | --- | --- |
| 6200 | Operator (>) or Manager (#) | Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only. |

## show capacities-status rip

```
show capacities-status rip
```

## Description

Displays number of RIP interfaces, routes and process configured in the system.

## Examples

Displaying number of RIP interfaces, routes and process:

```
switch# show capacities-status rip

System Capacities Status: Filter RIP
Capacities Name                                                  Value Maximum
----------------------------------------------------------------------------
Number of RIP interfaces configured in the system                   0      32
```

## Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

## Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | Operator (>) or Manager (#) | Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only. |

# show ip rip

`show ip rip [<PROCESS-ID>] [all-vrfs | vrf <VRF-NAME>]`

## Description

Displays general RIP configuration.

| Parameter | Description |
|---|---|
| *<PROCESS-ID>* | Specifies RIP process ID. Range: 1-63. |
| `all vrfs` | Displays general RIP information for all VRFs. |
| `vrf` | Selects VRF to display general RIP information for. |
| *<VRF-NAME>* | Specifies VRF name. |

## Usage

- Parameters display general RIP information for a specific RIP process.
- Parameters display general RIP information for a specific or all VRFs.
- If a VRF is not mentioned, information for the default VRF is displayed.

## Examples

Displaying general RIP configuration for all VRFs:

```
switch# show ip rip 34 all-vrfs
VRF : Default                        Process-ID : 34
---------------------------------------------------------------------
RIP Version             : RIPv2     Protocol Status : Enabled
Update Time             : 60  sec   Timeout Time    : 240 sec
Garbage Collection Time : 250 sec   ECMP            : 6
Distance                : 100       Redistribution  : static,
                                                      ospf 1

VRF : vrf_1                          Process-ID : 34
---------------------------------------------------------------------
RIP Version             : RIPv2     Protocol Status : Enabled
Update Time             : 30  sec   Timeout Time    : 180 sec
Garbage Collection Time : 120 sec   ECMP            : 4
Distance                : 120       Redistribution  : None
```

### Command History

| Release | Modification |
| --- | --- |
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
| --- | --- | --- |
| 6200 | Operator (>) or Manager (#) | Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only. |

## show ip rip interface

`show ip rip [<PROCESS-ID>] interface [<INTERFACE-NAME>] [brief] [all-vrfs | vrf <VRF-NAME>]`

### Description

Displays information about RIP enabled interfaces.

| Parameter | Description |
| --- | --- |
| *<PROCESS-ID>* | Specifies RIP process ID. Range: 1-63. |
| *<INTERFACE-NAME>* | Specifies interface. |
| `brief` | Shows brief overview information for the RIP interface. |
| `all-vrfs` | Displays interface information for all VRFs. |
| `vrf` | Selects specific VRF. |
| *<VRF-NAME>* | Specifies VRF. |

### Usage

- Parameters display general RIP information for a specific RIP process.
- If a VRF is not mentioned, information for the default VRF is displayed.

## Examples

```
switch# show ip rip interface
Interface 1/1/1 is up, IP Address is 10.10.10.1/24
-----------------------------------------------------------------------
VRF             : Default           Process-ID     : 1
Status          : Oper Up           Mode           : Send and Receive
MTU             : 500               Version        : RIPv2
Poision Reverse : Enabled

Interface 1/1/2 is up, IP Address is 20.10.10.1/24
-----------------------------------------------------------------------
VRF             : Default           Process-ID     : 1
Status          : Admin Down        Mode           : Receive
MTU             : 500               Version        : RIPv2
Poision Reverse : Enabled

Interface 1/1/3 is up, IP Address is 30.10.10.1/24
-----------------------------------------------------------------------
VRF             : Default           Process-ID     : 1
Status          : Admin Down        Mode           : Send
MTU             : 500               Version        : RIPv2
Poision Reverse : Enabled

switch# show ip rip interface brief
VRF : default   Process-ID : 1
-------------------------------

Total Number of Interfaces: 2

Interface     IP-Address/Mask    Status    MTU
--------------------------------------------------
1/1/1         10.10.10.1/24       up       500
1/1/2         20.10.10.1/24       up       500
1/1/3         30.10.10.1/24       up       500
```

## Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

## Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | Operator (>) or Manager (#) | Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only. |

# show ip rip neighbors

```
show ip rip [<PROCESS-ID>] neighbors [<IP-ADDRESS>] [all-vrfs | vrf <VRF-NAME>]
```

## Description

Displays information about RIP neighbors.

| Parameter | Description |
|---|---|
| *<PROCESS-ID>* | Specifies RIP process ID. Range: 1-63. |
| *<IP-ADDRESS>* | Specifies IP address of a specific neighbor to display information on. |
| `all-vrfs` | Displays neighbor information for all VRFs. |
| `vrf` | Selects VRF to display neighbor information. |
| *<VRF-NAME>* | Specifies VRF name. |

### Usage

- Parameters display RIP neighbor information for a specific RIP process.
- Parameters display RIP neighbor information for a specific neighbor.
- If a VRF is not mentioned, information for the default VRF is displayed.

### Examples

Displaying RIP neighbor information for all VRFs:

```
switch# show ip rip neighbors all-vrfs
VRF : default            Process-ID : 1
----------------------------------------

Total Number of Neighbors: 1

Peer-Address   Type    Last-Update  Rcvd-Bad-Pkts    Rcvd-Bad-Routes
------------------------------------------------------------------
1.1.1.2        RIPv2   0:0:7        4                5
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | Operator (>) or Manager (#) | Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only. |

## show ip rip routes

`show ip rip [<PROCESS-ID>] routes [<PREFIX/LENGTH>] [all-vrfs | vrf <VRF-NAME>]`

### Description

Displays RIP routing table for a specific RIP process.

| Parameter | Description |
|---|---|
| *<PROCESS-ID>* | Specifies RIP process ID to display information for a specific RIP process. Range: 1-63. |
| *<PREFIX/LENGTH>* | Specifies the network prefix. |
| all-vrfs | Displays RIP routing information for all VRFs. |
| vrf | Selects VRF to display RIP routing information. |
| *<VRF-NAME>* | Specifies VRF name. |

## Usage

- *<PREFIX/LENGTH>* is an optional parameter that displays RIP routing table information for a specific subnet.
- If a VRF is not mentioned, information for the default VRF is displayed.

## Examples

Displaying RIP routing table for all VRFs:

```
switch# show ip rip routes all-vrfs
VRF : default              Process-ID : 1
----------------------------------------
Total Number of Routes : 6

Prefix              Metric     Interface      Nexthop
--------------------------------------------------------
10.1.0.0/16           2          1/1/1          30.1.1.2
20.1.2.0/24           3          1/1/1          30.1.1.2
30.1.1.0/24           1          1/1/1

VRF : vrf_1                Process-ID : 34
----------------------------------------

Prefix               Metric     Interface      Nexthop
--------------------------------------------------------
20.1.0.0/16           10         1/1/2          50.1.1.2
40.1.2.0/24           14         1/1/2          50.1.1.2
50.1.1.0/24            1         1/1/2
```

## Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

## Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | Operator (>) or Manager (#) | Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only. |

## show ip rip statistics

`show ip rip [<PROCESS-ID>] statistics [all-vrfs | vrf <VRF-NAME>]`

### Description

Displays RIP statistics.

| Parameter | Description |
|---|---|
| *<PROCESS-ID>* | Specifies RIP process ID. Range: 1-63. |
| all-vrfs | Displays statistics information for all VRFs. |
| vrf | Selects VRF to display RIP statistics information for. |
| *<VRF-NAME>* | Specifies VRF name. |

### Usage

- Parameters can display information for all VRFs or a specific VRF.
- If a VRF is not mentioned, information for the default VRF is displayed.

### Examples

Displaying RIP statistics for all VRFs:

```
switch# show ip rip statistics all-vrfs
VRF : default  Process-ID : 1
-------------------------------------

Global Route Changes : 50
Global Queries       : 2
Last Cleared         : 0h 30m 28s ago

VRF : vrf_1    Process-ID : 34
-------------------------------------

Global Route Changes : 20
Global Queries       : 0
Last Cleared         : 0h 30m 28s ago
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | Operator (>) or Manager (#) | Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only. |

## show ip rip statistics interface

```
show ip rip [<PROCESS-ID>] statistics interface [<INTERFACE-NAME>] [all-vrfs | vrf <VRF-NAME>]
```

## Description

Displays RIP statistics for RIP enabled interfaces.

| Parameter | Description |
|---|---|
| *<PROCESS-ID>* | Specifies RIP process ID. Range: 1-63. |
| *<INTERFACE-NAME>* | Specifies name of interface. |
| all-vrfs | Displays RIP interface statistics for all VRFs. |
| vrf | Selects VRF to display RIP interface statistics. |
| *<VRF-NAME>* | Specifies VRF name. |

## Usage

- Parameters can display information for all VRFs or a specific VRF.
- If a VRF is not mentioned, information for the default VRF is displayed.

## Examples

Displaying RIP statistics for a RIP enabled interface:

```
switch# show ip rip statistics interface 1/1/1
VRF : default  Process-ID : 1      interface 1/1/1
--------------------------------------------------

IP-Address    Trigger-Updates    Rcvd-Bad-Packets    Rcvd-Bad-Routes
-----------------------------------------------------------------------
10.1.1.1      15                 3                   4

Last Cleared : 0h 30m 28s ago
```

## Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

## Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | Operator (>) or Manager (#) | Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only. |

# show running-config

```
show running config
```

## Description

Displays all running configurations for all protocols including RIP.

## Examples

Displaying all running configurations for all protocols including RIP:

```
switch# show running-config
Current configuration:
!
!Version Halon 0.1.0 (Build: genericx86-64-Halon-0.1.0-master-20170309054955-dev)
!Schema version 0.1.8
lldp enable
timezone set utc
vrf blue
vrf green
vrf red
led base-loc_fdc on
led base-loc on
led base-hlth_fdc fast_blink
led base-pwr_fdc on
!
!
!
!
!
!
aaa authentication login default local
aaa authorization commands default none
!
!
!
!
router ospf 1 vrf red
router rip 1
    maximum-paths 5
    distance 1
router rip 1 vrf red
    default-information originate always
    maximum-paths 7
    distance 5
    redistribute ospf 1
    timers update 40 timeout 200 garbage-collection 120
vlan 1
    no shutdown
interface lag 44
    no shutdown
    ip address 33.1.1.1/24
    ip rip 1 33.1.1.1
        send disable
interface 1/1/1
    no shutdown
    ip address 33.44.1.1/24
    ip address 44.44.1.1/24 secondary
    ip rip 1 33.44.1.1
        send disable
    ip rip 1 44.44.1.1
        send disable
interface 1/1/2
interface loopback 2
    ip address 55.55.55.55/32
    ip rip 1 55.55.55.55
```

## Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | Operator (>) or Manager (#) | Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only. |

# RIPng (IPv6) commands

## Configuration commands

### router ripng

```
router ripng <PROCESS-ID> [vrf <VRF-NAME>]
no router ripng <PROCESS-ID> [vrf <VRF-NAME>]
```

### Description

Creates RIPng process if not already created and enters the `router ripng <PROCESS-ID>` context for the VRF mentioned. If no VRF is mentioned, a default is used. Only one RIPng process is allowed per VRF.

The `no` form of this command deletes the RIPng instance for the VRF. If no VRF is mentioned the default is deleted.

| Parameter | Description |
|---|---|
| *<PROCESS-ID>* | Specifies name of the RIPng process ID. Range: 1-63. |
| vrf | Sets VRF name for RIPng process. |
| *<VRF-NAME>* | VRF name for VRF. |

### Examples

Creating RIPng process and naming the VRF:

```
switch(config)# router ripng 2 vrf red
```

Deleting RIPng process:

```
switch(config)# no router ripng 2 vrf red
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config` | Administrators or local user group members with execution rights for this command. |

## Interface commands

### ipv6 ripng

```
ipv6 ripng <PROCESS-ID>
no ipv6 ripng <PROCESS-ID>
```

### Description

Enables RIPng process on interface and creates a new context.

The `no` form of this command deletes RIPng process on interface.

| Parameter | Description |
|---|---|
| *<PROCESS-ID>* | Specifies RIPng process ID. Range: 1-63. |

### Examples

Enabling RIPng process on an interface:

```
switch(config)# interface 1/1/1
switch (config-if)# ipv6 ripng 1
switch (config-if-ripng)#
```

Deleting RIPng process on an interface:

```
switch(config)# interface 1/1/1
switch (config-if)# no ipv6 ripng 1
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config-if` | Administrators or local user group members with execution rights for this command. |

## Routing commands

### enable

```
enable
no enable
```

### Description

Enables RIPng process if disabled. By default RIPng process is enabled.

The `no` form of this command disables the RIPng process.

### Examples

Enabling RIPng process when disabled:

```
switch(config)# router ripng 1
switch(config-ripng-1)# enable
```

Disabling RIPng process when enabled:

```
switch(config)# router ripng 1
switch(config-ripng-1)# no enable
```

### Command History

| Release | Modification |
| --- | --- |
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
| --- | --- | --- |
| 6200 | `config` | Administrators or local user group members with execution rights for this command. |

## disable

```
disable
no disable
```

### Description

Disables RIPng process.

The `no` form of this command enables the RIPng process.

### Examples

Disabling RIPng process:

```
switch(config)# router ripng 1
switch(config-ripng-1)# disable
```

Enabling RIPng process:

```
switch(config)# router ripng 1
switch(config-ripng-1)# no disable
```

## Command History

| Release | Modification |
| --- | --- |
| 10.07 or earlier | -- |

## Command Information

| Platforms | Command context | Authority |
| --- | --- | --- |
| 6200 | `config` | Administrators or local user group members with execution rights for this command. |

# distance

```
distance <DISTANCE>
no distance
```

## Description

Configures administrative distance for RIPng. Administrative distance is used as criteria to select the best route when multiple protocols have the same route.

The `no` form of this command sets the RIPng administrative distance to the default. Default: 120.

| Parameter | Description |
| --- | --- |
| *<DISTANCE>* | Specifies RIPng administrative distance. Range: 1 to 255. |

## Examples

Configuring administrative distance for RIPng:

```
switch(config)# router ripng 1
switch(config-ripng-1)# distance 100
```

Setting administrative distance for RIPng to default values:

```
switch(config)# router ripng 1
switch(config-ripng-1)# no distance
```

## Command History

| Release | Modification |
| --- | --- |
| 10.07 or earlier | -- |

## Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config` | Administrators or local user group members with execution rights for this command. |

## maximum-paths

```
maximum-paths <MAX-VALUE>
no maximum-paths
```

### Description

Sets the maximum number of ECMP routes that RIPng can support.

The `no` form of this command sets the maximum number of ECMP routes to the default value of 4.

| Parameter | Description |
|---|---|
| *<MAX-VALUE>* | Sets the number of RIPng ECMP routes. Range: 1-8. |

### Examples

Setting maximum number of RIPng ECMP routes:

```
switch(config)# router ripng 1
switch (config-ripng-1)# maximum-paths 8
```

Setting maximum number of RIPng ECMP routes to default:

```
switch(config)# router ripng 1
switch (config-ripng-1)# no maximum-paths
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config` | Administrators or local user group members with execution rights for this command. |

## redistribute

```
redistribute {bgp | connected | ospfv3 <PROCESS-ID> | static}
no redistribute {bgp | connected | ospfv3 <PROCESS-ID> | static}
```

### Description

Redistributes routes originating from other protocols into RIPng.

The `no` form of this command disables redistribution of routes originating from other protocols into RIPng.

| Parameter | Description |
| --- | --- |
| `bgp` | Specifies BGP routes to redistribute into RIPng. |
| `connected` | Specifies connected routes (directly attached subnet or host) to redistribute into RIPng. |
| `ospfv3 <PROCESS-ID>` | Specifies OSPFv3 route to redistribute into RIPng. |
| `static` | Specifies static route to redistribute into RIPng. |

### Examples

Redistributing BGP routes into RIPng:

```
switch(config)# router ripng 1
switch(config-ripng-1)# redistribute bgp
```

Disabling BGP routes that originate from other protocols and redistribute into RIPng:

```
switch(config)# router ripng 1
switch(config-ripng-1)# no redistribute bgp
```

### Command History

| Release | Modification |
| --- | --- |
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
| --- | --- | --- |
| 6200 | `config` | Administrators or local user group members with execution rights for this command. |

## timers update

```
timers update <INTERVAL> timeout <DURATION> garbage-collection <PERIOD>
no timers
```

### Description

Configures RIPng timers with specific values.

The `no` form of this command sets all RIPng timers to default values.

| Parameter | Description |
| --- | --- |
| `timers update <INTERVAL>` | Specifies frequency at which RIPng sends updates to all of its peers. Range: 1 to 2147484. Default: 30. |
| `timeout <DURATION>` | Specifies timeout duration from the point of the last refresh after a route is received from a peer timeout and is marked as expired. |

| Parameter | Description |
|---|---|
| | Range: 1 to 255. Default: 180. |
| `garbage-collection <PERIOD>` | Specifies amount of time route remains in routing table after route expiration. Range: 1 to 255. Default: 120. |

### Examples

Configuring RIPng timers with specific values:

```
switch(config)# router ripng 1
switch(config-ripng-1)# timers update 40 timeout 200 garbage-collection 150
```

Configuring RIPng timers with default values:

```
switch(config)# router ripng 1
switch(config-ripng-1)# no timers
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config` | Administrators or local user group members with execution rights for this command. |

# RIPng clear commands

## clear ipv6 ripng statistics

`clear ipv6 ripng [<PROCESS-ID>] statistics [all-vrfs | vrf <VRF-NAME>]`

### Description

Clears RIPng event statistics.

| Parameter | Description |
|---|---|
| `<PROCESS-ID>` | Specifies RIPng process ID. Range: 1-63 |
| `all-vrfs` | Clears statistics for all VRFs. |
| `vrf` | Selects VRF to clear statistics for. |
| `<VRF-NAME>` | Specifies VRF name. |

### Examples

Clearing RIPng event statistics:

```
switch# clear ipv6 ripng statistics
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | Operator (>) or Manager (#) | Administrators or local user group members with execution rights for this command. |

## RIPng interface commands

### enable

```
enable
no enable
```

### Description

Enables RIPng process on interface.

The `no` form of this command disables RIPng process on interface.

### Examples

Enabling RIPng process on interface:

```
switch(config)# interface 1/1/1
switch(config-if)# ipv6 ripng 1
switch(config-if-ripng)# enable
```

Disabling RIPng process on interface:

```
switch(config)# interface 1/1/1
switch(config-if)# ipv6 ripng 1
switch(config-if-ripng)# no enable
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config-if` | Administrators or local user group members with execution rights for this command. |

## disable

```
disable
no disable
```

### Description

Disables RIPng process on interface.

The `no` form of this command enables RIPng process on interface.

### Examples

Disabling RIP process for all RIP enabled IP addresses on interface:

```
switch(config)# interface 1/1/1
switch(config-if)# ipv6 ripng 1
switch(config-if-ripng)# disable
```

Enabling RIP process for all RIP enabled IP addresses on interface:

```
switch(config)# interface 1/1/1
switch(config-if)# ipv6 ripng 1
switch(config-if-ripng)# no disable
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config-if` | Administrators or local user group members with execution rights for this command. |

## send disable

```
send disable
no send disable
```

### Description

Disables interface from sending RIPng packets. An interface can send RIPng packets by default.

The `no` form of this command enables interface to send RIPng packets, if disabled.

### Examples

Disabling interface from sending RIPng packets:

```
switch(config)# interface 1/1/1
switch (config-if)# ipv6 ripng 1
switch (config-if-ripng)# send disable
```

Enabling interface to send RIPng packets:

```
switch(config)# interface 1/1/1
switch (config-if)# ipv6 ripng 1
switch (config-if-ripng)# no send disable
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config-if` | Administrators or local user group members with execution rights for this command. |

## receive disable

```
receive disable
no receive disable
```

### Description

Disables interface from receiving RIPng packets for all enabled IP addresses. An interface can receive RIPng packets by default.

The `no` form of this command enables interface to receive RIPng packets, if disabled.

### Examples

Disabling interface from receiving RIPng packets:

```
switch(config)# interface 1/1/1
switch(config-if)# ipv6 ripng 1
switch (config-if-ripng)# receive disable
```

Enabling interface to receive RIPng packets when disabled:

```
switch(config)# interface 1/1/1
switch(config-if)# ipv6 ripng 1
switch (config-if-ripng)# no receive disable
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | `config-if` | Administrators or local user group members with execution rights for this command. |

# RIPng show commands

## show capacities ripng

```
show capacities ripng
```

### Description

Displays the maximum number of RIPng interfaces, routes and process.

### Examples

Displaying maximum number of RIPng interfaces, routes and process:

```
switch# show capacities ripng

System Capacities: Filter RIPng
Capacities Name                                                        Value
-----------------------------------------------------------------------------
Maximum number of RIPng interfaces configurable in the system         32
Maximum number of RIPng processes supported across each VRF           1
Maximum number of routes in RIPng supported across all VRFs           2540
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | Operator (>) or Manager (#) | Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only. |

## show capacities-status ripng

```
show capacities-status ripng
```

### Description

Displays number of RIPng interfaces, routes and process configured in the system.

### Examples

Displaying number of RIPng interfaces, routes and process:

```
switch# show capacities-status ripng

System Capacities Status: Filter RIPng
Capacities Name
        Value  Maximum
------------------------------------------------------------------------------------
-----------------------
Number of RIPng interfaces configured in the system
            0       32
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | Operator (>) or Manager (#) | Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only. |

## show ipv6 ripng

`show ipv6 ripng [<PROCESS-ID>] [all-vrfs | vrf <VRF-NAME>]`

### Description

Displays general RIPng configuration.

| Parameter | Description |
|---|---|
| `<PROCESS-ID>` | Specifies RIPng process ID. Range: 1-63. |
| `all vrfs` | Displays general RIPng information for all VRFs. |
| `vrf` | Selects VRF to display general RIPng information for. |
| `<VRF-NAME>` | Specifies VRF name. |

### Usage

- Parameters display general RIPng information for a specific RIPng process.
- Parameters display general RIPng information for a specific or all VRFs.
- If a VRF is not mentioned, information for the default VRF is displayed.

### Examples

Displaying general RIPng configuration for all VRFs:

```
switch# show ipv6 ripng 34 all-vrfs
VRF : Default                        Process-ID : 34
---------------------------------------------------------------------
Protocol Status          : Enabled   ECMP             : 6
Update Time              : 60  sec   Timeout Time     : 240 sec
Garbage Collection Time : 250 sec   Distance         : 100
Redistribution           : static,
                           ospfv3 1

VRF : vrf_1                          Process-ID : 34
---------------------------------------------------------------------
Protocol Status          : Enabled   ECMP             : 4
Update Time              : 30  sec   Timeout Time     : 180 sec
Garbage Collection Time : 120 sec   Distance         : 120
Redistribution           : None
```

### Command History

| Release | Modification |
| --- | --- |
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
| --- | --- | --- |
| 6200 | Operator (>) or Manager (#) | Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only. |

## show ipv6 ripng interface

`show ipv6 ripng [<PROCESS-ID>] interface [<INTERFACE-NAME>] [brief] [all-vrfs | vrf <VRF-NAME>]`

### Description

Displays information about RIPng enabled interfaces.

| Parameter | Description |
| --- | --- |
| *<PROCESS-ID>* | Specifies RIPng process ID. Range: 1-63. |
| *<INTERFACE-NAME>* | Specifies interface. |
| brief | Shows brief overview information for RIPng interface. |
| all-vrfs | Displays interface information for all VRFs. |
| vrf | Selects specific VRF. |
| *<VRF-NAME>* | Specifies VRF. |

### Usage

- Parameters display general RIPng information for a specific RIPng process.
- Parameters display general RIPng information for a specific or all VRFs.
- If a VRF is not mentioned, information for the default VRF is displayed.

### Examples

```
switch# show ipv6 ripng interface
Interface 1/1/1 is up, IPv6 Address is fe80::7272:cfff:fe70:67a
-----------------------------------------------------------------------
VRF              : Default            Process-ID      : 1
Status           : Oper Up            Mode            : Send and Receive
MTU              : 500                Poision Reverse : Enabled

Interface 1/1/2 is up, IPv6 Address is fe80::7272:cfff:fe70:67a
-----------------------------------------------------------------------
VRF              : Default            Process-ID      : 1
Status           : Admin Down         Mode            : Receive
MTU              : 500                Poision Reverse : Enabled

Interface 1/1/3 is up, IPv6 Address is fe80::7272:cfff:fe70:67a
-----------------------------------------------------------------------
VRF              : Default            Process-ID      : 1
Status           : Admin Down         Mode            : Send
MTU              : 500                Poision Reverse : Enabled

switch# show ipv6 ripng interface brief
VRF : default   Process-ID : 1
-------------------------------

Total Number of Interfaces: 2

Interface     IPv6-Address                      Status    MTU
-----------------------------------------------------------
1/1/1         fe80::7272:cfff:fe70:67a          up        500
1/1/2         fe80::7272:cfff:fe71:67a          up        500
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | Operator (>) or Manager (#) | Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only. |

## show ipv6 ripng neighbors

`show ipv6 ripng [<PROCESS-ID>] neighbors [<LINK-LOCAL-ADDRESS>] [all-vrfs | vrf <VRF-NAME>]`

### Description

Displays information about RIPng neighbors.

| Parameter | Description |
|---|---|
| *<PROCESS-ID>* | Specifies RIPng process ID. Range: 1-63. |
| neighbors | Specifies neighbor IP address. |
| *<LINK-LOCAL-ADDRESS>* | Specifies link-local address. |
| all-vrfs | Displays neighbor information for all VRFs. |
| vrf | Selects VRF to display neighbor information. |
| *<VRF-NAME>* | Specifies VRF name. |

### Usage

- Parameters display RIPng neighbor information for a specific RIPng process.
- Parameters display RIPng neighbor information for a specific neighbor.
- Parameters display general RIPng information for a specific or all VRFs.
- If a VRF is not mentioned, information for the default VRF is displayed.

### Examples

Displaying RIPng neighbor information for all VRFs:

```
switch# show ipv6 ripng neighbors all-vrfs
VRF : default            Process-ID : 1
----------------------------------------

Total Number of Neighbors: 1

Peer-Address      Type    Last-Update  Rcvd-Bad-Pkts    Rcvd-Bad-Routes
----------------------------------------------------------------------
fe80::7272:cfff:fe70:86ae
                  RIPng   0:0:7        4                5
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | Operator (>) or Manager (#) | Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only. |

## show ipv6 ripng routes

```
show ipv6 ripng [<PROCESS-ID>] routes [<PREFIX/LENGTH>] [all-vrfs | vrf <VRF-NAME>]
```

### Description

Displays RIPng routing table for a specific RIPng process.

| Parameter | Description |
|---|---|
| *<PROCESS-ID>* | Specifies RIPng process ID to display information for a specific RIPng process. Range: 1-63. |
| *<PREFIX/LENGTH>* | Specifies the network prefix. |
| `all-vrfs` | Displays RIPng routing information for all VRFs. |
| `vrf` | Selects VRF to display RIPng routing information. |
| *<VRF-NAME>* | Specifies VRF name. |

### Usage

- *<PREFIX/LENGTH>* is an optional parameter that displays RIPng routing table information for a specific subnet.
- If a VRF is not mentioned, information for the default VRF is displayed.

### Examples

Displaying RIPng routing table for all VRFs:

```
switch# show ipv6 ripng routes all-vrfs
VRF : default              Process-ID : 1
------------------------------------------

Prefix              Metric     Interface      Nexthop
-----------------------------------------------------------------------
2001:DB8:10::/64      2          1/1/1          FE80::2E0:E6FF:FE1B:8242
2002:DB8:10::/64      3          1/1/1          FE80::2E0:E6FF:FE1B:8242
2003:DB8:10::/64      1          1/1/1

VRF : vrf_1                Process-ID : 34
-------------------------------------------

Prefix               Metric     Interface      Nexthop
-----------------------------------------------------------------------
3001:DB8:10::/64      10         1/1/2          FE80::2E0:E6FF:FE1B:8232
3002:DB8:10::/64      14         1/1/2          FE80::2E0:E6FF:FE1B:8232
3003:DB8:10::/64       1         1/1/2
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | Operator (>) or Manager | Auditors or Administrators or local user group members with |

| Platforms | Command context | Authority |
| --- | --- | --- |
| | (#) | execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only. |

## show ipv6 ripng statistics

`show ipv6 ripng [<PROCESS-ID>] statistics [all-vrfs | vrf <VRF-NAME>]`

### Description

Displays RIPng statistics.

| Parameter | Description |
| --- | --- |
| *<PROCESS-ID>* | Specifies RIPng process ID. Range: 1-63. |
| `all-vrfs` | Displays statistics information for all VRFs. |
| `vrf` | Selects VRF and displays RIPng statistics for it. |
| *<VRF-NAME>* | Specifies VRF name. |

### Usage

- Parameters can display information for all VRFs or a specific VRF.
- If a VRF is not mentioned, information for the default VRF is displayed.

### Examples

Displaying RIPng statistics for all VRFs:

```
switch# show ipv6 ripng statistics all-vrfs
VRF : default   Process-ID : 1
-------------------------------------

Global Route Changes : 50
Global Queries       : 2
Last Cleared         : 0h 30m 28s ago

VRF : vrf_1     Process-ID : 34
-------------------------------------

Global Route Changes : 20
Global Queries       : 0
Last Cleared         : 0h 30m 28s ago
```

### Command History

| Release | Modification |
| --- | --- |
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | Operator (>) or Manager (#) | Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only. |

## show ipv6 ripng statistics interface

```
show ipv6 ripng [<PROCESS-ID>] statistics interface [<INTERFACE-NAME>] [all-vrfs | vrf <VRF-NAME>]
```

### Description

Displays RIPng statistics for RIPng enabled interfaces.

| Parameter | Description |
|---|---|
| *<PROCESS-ID>* | Specifies RIPng process ID. Range: 1-63. |
| *<INTERFACE-NAME>* | Specifies name of interface. |
| `all-vrfs` | Displays RIPng interface statistics for all VRFs. |
| `vrf` | Selects VRF to display RIPng interface statistics. |
| *<VRF-NAME>* | Specifies VRF name. |

### Usage

- Parameters can display information for all VRFs or a specific VRF.
- If a VRF is not mentioned, information for the default VRF is displayed.

### Examples

Displaying RIPng statistics for a RIPng enabled interface:

```
switch# show ipv6 ripng statistics interface 1/1/1
VRF : default  Process-ID : 1      interface 1/1/1
--------------------------------------------------

IPv6-Address         Trigger-Updates    Rcvd-Bad-Packets    Rcvd-Bad-Routes
-----------------------------------------------------------------------------
fe80::7272:cfff:fe70:86ae
                     15                 3                   4

Last Cleared: 0h 30m 28s ago
```

### Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | Operator (>) or Manager (#) | Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only. |

## show running-config

`show running config`

### Description

Displays all running configurations for all protocols including RIPng.

### Examples

Displaying all running configurations for all protocols including RIPng:

```
switch# show running-config
Current configuration:
!
!Version Halon 0.1.0 (Build: genericx86-64-Halon-0.1.0-master-20170309054955-dev)
!Schema version 0.1.8
lldp enable
timezone set utc
vrf green
vrf red
led base-loc_fdc on
led base-loc on
led base-hlth_fdc fast_blink
led base-pwr_fdc on
!
!
!
!
!
!
aaa authentication login default local
aaa authorization commands default none
!
!
!
!
router ospfv3 1
router ripng 1
    maximum-paths 5
    distance 1
    redistribute ospfv3 1
    timers update 40 timeout 200 garbage-collection 150
router ripng 1 vrf red
    default-information originate always
    maximum-paths 7
    distance 5
vlan 1
    no shutdown
interface lag 44
    no shutdown
    ipv6 address link-local
    ipv6 ripng 1
        send disable
interface 1
    no shutdown
```

```
    ipv6 address link-local
    ipv6 ripng 1
        receive disable
```

## Command History

| Release | Modification |
|---|---|
| 10.07 or earlier | -- |

## Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 6200 | Operator (>) or Manager (#) | Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only. |

# Accessing Aruba Support

| | |
|---|---|
| Aruba Support Services | https://www.arubanetworks.com/support-services/ |
| Aruba Support Portal | https://asp.arubanetworks.com/ |
| North America telephone | 1-800-943-4526 (US & Canada Toll-Free Number)<br>+1-408-754-1200 (Primary - Toll Number)<br>+1-650-385-6582 (Backup - Toll Number - Use only when all other numbers are not working) |
| International telephone | https://www.arubanetworks.com/support-services/contact-support/ |

Be sure to collect the following information before contacting Support:

- Technical support registration number (if applicable)
- Product name, model or version, and serial number
- Operating system name and version
- Firmware version
- Error messages
- Product-specific reports and logs
- Add-on products or components
- Third-party products or components

### Other useful sites

Other websites that can be used to find information:

| | |
|---|---|
| Airheads social forums and Knowledge Base | https://community.arubanetworks.com/ |
| Software licensing | https://lms.arubanetworks.com/ |
| End-of-Life information | https://www.arubanetworks.com/support-services/end-of-life/ |
| Aruba software and documentation | https://asp.arubanetworks.com/downloads |

# Accessing Updates

You can access updates from the Aruba Support Portal or the HPE My Networking Website.

## Aruba Support Portal

https://asp.arubanetworks.com/downloads

If you are unable to find your product in the Aruba Support Portal, you may need to search My Networking, where older networking products can be found:

## My Networking

https://www.hpe.com/networking/support

To view and update your entitlements, and to link your contracts and warranties with your profile, go to the Hewlett Packard Enterprise Support Center **More Information on Access to Support Materials** page:

https://support.hpe.com/portal/site/hpsc/aae/home/

Access to some updates might require product entitlement when accessed through the Hewlett Packard Enterprise Support Center. You must have an HP Passport set up with relevant entitlements.

Some software products provide a mechanism for accessing software updates through the product interface. Review your product documentation to identify the recommended software update method.

To subscribe to eNewsletters and alerts:

https://asp.arubanetworks.com/notifications/subscriptions (requires an active Aruba Support Portal (ASP) account to manage subscriptions). Security notices are viewable without an ASP account.

# Warranty Information

To view warranty information for your product, go to https://www.arubanetworks.com/support-services/product-warranties/.

# Regulatory Information

To view the regulatory information for your product, view the *Safety and Compliance Information for Server, Storage, Power, Networking, and Rack Products*, available at https://www.hpe.com/support/Safety-Compliance-EnterpriseProducts

### Additional regulatory information

Aruba is committed to providing our customers with information about the chemical substances in our products as needed to comply with legal requirements, environmental data (company programs, product recycling, energy efficiency), and safety information and compliance data, (RoHS and WEEE). For more information, see https://www.arubanetworks.com/company/about-us/environmental-citizenship/.

# Documentation Feedback

Aruba is committed to providing documentation that meets your needs. To help us improve the documentation, send any errors, suggestions, or comments to Documentation Feedback (docsfeedback-switching@hpe.com). When submitting your feedback, include the document title, part number, edition, and publication date located on the front cover of the document. For online help content, include the product name, product version, help edition, and publication date located on the legal notices page.